Question #812 *Topic 1*

A company hosts a data lake on Amazon S3. The data lake ingests data in Apache Parquet format from various data sources. The company uses multiple transformation steps to prepare the ingested data. The steps include filtering of anomalies, normalizing of data to standard date and time values, and generation of aggregates for analyses.

The company must store the transformed data in S3 buckets that data analysts access. The company needs a prebuilt solution for data transformation that does not require code. The solution must provide data lineage and data profiling. The company needs to share the data transformation steps with employees throughout the company.

Which solution will meet these requirements?

A. Configure an AWS Glue Studio visual canvas to transform the data. Share the transformation steps with employees by using AWS Glue jobs.

B. Configure Amazon EMR Serverless to transform the data. Share the transformation steps with employees by using EMR Serverless jobs.

C. Configure AWS Glue DataBrew to transform the data. Share the transformation steps with employees by using DataBrew recipes. **Most Voted**

D. Create Amazon Athena tables for the data. Write Athena SQL queries to transform the data. Share the Athena SQL queries with employees.

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

A - AWS Glue doesn't focus on data profiling or provide an easy way for non-technical employees to collaborate on transformation steps.
B - You need to code.
C - AWS Glue DataBrew is a no-code tool specifically designed for data preparation and transformation.
D - Let's say writing SQLs are not coding, still, Athena lacks data lineage, data profiling, and the ability to reuse transformations as recipes.

upvoted 2 times

**Lingo43** 1 year, 11 months ago

**Selected Answer: C**

AWS Glue DataBrew: This is a visual data preparation tool that allows you to clean and normalize data without writing code. It has built-in transformations for common tasks like filtering anomalies, normalizing dates, and generating aggregates. It also provides data lineage and profiling capabilities, which are required by the company.
DataBrew Recipes: These are reusable workflows that define the data transformation steps. They can be easily shared with other employees, making i simple to collaborate on data preparation tasks.

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC

AWS Glue DataBrew is a visual data preparation tool that enables users to clean and normalize data without writing any code. Using DataBrew helps reduce the time it takes to prepare data for analytics and machine learning (ML) by up to 80 percent, compared to custom developed data preparation. You can choose from over 250 ready-made transformations to automate data preparation tasks, such as filtering anomalies, converting data to standard formats, and correcting invalid values.
https://docs.aws.amazon.com/databrew/latest/dg/what-is.html

upvoted 4 times

**Linuslin** 2 years ago

Selected Answer: C

C is correct.
https://docs.aws.amazon.com/databrew/latest/dg/recipes.html

upvoted 1 times

**seetpt** 2 years, 3 months ago

Selected Answer: C

Agree with C

upvoted 2 times

**asdfcdsxdfc** 2 years, 3 months ago

Selected Answer: C

Should be C

upvoted 4 times

Question #813 *Topic 1*

A solutions architect runs a web application on multiple Amazon EC2 instances that are in individual target groups behind an Application Load Balancer (ALB). Users can reach the application through a public website.

The solutions architect wants to allow engineers to use a development version of the website to access one specific development EC2 instance to test new features for the application. The solutions architect wants to use an Amazon Route 53 hosted zone to give the engineers access to the development instance. The solution must automatically route to the development instance even if the development instance is replaced.

Which solution will meet these requirements?

A. Create an A Record for the development website that has the value set to the ALB. Create a listener rule on the ALB that forwards requests for the development website to the target group that contains the development instance. **Most Voted**

B. Recreate the development instance with a public IP address. Create an A Record for the development website that has the value set to the public IP address of the development instance.

C. Create an A Record for the development website that has the value set to the ALB. Create a listener rule on the ALB to redirect requests for the development website to the public IP address of the development instance.

D. Place all the instances in the same target group. Create an A Record for the development website. Set the value to the ALB. Create a listener rule on the ALB that forwards requests for the development website to the target group.

**Correct Answer:** A

*Community vote distribution*

A (100%)



## Comments

**Mikado211** Highly Voted 2 years, 2 months ago

Both A and C look correct but with the C you pass through the ALB to be redirected to a public IP (so go outside) to come back again through this public IP which is not ideal.

The answer A is much cleaner and simplier with a dedicated target group and a listener rule pointing it.

upvoted 8 times

**elmyth** 1 year, 9 months ago

Public IP is not permanent, so definitely not B and C

upvoted 3 times

**MatAlves** 1 year, 8 months ago

B: Directly using a public IP address ties the A Record to a specific instance, which is not ideal since replacing the instance would require manually updating the Route 53 record.

C: Redirecting to a public IP address of the development instance is similar to option B, and would also require manual updates if the instance changes.

upvoted 2 times

**gdf54634** Highly Voted 2 years, 3 months ago

Selected Answer: A

Should be A as it points to the target group for easy replacement etc

upvoted 7 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

Selected Answer: A

B - If the development instance is replaced, the public IP address changes, requiring manual DNS updates in Route 53.
C - The ALB listener rule cannot directly redirect traffic based on the instance's public IP address without using a target group.
D - Without traffic isolation, you could easily route requests for the development website to production instances. Dangerous and just plain wrong.
upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago
FYI, I copied this after googling.
What is a DNS "A record"?
The "A" stands for "address" and this is the most fundamental type of DNS record: it indicates the IP address of a given domain. For example, if you pull the DNS records of cloudflare.com, the A record currently returns an IP address of: 104.17.210.9.

A records only hold IPv4 addresses. If a website has an IPv6 address, it will instead use an "AAAA" record.
upvoted 1 times

**Cpso** 1 year, 6 months ago
Selected Answer: A
A.But not totally correct. ALB IP can be change. should use CNAME record.
upvoted 2 times

**GOTJ** 1 year, 2 months ago
That's right! In fact, you shouldn't associate an "A" record to an ALB. You must use a CNAME record instead
upvoted 1 times

**JA2018** 1 year, 6 months ago
Selected Answer: A
By setting the A Record to point to the ALB, you ensure that even if the development instance is replaced, the DNS record will still direct traffic to the correct target group which contains the new development instance, maintaining accessibility for engineers.
upvoted 1 times

**JA2018** 1 year, 6 months ago
Why we have to rule out the other options:
B: Using the public IP address of the development instance directly would break access if the instance is replaced, as the IP address would change.
C: Redirecting requests from the ALB to the development instance's public IP is not a good practice as it bypasses the load balancing functionality of the ALB.
D: Placing all instances in the same target group would not allow for separate access to the development instance as the load balancer would distribute traffic across all instances, including production ones.
upvoted 1 times

**JA2018** 1 year, 6 months ago
To conclude, option A provides the necessary flexibility and reliability to access the development instance while utilizing the ALB for load balancing and automatic failover capabilities.
upvoted 1 times

**Scheldon** 2 years ago
Selected Answer: A
AnswerA,

With ALB which will point to appopriate dev group we will be able easy to create HA for dev servers.
upvoted 3 times

**asdfcdsxdfc** 2 years, 3 months ago
Selected Answer: A
I think its A
upvoted 2 times

Question #814 Topic 1

A company runs a container application on a Kubernetes cluster in the company's data center. The application uses Advanced Message Queuing Protocol (AMQP) to communicate with a message queue. The data center cannot scale fast enough to meet the company's expanding business needs. The company wants to migrate the workloads to AWS.

Which solution will meet these requirements with the LEAST operational overhead?

A. Migrate the container application to Amazon Elastic Container Service (Amazon ECS). Use Amazon Simple Queue Service (Amazon SQS) to retrieve the messages.

B. Migrate the container application to Amazon Elastic Kubernetes Service (Amazon EKS). Use Amazon MQ to retrieve the messages. Most Voted

C. Use highly available Amazon EC2 instances to run the application. Use Amazon MQ to retrieve the messages.

D. Use AWS Lambda functions to run the application. Use Amazon Simple Queue Service (Amazon SQS) to retrieve the messages.

**Correct Answer:** B

*Community vote distribution*

B (95%) 5%

## Comments

Mikado211 Highly Voted 2 years, 2 months ago

Selected Answer: B

This question is a trap because A is definitely the answer for a Least overhead (ECS + SQS) and in a real life scenario could be good in 99% of cases.

However SQS do not implement AMQP (SQS is only a simple queueing system very basic) so we have to use Amazon MQ.

In terms of containers EKS will always be a better solution than a manual setup of Docker.

Good solution would have been ECS+AmazonMQ not given here

Lambda can work with containers, but since there are limitations like 15 minutes limit we can't really consider it as a good solution.

So B is the least bad solution.

upvoted 16 times

Dantecito Most Recent 1 year, 3 months ago

Selected Answer: B

We were asked for the LEAST operational overhead, so the company already has a managed message broker and thats why we use Amazon MQ and they are already using Kubernetes.

upvoted 1 times

Scheldon 2 years ago

Selected Answer: B

AnswerB

For me B is a correct solution. In question AMQP is mentioned and Amazon on his doc page about MQ is providing such information:
"Amazon MQ is a managed message broker service that provides compatibility with many popular message brokers. We recommend Amazon MQ for migrating applications from existing message brokers that rely on compatibility with APIs such as JMS or protocols such as AMQP 0-9-1, AMQP 1.0, MQTT, OpenWire, and STOMP."
I cannot be coincidence that documentation is mentioning about AMQP.

https://docs.aws.amazon.com/amazon-mq/latest/developer-guide/welcome.html

upvoted 1 times

**sandordini** 2 years, 1 month ago

**Selected Answer: A**

I'd go for A.
Although the ideal solution and least modification would require B, with heavy rework the application can be (most likely) adopted to ECS+SQS. As it is an AWS exam, not a vendor-agnostic SA exam, A will be the correct answer.

upvoted 1 times

**seetpt** 2 years, 3 months ago

**Selected Answer: B**

B because only solution with Kubernetes

upvoted 1 times

**asdfcdsxdfc** 2 years, 3 months ago

**Selected Answer: B**

Should be B

upvoted 2 times

Question #815 Topic 1

An online gaming company hosts its platform on Amazon EC2 instances behind Network Load Balancers (NLBs) across multiple AWS Regions. The NLBs can route requests to targets over the internet. The company wants to improve the customer playing experience by reducing end-to-end load time for its global customer base.

Which solution will meet these requirements?

A. Create Application Load Balancers (ALBs) in each Region to replace the existing NLBs. Register the existing EC2 instances as targets for the ALBs in each Region.

B. Configure Amazon Route 53 to route equally weighted traffic to the NLBs in each Region.

C. Create additional NLBs and EC2 instances in other Regions where the company has large customer bases.

D. Create a standard accelerator in AWS Global Accelerator. Configure the existing NLBs as target endpoints. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (100%)

## Comments

**Mikado211** Highly Voted 2 years, 2 months ago

**Selected Answer: D**

In such situation if you had an ALB you would use Cloudfront
Since you have a NLB you use AWS Global Accelerator
So D.

upvoted 8 times

**asdfcdsxdfc** Highly Voted 2 years, 3 months ago

**Selected Answer: D**

Should be D

upvoted 5 times

**Scheldon** Most Recent 2 years ago

**Selected Answer: D**

AnswerD

Usage of Global Accelerator should help here.
"
Acceleration for latency-sensitive applications
Many applications, especially in areas such as gaming, media, mobile apps, ad-tech, and financials, require very low latency for a great user experience. To improve the user experience, Global Accelerator directs user traffic to the application endpoint that is nearest to the client, which reduces internet latency and jitter. Global Accelerator routes traffic to the closest edge location by using Anycast, and then routes it to the closest regional endpoint over the AWS global network. Global Accelerator quickly reacts to changes in network performance to improve your users' application performance.
"
https://docs.aws.amazon.com/global-accelerator/latest/dg/introduction-benefits-of-migrating.html

upvoted 3 times

**seetpt** 2 years, 3 months ago

**Selected Answer: D**

Agree with D

upvoted 2 times

Question #816 Topic 1

A company has an on-premises application that uses SFTP to collect financial data from multiple vendors. The company is migrating to the AWS Cloud. The company has created an application that uses Amazon S3 APIs to upload files from vendors.

Some vendors run their systems on legacy applications that do not support S3 APIs. The vendors want to continue to use SFTP-based applications to upload data. The company wants to use managed services for the needs of the vendors that use legacy applications.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an AWS Database Migration Service (AWS DMS) instance to replicate data from the storage of the vendors that use legacy applications to Amazon S3. Provide the vendors with the credentials to access the AWS DMS instance.

B. Create an AWS Transfer Family endpoint for vendors that use legacy applications. Most Voted

C. Configure an Amazon EC2 instance to run an SFTP server. Instruct the vendors that use legacy applications to use the SFTP server to upload data.

D. Configure an Amazon S3 File Gateway for vendors that use legacy applications to upload files to an SMB file share.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

asdfcdsxdfc Highly Voted 2 years, 3 months ago

Selected Answer: B

B is correct

upvoted 8 times

LeonSauveterre Most Recent 1 year, 5 months ago

Selected Answer: B

A - DMS is designed for database migrations, not for file-based SFTP uploads.
B - AWS Transfer Family is a fully managed service that supports SFTP, FTPS, and FTP protocols.
C - OK but significant operational overhead.
D - S3 File Gateway supports SMB and NFS, not SFTP.

upvoted 2 times

Dantecito 1 year, 3 months ago

also C - Not a managed services.

upvoted 1 times

Johnoppong101 1 year, 9 months ago

Selected Answer: B

Answer is B

upvoted 1 times

Sergiuss95 2 years, 1 month ago

Selected Answer: B

Explanation:

AWS Transfer Family is a fully managed service that allows you to set up SFTP, FTPS, and FTP endpoints for accessing Amazon S3 and Amazon EFS storage.
By creating an AWS Transfer Family endpoint, the company can provide vendors with the familiar SFTP interface to upload data directly to Amazon S:
without requiring them to make any changes to their legacy applications.

This solution eliminates the need for the company to manage and maintain additional infrastructure such as EC2 instances or file gateways. AWS Transfer Family handles scalability, availability, and security, reducing operational overhead for the company.

upvoted 4 times

**seetpt** 2 years, 3 months ago

**Selected Answer: B**

B is correct

upvoted 2 times

Question #817 *Topic 1*

A marketing team wants to build a campaign for an upcoming multi-sport event. The team has news reports from the past five years in PDF format. The team needs a solution to extract insights about the content and the sentiment of the news reports. The solution must use Amazon Textract to process the news reports.

Which solution will meet these requirements with the LEAST operational overhead?

A. Provide the extracted insights to Amazon Athena for analysis. Store the extracted insights and analysis in an Amazon S3 bucket.

B. Store the extracted insights in an Amazon DynamoDB table. Use Amazon SageMaker to build a sentiment model.

C. Provide the extracted insights to Amazon Comprehend for analysis. Save the analysis to an Amazon S3 bucket. **Most Voted**

D. Store the extracted insights in an Amazon S3 bucket. Use Amazon QuickSight to visualize and analyze the data.

**Correct Answer:** C

*Community vote distribution*

C (100%)



## Comments

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC

"

Amazon Comprehend uses natural language processing (NLP) to extract insights about the content of documents. It develops insights by recognizing the entities, key phrases, language, sentiments, and other common elements in a document. Use Amazon Comprehend to create new products based on understanding the structure of documents. For example, using Amazon Comprehend you can search social networking feeds for mentions of products or scan an entire document repository for key phrases.

"

https://docs.aws.amazon.com/comprehend/latest/dg/what-is.html

upvoted 3 times

**zinabu** 2 years, 1 month ago

**Selected Answer: C**

Selected Answer: C

amazon comprehend= sentiment analysis

upvoted 2 times

**zinabu** 2 years, 2 months ago

Selected Answer: C

amazon comprehend= sentiment analysis

upvoted 4 times

**alawada** 2 years, 2 months ago

**Selected Answer: C**

Whenever new PDF files are uploaded to the designated S3 bucket, the Lambda function will be triggered to extract insights using Textract and Comprehend.

upvoted 3 times

**Mikado211** 2 years, 2 months ago

**Selected Answer: C**

When you have words like "sentiment" in a sentence, it's related to Comprehend

So C.

upvoted 2 times

**seetpt** 2 years, 3 months ago

**Selected Answer: C**

Maybe C?

upvoted 2 times

**asdfcdsxdfc** 2 years, 3 months ago

Shouldn't it be C?

upvoted 4 times

Question #818 *Topic 1*

A company's application runs on Amazon EC2 instances that are in multiple Availability Zones. The application needs to ingest real-time data from third-party applications.

The company needs a data ingestion solution that places the ingested raw data in an Amazon S3 bucket.

Which solution will meet these requirements?

A. Create Amazon Kinesis data streams for data ingestion. Create Amazon Kinesis Data Firehose delivery streams to consume the Kinesis data streams. Specify the S3 bucket as the destination of the delivery streams. **Most Voted**

B. Create database migration tasks in AWS Database Migration Service (AWS DMS). Specify replication instances of the EC2 instances as the source endpoints. Specify the S3 bucket as the target endpoint. Set the migration type to migrate existing data and replicate ongoing changes.

C. Create and configure AWS DataSync agents on the EC2 instances. Configure DataSync tasks to transfer data from the EC2 instances to the S3 bucket.

D. Create an AWS Direct Connect connection to the application for data ingestion. Create Amazon Kinesis Data Firehose delivery streams to consume direct PUT operations from the application. Specify the S3 bucket as the destination of the delivery streams.

**Correct Answer:** A

*Community vote distribution*

A (78%) C (22%)

## Comments

**asdfcdsxdfc** Highly Voted 2 years, 3 months ago

**Selected Answer: A**

A is correct

upvoted 9 times

**JA2018** Most Recent 1 year, 6 months ago

**Selected Answer: A**

Why A?

Kinesis Data Streams: This is designed for real-time data ingestion, which is exactly what the scenario requires.

Kinesis Data Firehose: This service can then be used to efficiently deliver the ingested data from Kinesis streams to an S3 bucket, fulfilling the need tc store raw data in S3

upvoted 1 times

**JA2018** 1 year, 6 months ago

Why the other options are incorrect:
B. AWS Database Migration Service (DMS):
DMS is primarily used for migrating databases, not for real-time data ingestion from third-party applications.
C. AWS DataSync:
While DataSync can transfer data to S3, it's not optimized for real-time data ingestion and would likely not be the best choice for this scenario.
D. AWS Direct Connect:
Direct Connect is used for dedicated private network connections between your on-premises network and AWS, not for real-time data ingestion from third-party applications.

upvoted 1 times

**JA2018** 1 year, 6 months ago

Key points to remember:
When dealing with real-time data ingestion, Kinesis Data Streams is generally the preferred solution on AWS.

Kinesis Data Firehose can be used to easily stream data from Kinesis to various destinations like S3 buckets.
upvoted 1 times

**mk168898** 1 year, 7 months ago

**Selected Answer: A**

real time data => amazon kinesis data stream
upvoted 2 times

**MatAlves** 1 year, 8 months ago

**Selected Answer: A**

- Amazon Kinesis Data Streams: designed for real-time data ingestion.
- Kinesis Data Firehose: can consume data from Kinesis Data Streams and automatically deliver it to Amazon S3.

A is the answer.
upvoted 2 times

**Mayank0502** 1 year, 11 months ago

**Selected Answer: C**

each ec2 needs to proceed data separately
upvoted 1 times

**xBUGx** 2 years, 2 months ago

**Selected Answer: C**

A is best solution, but i think the question is saying "The application needs to ingest real-time data from third-party applications." and the applicatio
is run on EC2.
so i think we need a solution that works with the application on EC2 for this question?
upvoted 3 times

**Sergiuss95** 2 years, 1 month ago

DataSync is more suitable for transferring data between on-premises storage systems and AWS, rather than ingesting real-time data. Best solution i
A
upvoted 5 times

**seetpt** 2 years, 3 months ago

Agree with A
upvoted 2 times

Question #819 Topic 1

A company's application is receiving data from multiple data sources. The size of the data varies and is expected to increase over time. The current maximum size is 700 KB. The data volume and data size continue to grow as more data sources are added.

The company decides to use Amazon DynamoDB as the primary database for the application. A solutions architect needs to identify a solution that handles the large data sizes.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Create an AWS Lambda function to filter the data that exceeds DynamoDB item size limits. Store the larger data in an Amazon DocumentDB (with MongoDB compatibility) database.

B. Store the large data as objects in an Amazon S3 bucket. In a DynamoDB table, create an item that has an attribute that points to the S3 URL of the data. Most Voted

C. Split all incoming large data into a collection of items that have the same partition key. Write the data to a DynamoDB table in a single operation by using the BatchWriteItem API operation.

D. Create an AWS Lambda function that uses gzip compression to compress the large objects as they are written to a DynamoDB table.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

**Neung983** Highly Voted 2 years, 3 months ago

Selected Answer: B

option B is the most operationally efficient solution for handling large data sizes in Amazon DynamoDB.

upvoted 9 times

**seetpt** Highly Voted 2 years, 3 months ago

Selected Answer: B

B is correct

upvoted 5 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

Selected Answer: B

A - Introducing another database. No need.
B - Decoupling the large data storage (S3) from the metadata or smaller structured data (DynamoDB) is an AWS-recommended pattern.
C - OK but this is too complicated.
D - Compression does not guarantee that the data will always fit within DynamoDB's 400 KB limit.

upvoted 2 times

**Scheldon** 2 years ago

Selected Answer: B

AnswerB

Compresion of data in DynamoDB is a good idea especially for text data link from forum, but to do that we do not need AWS Lambda if I'm not wrong.
In other head Storing big object on S3 and seving URL to it in DynamoDB is one of best practices mentioned by Amazon. Hence we do not know what kind of data we are storing in DB and how big objects will be in the future option B looks like the best solution.
https://aws.amazon.com/blogs/database/large-object-storage-strategies-for-amazon-dynamodb/ <<<< Read Option 2
https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/bp-use-s3-too.html

upvoted 1 times

**Sergiuss95** 2 years, 1 month ago

**Selected Answer: B**

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/bp-use-s3-too.html

upvoted 2 times

Question #820 *Topic 1*

A company is migrating a legacy application from an on-premises data center to AWS. The application relies on hundreds of cron jobs that run between 1 and 20 minutes on different recurring schedules throughout the day.

The company wants a solution to schedule and run the cron jobs on AWS with minimal refactoring. The solution must support running the cron jobs in response to an event in the future.

Which solution will meet these requirements?

A. Create a container image for the cron jobs. Use Amazon EventBridge Scheduler to create a recurring schedule. Run the cron job tasks as AWS Lambda functions.

B. Create a container image for the cron jobs. Use AWS Batch on Amazon Elastic Container Service (Amazon ECS) with a scheduling policy to run the cron jobs.

C. Create a container image for the cron jobs. Use Amazon EventBridge Scheduler to create a recurring schedule. Run the cron job tasks on AWS Fargate. **Most Voted**

D. Create a container image for the cron jobs. Create a workflow in AWS Step Functions that uses a Wait state to run the cron jobs at a specified time. Use the RunTask action to run the cron job tasks on AWS Fargate.

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**Kezuko** Highly Voted 1 year, 8 months ago

Give yourself a pat on the back when you reach this question, its been a long run

upvoted 21 times

**Drew3000** 1 year, 8 months ago

I finally managed to get through the last question, then refreshed the page , and they have added more questions.

upvoted 8 times

**LeonSauveterre** 1 year, 5 months ago

I know right. Where did they get so many questions? It's been a long run but too long now...

upvoted 2 times

**Linuslin** Highly Voted 1 year, 6 months ago

**Selected Answer: C**

Lambda has 15 mins limit, so A is out.

B works, but you have to run highly-available virtual machines or containers waiting for the event to happen.

C is the best answer in this question, with AWS Fargate allows you to pay for only what you use and free you from provisioning, configuring, and scaling clusters of Amazon EC2 instances.

https://aws.amazon.com/blogs/containers/migrate-cron-jobs-to-event-driven-architectures-using-amazon-elastic-container-service-and-amazon-eventbridge/

upvoted 5 times

**FlyingHawk** Most Recent 1 year, 4 months ago

**Selected Answer: C**

As an example, one customer had hundreds of cron jobs configured on a single on-premise server. They chose to run these various cron jobs as separate scheduled tasks on Amazon Elastic Container Service (Amazon ECS) using AWS Fargate. They used Amazon EventBridge to schedule the tasks because it triggers an Amazon ECS task at scheduled times rather than having to run highly-available virtual machines or containers waiting for the event to happen. They chose Amazon ECS because of its simple container orchestration. And they chose AWS Fargate because it allows them to pay for only what they use and frees them from provisioning, configuring, and scaling clusters of Amazon EC2 instances.

upvoted 1 times

**cvoiceip** 1 year, 9 months ago

Ans : C

https://aws.amazon.com/blogs/containers/migrate-cron-jobs-to-event-driven-architectures-using-amazon-elastic-container-service-and-amazon-eventbridge/

upvoted 1 times

**seetpt** 1 year, 9 months ago

**Selected Answer: C**

C because lambda has 15min time limit.

upvoted 2 times

**asdfcdsxdfc** 1 year, 9 months ago

**Selected Answer: C**

its either A or C. C looks correct because lambda works for 15 mins and the question says between 1-20

upvoted 4 times

Question #821 *Topic 1*

A company uses Salesforce. The company needs to load existing data and ongoing data changes from Salesforce to Amazon Redshift for analysis. The company does not want the data to travel over the public internet.

Which solution will meet these requirements with the LEAST development effort?

A. Establish a VPN connection from the VPC to Salesforce. Use AWS Glue DataBrew to transfer data.

B. Establish an AWS Direct Connect connection from the VPC to Salesforce. Use AWS Glue DataBrew to transfer data.

C. Create an AWS PrivateLink connection in the VPC to Salesforce. Use Amazon AppFlow to transfer data. **Most Voted**

D. Create a VPC peering connection to Salesforce. Use Amazon AppFlow to transfer data.

**Correct Answer:** C

*Community vote distribution*

C (100%)



## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

A - VPN connection, public internet, no.
B - Significant setup effort and cost. Also, DataBrew is not suitable for loading ongoing data changes.
C - Amazon AppFlow is a fully managed integration service that natively supports Salesforce and Amazon Redshift.
D - VPC peering is for connecting two AWS VPCs. Salesforce is external to AWS, so no.

upvoted 3 times

**LeonSauveterre** 1 year, 5 months ago

Also, maybe this will help you along the way (things after "==>" are their disadvantages):

1. VPN Connection: For secure, encrypted traffic between on-premises and AWS over the public internet. ==> Relies on public internet (higher latency)

2. Direct Connect: For dedicated private network connectivity between on-premises and AWS (like data centers with large workloads). ==> Higher cost, longer setup time.

3. PrivateLink: For secure private connectivity to AWS services or third-party services (like Salesforce) within AWS. ==> Simple setup but limited to services configured with PrivateLink (not general connectivity).

4. VPC Peering: For direct connectivity between two AWS VPCs (like multi-account or region scenarios). ==> Does not support transitive connections; not suitable for external services like Salesforce.

upvoted 2 times

**mk168898** 1 year, 7 months ago

3rd party SaaS salesforce integration => Use AWS AppFlow
So left C and D
Not D because VPC Peering need 2 VPC and 3rd party SaaS does not have a VPC

upvoted 2 times

**MatAlves** 1 year, 8 months ago

**Selected Answer: C**

AWS PrivateLink: This service enables private connectivity between VPCs and supported AWS services, effectively keeping data off the public internet
It allows secure communication without exposing your data to the internet.

Amazon AppFlow: This is a fully managed integration service that simplifies data transfer between SaaS applications (like Salesforce) and AWS services (like Amazon Redshift).

upvoted 2 times

**Johnoppong101** 1 year, 9 months ago

Selected Answer: C

C is the answer

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

Selected Answer: C

Private link for sure

upvoted 2 times

**Scheldon** 2 years ago

Selected Answer: C

AnswerC
To connect your own VPC with third-party VPC we need to use ProvateLink.

AWS PrivateLink is a highly available, scalable technology that you can use to privately connect your VPC to services as if they were in your VPC. You do not need to use an internet gateway, NAT device, public IP address, AWS Direct Connect connection, or AWS Site-to-Site VPN connection to allow communication with the service from your private subnets. Therefore, you control the specific API endpoints, sites, and services that are reachable from your VPC.

https://docs.aws.amazon.com/vpc/latest/privatelink/what-is-privatelink.html

upvoted 4 times

**Kaula** 2 years, 2 months ago

Selected Answer: C

Should be C

upvoted 2 times

**Kaula** 2 years, 2 months ago

C
https://docs.aws.amazon.com/connect/latest/adminguide/integrate-salesforce-tasks.html
https://docs.aws.amazon.com/connect/latest/adminguide/vpc-interface-endpoints.html

upvoted 3 times

Question #822 Topic 1

A company recently migrated its application to AWS. The application runs on Amazon EC2 Linux instances in an Auto Scaling group across multiple Availability Zones. The application stores data in an Amazon Elastic File System (Amazon EFS) file system that uses EFS Standard-Infrequent Access storage. The application indexes the company's files. The index is stored in an Amazon RDS database.

The company needs to optimize storage costs with some application and services changes.

Which solution will meet these requirements MOST cost-effectively?

A. Create an Amazon S3 bucket that uses an Intelligent-Tiering lifecycle policy. Copy all files to the S3 bucket. Update the application to use Amazon S3 API to store and retrieve files. Most Voted

B. Deploy Amazon FSx for Windows File Server file shares. Update the application to use CIFS protocol to store and retrieve files.

C. Deploy Amazon FSx for OpenZFS file system shares. Update the application to use the new mount point to store and retrieve files.

D. Create an Amazon S3 bucket that uses S3 Glacier Flexible Retrieval. Copy all files to the S3 bucket. Update the application to use Amazon S3 API to store and retrieve files as standard retrievals.

**Correct Answer:** A

*Community vote distribution*

A (100%)

## Comments

**xBUGx** Highly Voted 2 years, 2 months ago

**Selected Answer: A**

i go with A since there is no other better options

upvoted 6 times

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: A**

B, C, D off.
B - Windows
C - OpenZFS Costs ~ NFS Costs *3 (optimized mainly for high-performance Data analysis)
D - Glacier Standard retrieval: 12 Hrs

upvoted 5 times

**Dantecito** 1 year, 3 months ago

Expedited retrievals: Typically complete in 1–5 minutes
Standard retrievals: Typically complete in 3–5 hours
Free bulk retrievals: Typically complete in 5–12 hours

upvoted 1 times

**TeamACT** 1 year, 11 months ago

Glacier standard retrieval is 5 hrs and not 12. It is glacier deep archive standard that is 12 hours.

upvoted 2 times

**mk168898** Most Recent 1 year, 7 months ago

**Selected Answer: A**

Chose A because optimise storage costs => s3 bucket intelligent tiering

upvoted 2 times

**MatAlves** 1 year, 8 months ago

**Selected Answer: A**

Since the company is ok with "some application and services changes", then A is definitely the most cost-effective option.

D can take up to few hours to complete retrievals.

upvoted 2 times

**Johnoppong101** 1 year, 9 months ago

**Selected Answer: A**

EFS Infrequent access -> milliseconds retrieval time, can`t replace with 12hrs for Glacier.

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: A**

optimize storage costs with some application and services changes -> Intelligent Tiering

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: A**

AnswerA
https://docs.aws.amazon.com/AmazonS3/latest/userguide/intelligent-tiering-overview.html

upvoted 3 times

**TruthWS** 2 years, 2 months ago

A is correct

upvoted 2 times

**Kenneth99** 2 years, 2 months ago

should be A?

upvoted 3 times

Question #823 *Topic 1*

A robotics company is designing a solution for medical surgery. The robots will use advanced sensors, cameras, and AI algorithms to perceive their environment and to complete surgeries.

The company needs a public load balancer in the AWS Cloud that will ensure seamless communication with backend services. The load balancer must be capable of routing traffic based on the query strings to different target groups. The traffic must also be encrypted.

Which solution will meet these requirements?

A. Use a Network Load Balancer with a certificate attached from AWS Certificate Manager (ACM). Use query parameter-based routing.

B. Use a Gateway Load Balancer. Import a generated certificate in AWS Identity and Access Management (IAM). Attach the certificate to the load balancer. Use HTTP path-based routing.

C. Use an Application Load Balancer with a certificate attached from AWS Certificate Manager (ACM). Use query parameter-based routing. **Most Voted**

D. Use a Network Load Balancer. Import a generated certificate in AWS Identity and Access Management (IAM). Attach the certificate to the load balancer. Use query parameter-based routing.

**Correct Answer:** C

*Community vote distribution*

C (100%)



## Comments

**TruthWS** Highly Voted 2 years, 2 months ago

Option C - parameter is not a thing NLB can process

upvoted 6 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

**Selected Answer: C**

Neither Network Load Balancer nor Gateway Load Balancer supports query parameter-based routing.

upvoted 1 times

**mk168898** 1 year, 7 months ago

**Selected Answer: C**

only ALB can route traffic based on query parameter, NLB cannot.
So C

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC

I was thinking that it will be network LB but after checks it occurs that only Application LB is able to redirect/forward traffic based on query string. and for encrypted traffic ACM is needed

We recommend that you create certificates for your load balancer using AWS Certificate Manager (ACM). ACM supports RSA certificates with 2048, 3072, and 4096-bit key lengths, and all ECDSA certificates. ACM integrates with Elastic Load Balancing so that you can deploy the certificate on your load balancer. For more information, see the AWS Certificate Manager User Guide.

https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-listeners.html#rule-condition-types
https://docs.aws.amazon.com/elasticloadbalancing/latest/application/create-https-listener.html

upvoted 4 times

**alawada** 2 years, 2 months ago

**Selected Answer: C**

Provision an Application Load Balancer (ALB) in the AWS Cloud. ALB is a Layer 7 load balancer that supports advanced routing features, including path-based routing.

upvoted 3 times

Question #824 Topic 1

A company has an application that runs on a single Amazon EC2 instance. The application uses a MySQL database that runs on the same EC2 instance. The company needs a highly available and automatically scalable solution to handle increased traffic.

Which solution will meet these requirements?

A. Deploy the application to EC2 instances that run in an Auto Scaling group behind an Application Load Balancer. Create an Amazon Redshift cluster that has multiple MySQL-compatible nodes.

B. Deploy the application to EC2 instances that are configured as a target group behind an Application Load Balancer. Create an Amazon RDS for MySQL cluster that has multiple instances.

C. Deploy the application to EC2 instances that run in an Auto Scaling group behind an Application Load Balancer. Create an Amazon Aurora Serverless MySQL cluster for the database layer. Most Voted

D. Deploy the application to EC2 instances that are configured as a target group behind an Application Load Balancer. Create an Amazon ElastiCache for Redis cluster that uses the MySQL connector.

**Correct Answer:** C

*Community vote distribution*

C (100%)



## Comments

**haci** Highly Voted 2 years, 2 months ago

Selected Answer: C

Target groups are just a group of Ec2 instances. Target groups are closely associated with ELB and not ASG. We can just use ELB and Target groups tc route requests to EC2 instances. With this setup, there is no autoscaling which means instances cannot be added or removed when your load increases/decreases.

upvoted 8 times

**sheilawu** Highly Voted 2 years ago

Selected Answer: C

scalable solution= Amazon Aurora Serverless

upvoted 6 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

Selected Answer: C

I don't see how Redshift or ElastiCache relates to the question.

upvoted 1 times

**MatAlves** 1 year, 8 months ago

Selected Answer: C

Option B:
Target Group: it doesn't inherently imply automatic scaling. You would need to manage scaling separately, either manually or through other mechanisms like scheduled actions.

Option C:
Auto Scaling Group: This ensures that the EC2 instances can automatically scale in or out based on traffic and demand.

"The company needs a highly available and automatically scalable solution" => C

upvoted 2 times

**Scheldon** 2 years ago

Selected Answer: C

AnswerC

highly available and automatically scalable solution = Auto Scale for EC2 in front which we will have ALB and Aurora Server less will give us perfect decoupled solution in which we can increase amount of servers per need and in case of server failure AutoScale will run new EC2 instance

upvoted 3 times

**camps** 2 years, 2 months ago

It's C!

upvoted 2 times

**TruthWS** 2 years, 2 months ago

Option C - keywords HA, automatically scalable

upvoted 2 times

**alawada** 2 years, 2 months ago

**Selected Answer: C**

C Is what I will go for

upvoted 3 times

Question #825 Topic 1

A company is planning to migrate data to an Amazon S3 bucket. The data must be encrypted at rest within the S3 bucket. The encryption key must be rotated automatically every year.

Which solution will meet these requirements with the LEAST operational overhead?

A. Migrate the data to the S3 bucket. Use server-side encryption with Amazon S3 managed keys (SSE-S3). Use the built-in key rotation behavior of SSE-S3 encryption keys.

B. Create an AWS Key Management Service (AWS KMS) customer managed key. Enable automatic key rotation. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Migrate the data to the S3 bucket. **Most Voted**

C. Create an AWS Key Management Service (AWS KMS) customer managed key. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Migrate the data to the S3 bucket. Manually rotate the KMS key every year.

D. Use customer key material to encrypt the data. Migrate the data to the S3 bucket. Create an AWS Key Management Service (AWS KMS) key without key material. Import the customer key material into the KMS key. Enable automatic key rotation.

**Correct Answer:** B

*Community vote distribution*

B (75%) A (25%)



## Comments

**EllenLiu** 1 year, 5 months ago

**Selected Answer: B**

https://repost.aws/questions/QUES_1VN01TU-eRSO3LXergA/s3-managed-key-sse-s3-rotation-period

upvoted 2 times

**JoeTromundo** 1 year, 8 months ago

**Selected Answer: B**

The anwser can't be A. In addition to other justifications written here in the comments, if the data is copied before enabling encryption, this data will not be encrypted.

upvoted 4 times

**Johnoppong101** 1 year, 9 months ago

**Selected Answer: B**

B is the Answer

upvoted 1 times

**n999** 1 year, 10 months ago

**Selected Answer: A**

It's said should be encrypted within S3 not before so A its correct

upvoted 2 times

**Johnoppong101** 1 year, 9 months ago

Me: Does SSE-S3 allow custom key rotation scheduling?
Gemini: No, SSE-S3 does not allow for custom key rotation scheduling.
Gemini: If you require more granular control over key rotation, you should consider using Server-Side Encryption with AWS KMS (SSE-KMS)

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: B**

AnswerB

Looks like key rotation is only possible when KMS is in use. If we will use AWS managed keys Rotation is forced and if we will not provide any specifications regarding rotation time for key, KMS will rotate key every 365days.

https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingKMSEncryption.html
https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html
https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#key-mgmt
upvoted 4 times

**sheilawu** 2 years ago

Selected Answer: B

If you see rotation, SEE-SE is out
upvoted 3 times

**f07ed8f** 2 years ago

Selected Answer: B

SSE-S3 does not rotate the key EVERY YEAR and it is not fit the requirement
upvoted 3 times

**Linuslin** 2 years ago

Selected Answer: A

This question is flawed.
SSE-S3 is not SSE-KMS, so it will not automatic rotation every year, only KMS will. (check link below)
But the question says "LEAST operational overhead", I think it want us to choose SSE-S3, so I will pick option A.
upvoted 2 times

**Linuslin** 2 years ago

SSE-S3 is the simplest method to use as encryption keys are handled and managed by AWS. But is not what we're saying about "AWS managed key", so it will not automatic rotation every year.
https://catalog.us-east-1.prod.workshops.aws/workshops/aad9ff1e-b607-45bc-893f-121ea5224f24/en-US/s3/serverside/sses3

"AWS managed keys" are "KMS keys" in your account. And will (required) automatic rotation every year.
https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#key-mgmt
upvoted 2 times

**FlyingHawk** 1 year, 4 months ago

You are correct. SSE-S3 is not related to AWS KMS or AWS managed keys.
upvoted 1 times

**EdricHoang** 1 year, 10 months ago

there are several pages (not the official aws page) said its 365 days. But, the official page does not mention about the rotation period is 365 days
upvoted 1 times

**Linuslin** 2 years ago

SSE-KMS is similar to SSE-S3 but comes with some additional benefits over SSE-S3. And SSE-KMS is "AWS managed key."
So it will (required) automatic rotation every year.
https://catalog.us-east-1.prod.workshops.aws/workshops/aad9ff1e-b607-45bc-893f-121ea5224f24/en-US/s3/serverside/ssekms

Difference between AWS S3 Bucket Encryption SSE-C , SSE-S3, SSE-KMS.
https://awstip.com/5-minutes-to-aws-s3-bucket-encryption-sse-c-sse-s3-sse-kms-e2fb07b05cb3
upvoted 1 times

**TwinSpark** 2 years ago

Selected Answer: B

I will go for B.
A it's somehow wrong for coupl of reason:
1- Encription must be specified before to transfer the data (even if from 1/23 it's automatically for every bucket, so actualy make no sense to specify it)
2- SSE-S3 keys are regurarly rotated but aws do not specify when (https://docs.aws.amazon.com/AmazonS3/latest/userguide/serv-side-encryption.html )
IMO if need to be compliance with rotation period better use Costumer managed key as stated from aws support in 01/2024
https://repost.aws/questions/QUES_1VN01TU-eRSO3LXergA/s3-managed-key-sse-s3-rotation-period
upvoted 2 times

**bujuman** 2 years ago

Selected Answer: A

Considering the statement "the LEAST operational overhead" we could go for option A due to the following AWS managed keys capabilities
https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#master_keys
upvoted 1 times

**f04dc74** 2 years, 1 month ago

Option A

upvoted 1 times

**sandordini** 2 years, 1 month ago

**Selected Answer: A**

From May 2022 the scheduled rotation is 1 year (SSE-S3)

upvoted 2 times

**3b196fc** 2 years, 1 month ago

A is wrong because you need to set the encryp options before send the data to S3.

upvoted 1 times

**camps** 2 years, 2 months ago

It's B.

upvoted 1 times

**TruthWS** 2 years, 2 months ago

A is correct because SSE-S3 help decrease the management

upvoted 2 times

**Yushib** 2 years, 2 months ago

**Selected Answer: B**

B is the right one

upvoted 2 times

**haci** 2 years, 2 months ago

Same with Question #202, I'll go with B but not sure

upvoted 1 times

Question #826 Topic 1

A company is migrating applications from an on-premises Microsoft Active Directory that the company manages to AWS. The company deploys the applications in multiple AWS accounts. The company uses AWS Organizations to manage the accounts centrally.

The company's security team needs a single sign-on solution across all the company's AWS accounts. The company must continue to manage users and groups that are in the on-premises Active Directory.

Which solution will meet these requirements?

A. Create an Enterprise Edition Active Directory in AWS Directory Service for Microsoft Active Directory. Configure the Active Directory to be the identity source for AWS IAM Identity Center.

B. Enable AWS IAM Identity Center. Configure a two-way forest trust relationship to connect the company's self-managed Active Directory with IAM Identity Center by using AWS Directory Service for Microsoft Active Directory. **Most Voted**

C. Use AWS Directory Service and create a two-way trust relationship with the company's self-managed Active Directory.

D. Deploy an identity provider (IdP) on Amazon EC2. Link the IdP as an identity source within AWS IAM Identity Center.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: B**

AWS IAM Identity Center (formerly AWS SSO): Provides SSO across multiple AWS accounts in an organization. By configuring a 2-way forest trust relationship between the on-premises AD and MAD, IAM Identity Center can integrate seamlessly with the existing AD to manage authentication anc authorization, allowing the company to retain its existing on-premises AD as the primary identity source while extending to AWS.

A - This creates a separate directory in AWS, requiring migration of users and groups, which fails to continue using the on-premises AD.
C - IAM Identity Center is still needed for SSO (single sign-on) functionality.
D - OK but so much overhead.

upvoted 2 times

**LuongTo** 1 year, 7 months ago

why C is out?

upvoted 1 times

**EdricHoang** 1 year, 11 months ago

**Selected Answer: B**

"continue to manage users and groups that are in the on-premises Active Directory"
I go for B

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: B**

AnswerB

AWS Directory Service lets you run Microsoft Active Directory (AD) as a managed service. AWS Directory Service for Microsoft Active Directory, also referred to as AWS Managed Microsoft AD, is powered by Windows Server 2019.
With AWS Managed Microsoft AD, you can run directory-aware workloads in the AWS Cloud, including Microsoft SharePoint and custom .NET and SQL Server-based applications. You can also configure a trust relationship between AWS Managed Microsoft AD in the AWS Cloud and your existing on-premises Microsoft Active Directory, providing users and groups with access to resources in either domain, using AWS IAM Identity Center.

https://docs.aws.amazon.com/directoryservice/latest/admin-guide/directory_microsoft_ad.html

upvoted 2 times

**Kaula** 2 years, 2 months ago

Selected Answer: B

https://docs.aws.amazon.com/directoryservice/latest/admin-guide/ms_ad_setup_trust.html

upvoted 4 times

**haci** 2 years, 2 months ago

Selected Answer: B

Same with Q-28

upvoted 2 times

Question #827 Topic 1

A company is planning to deploy its application on an Amazon Aurora PostgreSQL Serverless v2 cluster. The application will receive large amounts of traffic. The company wants to optimize the storage performance of the cluster as the load on the application increases.

Which solution will meet these requirements MOST cost-effectively?

A. Configure the cluster to use the Aurora Standard storage configuration.

B. Configure the cluster storage type as Provisioned IOPS.

C. Configure the cluster storage type as General Purpose.

D. Configure the cluster to use the Aurora I/O-Optimized storage configuration. Most Voted

**Correct Answer:** D

*Community vote distribution*

D (81%) Other



## Comments

**Scheldon** Highly Voted 2 years ago

Selected Answer: D

AnswerD

For Aurora we have 2 storage type/options:
A - Standard
B -- I/O-Optimized
hence answers B and C options are incorrect.
Because customer inform us that there will be a big amount of traffic for their application I would go with I/O-Optimized. Maybe we are giving more $$ per GB-month but we are not paing for I/O operations/request.

https://aws.amazon.com/rds/aurora/pricing/

In general question is not precise and it is hard to say which option will be more beneficial (cost effective)

upvoted 6 times

**GOTJ** Most Recent 1 year, 3 months ago

Selected Answer: A

In my opinion, when the company "wants to optimize the storage performance of the cluster AS the load on the application increases" they mean that they EXPECT that the load increases somewhere in the future. If so, what's the point of paying more for the I/O optimised storage configuration from the very beginning? You can switch to this more expensive configuration AS the load increased, saving money before this increment occurs

upvoted 1 times

**mk168898** 1 year, 7 months ago

Selected Answer: D

Aurora only have:
-> Standard
-> I/O-Optimized (need optimise storage thats why i chose this)

upvoted 2 times

**Johnoppong101** 1 year, 9 months ago

Selected Answer: D

Answer is D

upvoted 2 times

**joseantoniopolo** 2 years, 2 months ago

Selected Answer: D

Aurora I/O-Optimized – Improved price performance and predictability for I/O-intensive applications. You pay only for the usage and storage of your DB clusters, with no additional charges for read and write I/O operations.

upvoted 4 times

**osmk** 2 years, 2 months ago

**Selected Answer: D**

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Overview.StorageReliability.html#aurora-storage-type

upvoted 2 times

**JCAWS** 2 years, 2 months ago

**Selected Answer: D**

D is more suitable

upvoted 2 times

**camps** 2 years, 2 months ago

I would choose D

upvoted 2 times

**TruthWS** 2 years, 2 months ago

I think A is true answer

upvoted 2 times

**xBUGx** 2 years, 2 months ago

**Selected Answer: D**

https://aws.amazon.com/about-aws/whats-new/2023/05/amazon-aurora-i-o-optimized/
Aurora I/O-Optimized offers up to 40% cost savings for I/O-intensive applications where I/O charges exceed 25% of the total Aurora database spend

upvoted 3 times

**Kaula** 2 years, 2 months ago

**Selected Answer: C**

Agree with haci

upvoted 2 times

**haci** 2 years, 2 months ago

**Selected Answer: C**

The traffic load is not defined well enough to decide which storage type to use.

General Purpose (SSD) storage suits many workloads, including small to medium-sized databases and it is cost-effective.

Provisioned IOPS (PIOPS) storage is the highest-performing option available for RDS instances. With Provisioned IOPS storage, you can provision a specific amount of IOPS (input/output operations per second) based on your application's needs. But here we don't know the amount of requests.

So since the question is asking for cost-effective I'll go with C

upvoted 2 times

Question #828 *Topic 1*

A financial services company that runs on AWS has designed its security controls to meet industry standards. The industry standards include the National Institute of Standards and Technology (NIST) and the Payment Card Industry Data Security Standard (PCI DSS).

The company's third-party auditors need proof that the designed controls have been implemented and are functioning correctly. The company has hundreds of AWS accounts in a single organization in AWS Organizations. The company needs to monitor the current state of the controls across accounts.

Which solution will meet these requirements?

A. Designate one account as the Amazon Inspector delegated administrator account from the Organizations management account. Integrate Inspector with Organizations to discover and scan resources across all AWS accounts. Enable Inspector industry standards for NIST and PCI DSS.

B. Designate one account as the Amazon GuardDuty delegated administrator account from the Organizations management account. In the designated GuardDuty administrator account, enable GuardDuty to protect all member accounts. Enable GuardDuty industry standards for NIST and PCI DSS.

C. Configure an AWS CloudTrail organization trail in the Organizations management account. Designate one account as the compliance account. Enable CloudTrail security standards for NIST and PCI DSS in the compliance account.

D. Designate one account as the AWS Security Hub delegated administrator account from the Organizations management account. In the designated Security Hub administrator account, enable Security Hub for all member accounts. Enable Security Hub standards for NIST and PCI DSS. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (100%)

## Comments

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: D**

Security Hub: assess your AWS environment against security industry standards and best practices.

upvoted 7 times

**mk168898** Most Recent 1 year, 7 months ago

**Selected Answer: D**

security industry standards => security hub

upvoted 2 times

**Johnoppong101** 1 year, 9 months ago

**Selected Answer: D**

NIST, PCI DSS Compliance + AWS accounts -> Security Hub

upvoted 2 times

**Kaula** 2 years, 2 months ago

**Selected Answer: D**

https://docs.aws.amazon.com/securityhub/latest/userguide/what-is-securityhub.html

upvoted 4 times

Question #829 *Topic 1*

A company uses an Amazon S3 bucket as its data lake storage platform. The S3 bucket contains a massive amount of data that is accessed randomly by multiple teams and hundreds of applications. The company wants to reduce the S3 storage costs and provide immediate availability for frequently accessed objects.

What is the MOST operationally efficient solution that meets these requirements?

A. Create an S3 Lifecycle rule to transition objects to the S3 Intelligent-Tiering storage class. **Most Voted**

B. Store objects in Amazon S3 Glacier. Use S3 Select to provide applications with access to the data.

C. Use data from S3 storage class analysis to create S3 Lifecycle rules to automatically transition objects to the S3 Standard-Infrequent Access (S3 Standard-IA) storage class.

D. Transition objects to the S3 Standard-Infrequent Access (S3 Standard-IA) storage class. Create an AWS Lambda function to transition objects to the S3 Standard storage class when they are accessed by an application.

**Correct Answer:** A

*Community vote distribution*

A (100%)

## Comments

**mk168898** 1 year, 7 months ago

**Selected Answer: A**

selected A because only option that don't violate the "need immediate access"

upvoted 2 times

**MatAlves** 1 year, 8 months ago

**Selected Answer: A**

"data that is accessed randomly" = S3 Intelligent-Tiering storage class.

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: A**

AnswerA

The Amazon S3 Intelligent-Tiering storage class automatically stores objects in three access tiers. One tier is optimized for frequent access, one lower-cost tier is optimized for infrequent access, and another very low-cost tier is optimized for rarely accessed data. For a low monthly object monitoring and automation charge, S3 Intelligent-Tiering monitors access patterns and automatically moves objects to the Infrequent Access tier when they haven't been accessed for 30 consecutive days. After 90 days of no access, the objects are moved to the Archive Instant Access tier without performance impact or operational overhead.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/intelligent-tiering-overview.html

upvoted 3 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: A**

File accessed randomly by multiple teams = intelligent tiering

upvoted 2 times

**Kaula** 2 years, 2 months ago

**Selected Answer: A**

https://docs.aws.amazon.com/AmazonS3/latest/userguide/intelligent-tiering-managing.html

upvoted 2 times

Question #830 Topic 1

A company has 5 TB of datasets. The datasets consist of 1 million user profiles and 10 million connections. The user profiles have connections as many-to-many relationships. The company needs a performance efficient way to find mutual connections up to five levels.

Which solution will meet these requirements?

A. Use an Amazon S3 bucket to store the datasets. Use Amazon Athena to perform SQL JOIN queries to find connections.

B. Use Amazon Neptune to store the datasets with edges and vertices. Query the data to find connections. **Most Voted**

C. Use an Amazon S3 bucket to store the datasets. Use Amazon QuickSight to visualize connections.

D. Use Amazon RDS to store the datasets with multiple tables. Perform SQL JOIN queries to find connections.

**Correct Answer:** B

*Community vote distribution*

B (100%)



## Comments

**Kaula** Highly Voted 2 years, 2 months ago

**Selected Answer: B**

https://docs.aws.amazon.com/neptune/latest/userguide/notebooks-visualization.html

upvoted 5 times

**mk168898** Most Recent 1 year, 7 months ago

**Selected Answer: B**

everytime i see some social network related qns i immediately look for amazon neptune

upvoted 2 times

**sandordini** 2 years, 1 month ago

**Selected Answer: B**

Neptune: A Graph database stores nodes and relationships instead of tables or documents

upvoted 3 times

**alawada** 2 years, 2 months ago

**Selected Answer: B**

Neptune automatically scales storage and compute resources based on workload demands, ensuring optimal performance even as the dataset grow over time.

upvoted 3 times

Question #831 *Topic 1*

A company needs a secure connection between its on-premises environment and AWS. This connection does not need high bandwidth and will handle a small amount of traffic. The connection should be set up quickly.

What is the MOST cost-effective method to establish this type of connection?

A. Implement a client VPN.

B. Implement AWS Direct Connect.

C. Implement a bastion host on Amazon EC2.

D. Implement an AWS Site-to-Site VPN connection. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (100%)

## Comments

**mk168898** 1 year, 7 months ago

**Selected Answer: D**

Not A because for individual conenction not company
Not B because overkill, usually for high bandwidth, but qns clearly stated no need for high bandwidth and handle small traffic
Not C because bastion host for remote access
D because ideal for small amt of traffic

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: D**

AnswerD

AWS site-to-site VPN is the best solution here.

https://docs.aws.amazon.com/vpn/latest/s2svpn/VPC_VPN.html

upvoted 4 times

**sandordini** 2 years, 1 month ago

**Selected Answer: D**

You can enable access to your remote (on-prem) network from your VPC by creating an AWS Site-to-Site VPN (Site-to-Site VPN) connection and configuring routing to pass traffic through the connection.

upvoted 3 times

**Kaula** 2 years, 2 months ago

**Selected Answer: D**

https://docs.aws.amazon.com/vpn/latest/s2svpn/VPC_VPN.html

upvoted 3 times

Question #832 *Topic 1*

A company has an on-premises SFTP file transfer solution. The company is migrating to the AWS Cloud to scale the file transfer solution and to optimize costs by using Amazon S3. The company's employees will use their credentials for the on-premises Microsoft Active Directory (AD) to access the new solution. The company wants to keep the current authentication and file access mechanisms.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure an S3 File Gateway. Create SMB file shares on the file gateway that use the existing Active Directory to authenticate.

B. Configure an Auto Scaling group with Amazon EC2 instances to run an SFTP solution. Configure the group to scale up at 60% CPU utilization.

C. Create an AWS Transfer Family server with SFTP endpoints. Choose the AWS Directory Service option as the identity provider. Use AD Connector to connect the on-premises Active Directory. **Most Voted**

D. Create an AWS Transfer Family SFTP endpoint. Configure the endpoint to use the AWS Directory Service option as the identity provider to connect to the existing Active Directory.

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**Scheldon** 2 years ago

**Selected Answer: C**

Answer C

We need Transfer Family SFTP enabled server (with SFTP endpoint). Additiionally AWS Directory Service with AD connector to reach on-premises AD for authentication and authorization.
https://docs.aws.amazon.com/transfer/latest/userguide/getting-started.html
https://docs.aws.amazon.com/transfer/latest/userguide/create-server-sftp.html
https://docs.aws.amazon.com/directoryservice/latest/admin-guide/what_is.html
https://docs.aws.amazon.com/directoryservice/latest/admin-guide/directory_ad_connector.html

upvoted 3 times

**phoenix2023** 2 years ago

what is the difference between C and D ???

upvoted 2 times

**EdricHoang** 1 year, 10 months ago

AD Connector - when the company already had the Active Directory, you need a connector

upvoted 6 times

**sandordini** 2 years, 1 month ago

**Selected Answer: C**

1. Create one or more AWS Managed Microsoft AD directories using the AWS Directory Service console.

2. Use the Transfer Family console to create a **server** that uses **AWS Managed Microsoft AD** as its identity provider.

3. Add access from one or more of your AWS Directory Service groups.

4. Although not required, we recommend that you test and verify user access.

upvoted 3 times

**Kaula** 2 years, 2 months ago

**Selected Answer: C**

https://docs.aws.amazon.com/directoryservice/latest/admin-guide/directory_ad_connector.html

upvoted 3 times

Question #833 Topic 1

A company is designing an event-driven order processing system. Each order requires multiple validation steps after the order is created. An idempotent AWS Lambda function performs each validation step. Each validation step is independent from the other validation steps. Individual validation steps need only a subset of the order event information.

The company wants to ensure that each validation step Lambda function has access to only the information from the order event that the function requires. The components of the order processing system should be loosely coupled to accommodate future business changes.

Which solution will meet these requirements?

A. Create an Amazon Simple Queue Service (Amazon SQS) queue for each validation step. Create a new Lambda function to transform the order data to the format that each validation step requires and to publish the messages to the appropriate SQS queues. Subscribe each validation step Lambda function to its corresponding SQS queue.

B. Create an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe the validation step Lambda functions to the SNS topic. Use message body filtering to send only the required data to each subscribed Lambda function.

C. Create an Amazon EventBridge event bus. Create an event rule for each validation step. Configure the input transformer to send only the required data to each target validation step Lambda function. **Most Voted**

D. Create an Amazon Simple Queue Service (Amazon SQS) queue. Create a new Lambda function to subscribe to the SQS queue and to transform the order data to the format that each validation step requires. Use the new Lambda function to perform synchronous invocations of the validation step Lambda functions in parallel on separate threads.

**Correct Answer:** C

*Community vote distribution*

C (94%) 6%

## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

A - Requires a separate Lambda function to preprocess and publish transformed data to multiple queues. Unnecessary.
B - SNS filtering operates at the subscription level, not the event transformation level, making it harder to customize data subsets for specific Lambda functions.
C - Amazon EventBridge: event-driven systems. Input Transformer: ensures each receives only the required subset of data.
D - Synchronous invocation adds tighter coupling between Lambda functions and the validation steps.

upvoted 2 times

**MatAlves** 1 year, 8 months ago

SNS and Message Filtering
- With SNS, message filtering allows you to control which subscribers receive messages based on attributes. However, the entire message is sent to each subscribed Lambda function; only those that match the filter criteria are processed.

EventBridge and Input Transformation
- EventBridge enables you to define rules that transform or modify events before they reach their targets. This allows you to customize the event payload, ensuring each validation step receives only the relevant information.

"The company wants to ensure that each validation step Lambda function has access to only the information from the order event that the function requires."

Therefore, C is the answer.

upvoted 4 times

**Johnoppong101** 1 year, 9 months ago

**Selected Answer: C**
Event-driven Architecture + Each validation step needs ONLY a subset of the order EVENT created. Best way to transform this order even is EB Transformer.
upvoted 2 times

**Scheldon** 2 years ago
**Selected Answer: C**
Answer C

I wasn't sure but looks like EB with Input Transformation will allow for sending data which were choosed per destination

https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-pipes-input-transformation.html
upvoted 3 times

**TwinSpark** 2 years ago
**Selected Answer: C**
not B because SNS cannot make messages manipulation, the option "message body filtering" will make discard or forward the FULL message if there is a matching field:
https://docs.aws.amazon.com/sns/latest/dg/sns-message-filtering.html
C - eventbus instead can manipulate event:
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-event-bus.html
D - Works, but too much operation IMO
upvoted 4 times

**88f8032** 2 years, 1 month ago
Why can't it be B?
upvoted 3 times

**BBR01** 2 years, 1 month ago
**Selected Answer: D**
It is D. It is one order event, not "events from many sources"
The main lambda parse the info to pieces, then makes synchronous invocations of the validation step Lambda functions on separate threads, and wait them to complete.
upvoted 1 times

**waldirlsantos** 2 years, 1 month ago
**Selected Answer: C**
IMO, C

"An event bus is a router that receives events and delivers them to zero or more destinations, or targets. Event buses are well-suited for routing events from many sources to many targets, with optional transformation of events prior to delivery to a target."
upvoted 2 times

**TruthWS** 2 years, 2 months ago
Option C
upvoted 2 times

**Kaula** 2 years, 2 months ago
**Selected Answer: C**
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-event-bus.html
upvoted 3 times

Question #834 *Topic 1*

A company is migrating a three-tier application to AWS. The application requires a MySQL database. In the past, the application users reported poor application performance when creating new entries. These performance issues were caused by users generating different real-time reports from the application during working hours.

Which solution will improve the performance of the application when it is moved to AWS?

A. Import the data into an Amazon DynamoDB table with provisioned capacity. Refactor the application to use DynamoDB for reports.

B. Create the database on a compute optimized Amazon EC2 instance. Ensure compute resources exceed the on-premises database.

C. Create an Amazon Aurora MySQL Multi-AZ DB cluster with multiple read replicas. Configure the application to use the reader endpoint for reports. Most Voted

D. Create an Amazon Aurora MySQL Multi-AZ DB cluster. Configure the application to use the backup instance of the cluster as an endpoint for the reports.

**Correct Answer:** C

*Community vote distribution*

C (100%)



## Comments

**xBUGx** Highly Voted 2 years, 2 months ago

**Selected Answer: C**

real-time reports - > read replica

upvoted 6 times

**KennethNg923** Most Recent 1 year, 12 months ago

**Selected Answer: C**

caused by users generating different real-time reports -> read replica

upvoted 1 times

**Scheldon** 2 years ago

**Selected Answer: C**

Answer C

Any replica of Amazon Aurora Db will be read-only replica (even backup one). Option C is better then D because we will use multiple replicas which when used will signicially allow to increase performance for creating reports. In the same time no write operation should be affected.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Replication.html

upvoted 3 times

Question #835 Topic 1

A company is expanding a secure on-premises network to the AWS Cloud by using an AWS Direct Connect connection. The on-premises network has no direct internet access. An application that runs on the on-premises network needs to use an Amazon S3 bucket.

Which solution will meet these requirements MOST cost-effectively?

A. Create a public virtual interface (VIF). Route the AWS traffic over the public VIF.

B. Create a VPC and a NAT gateway. Route the AWS traffic from the on-premises network to the NAT gateway.

C. Create a VPC and an Amazon S3 interface endpoint. Route the AWS traffic from the on-premises network to the S3 interface endpoint. **Most Voted**

D. Create a VPC peering connection between the on-premises network and Direct Connect. Route the AWS traffic over the peering connection.

**Correct Answer:** C

*Community vote distribution*

C (65%) A (35%)



## Comments

**c7e80cd** 8 months, 3 weeks ago

**Selected Answer: A**

In short: C only works if the traffic originates inside a VPC, but your requirement is from on-premises outside AWS. That's why A is the correct, cost-effective solution.

upvoted 1 times

**0c21057** 1 year, 1 month ago

**Selected Answer: A**

AWS Direct Connect links your internal network to an AWS Direct Connect location over a standard Ethernet fiber-optic cable. One end of the cable i connected to your router, and the other to an AWS Direct Connect router. With this connection, you can create virtual interfaces directly to public AWS services (for example, to Amazon S3) or to Amazon VPC, bypassing internet service providers in your network path. An AWS Direct Connect location provides access to AWS in the Region with which it is associated. You can use a single connection in a public Region or AWS GovCloud (US) to access public AWS services in all other public Regions.

upvoted 1 times

**Salilgen** 1 year, 5 months ago

**Selected Answer: A**

https://aws.amazon.com/blogs/networking-and-content-delivery/optimizing-amazon-s3-data-transfers-over-direct-connect/

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

A - Public VIFs route traffic over the public internet, but on-premises network has no direct internet access.
B - NAT gateways provide internet access for resources within a VPC, but, again, on-premises network has no direct internet access.
C - The traffic stays within AWS and the Direct Connect connection, ensuring security and compliance with the no-internet requirement. Also, this is cost-effective because it avoids the need for additional NAT gateways or public VIFs.
D - Peering connections don't even support access to AWS public services like S3.

upvoted 3 times

**EllenLiu** 1 year, 5 months ago

**Selected Answer: A**

public VIF allows on-premises networks to access AWS public services, like S3
it eliminates the need for internet access & interface endpoints while being cost-effective

upvoted 2 times

**EllenLiu** 1 year, 5 months ago
https://aws.amazon.com/privatelink/pricing/
upvoted 1 times

**AMEJack** 1 year, 6 months ago
Selected Answer: A
Option A is the correct answer:
1) I need to connect to public services only, thus public VIF will be enough.
2) Interface endpoints is charged per hour, not cost effective.
upvoted 2 times

**Scheldon** 2 years ago
Selected Answer: C
AnswerC

Amazon S3 interface endpoint seems to be the best and only option as we are forced to use Private IP addressation.

Interface endpoints for Amazon S3
Your network traffic remains on the AWS network.
Use private IP addresses from your VPC to access Amazon S3
Require endpoint-specific Amazon S3 DNS names
Allow access from on premises
Allow access from a VPC in another AWS Region by using VPC peering or AWS Transit Gateway

https://docs.aws.amazon.com/AmazonS3/latest/userguide/privatelink-interface-endpoints.html#types-of-vpc-endpoints-for-s3
upvoted 4 times

**0bdf3af** 2 years ago
A. https://repost.aws/knowledge-center/s3-bucket-access-direct-connect
upvoted 1 times

**elmyth** 1 year, 9 months ago
This article says "Use a private IP address over Direct Connect (with an interface VPC endpoint)" - C
upvoted 3 times

**0bdf3af** 2 years ago
A. public VIF is the way you can connect on-premise with S3 via DirectConnect
upvoted 1 times

**waldirlsantos** 2 years, 1 month ago
Selected Answer: C
B Need internet
A,D doesn't conect to the s3
IMO, C is the solution for this question.
upvoted 4 times

**TruthWS** 2 years, 2 months ago
Option C
upvoted 2 times

**Kaula** 2 years, 2 months ago
Selected Answer: C
https://docs.aws.amazon.com/AmazonS3/latest/userguide/privatelink-interface-endpoints.html
upvoted 2 times

Question #836 *Topic 1*

A company serves its website by using an Auto Scaling group of Amazon EC2 instances in a single AWS Region. The website does not require a database.

The company is expanding, and the company's engineering team deploys the website to a second Region. The company wants to distribute traffic across both Regions to accommodate growth and for disaster recovery purposes. The solution should not serve traffic from a Region in which the website is unhealthy.

Which policy or resource should the company use to meet these requirements?

A. An Amazon Route 53 simple routing policy

B. An Amazon Route 53 multivalue answer routing policy **Most Voted**

C. An Application Load Balancer in one Region with a target group that specifies the EC2 instance IDs from both Regions

D. An Application Load Balancer in one Region with a target group that specifies the IP addresses of the EC2 instances from both Regions

**Correct Answer:** B

*Community vote distribution*

B (100%)



## Comments

**waldirlsantos** Highly Voted 2 years, 1 month ago

**Selected Answer: B**

53 with multivalue is the best option for this case

Multivalue answer routing lets you configure Amazon Route 53 to return multiple values, such as IP addresses for your web servers, in response to DNS queries. You can specify multiple values for almost any record, but multivalue answer routing also lets you check the health of each resource, so Route 53 returns only values for healthy resources. It's not a substitute for a load balancer, but the ability to return multiple health-checkable IP addresses is a way to use DNS to improve availability and load balancing.

upvoted 5 times

**mk168898** Most Recent 1 year, 7 months ago

**Selected Answer: B**

C and D are wrong because need serve traffic across both region.
B seems correct

upvoted 2 times

**MatAlves** 1 year, 8 months ago

A - Doesn't provide health check
C and D - Only work within a single zone.

upvoted 1 times

**Sergiuss95** 2 years, 1 month ago

**Selected Answer: B**

Yes, is the option b.

upvoted 1 times

**TruthWS** 2 years, 2 months ago

Option B

upvoted 1 times

**Kaula** 2 years, 2 months ago

**Selected Answer: B**

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalue.html
upvoted 3 times

Question #837 *Topic 1*

A company runs its applications on Amazon EC2 instances that are backed by Amazon Elastic Block Store (Amazon EBS). The EC2 instances run the most recent Amazon Linux release. The applications are experiencing availability issues when the company's employees store and retrieve files that are 25 GB or larger. The company needs a solution that does not require the company to transfer files between EC2 instances. The files must be available across many EC2 instances and across multiple Availability Zones.

Which solution will meet these requirements?

A. Migrate all the files to an Amazon S3 bucket. Instruct the employees to access the files from the S3 bucket.

B. Take a snapshot of the existing EBS volume. Mount the snapshot as an EBS volume across the EC2 instances. Instruct the employees to access the files from the EC2 instances.

C. Mount an Amazon Elastic File System (Amazon EFS) file system across all the EC2 instances. Instruct the employees to access the files from the EC2 instances. **Most Voted**

D. Create an Amazon Machine Image (AMI) from the EC2 instances. Configure new EC2 instances from the AMI that use an instance store volume. Instruct the employees to access the files from the EC2 instances.

**Correct Answer:** C

*Community vote distribution*

C (100%)



## Comments

**xBUGx** Highly Voted 2 years, 2 months ago

**Selected Answer: C**

cross many EC2 instances and across multiple Availability Zones = EFS

upvoted 6 times

**freedafeng** Highly Voted 1 year, 10 months ago

my question is, why not A?
I am fine with C

upvoted 6 times

**JA2018** 1 year, 6 months ago

For Option A (S3 bucket): While S3 is great for storing large files, accessing files directly from S3 would require the employees to switch to a different interface, making it inconvenient and not aligned with the requirement to access files from the EC2 instances.

upvoted 2 times

**mk168898** Most Recent 1 year, 7 months ago

**Selected Answer: C**

"files must be available across many EC2 instances" means need some sort of shared system
immediately i look for EFS

upvoted 3 times

**muhammadahmer36** 1 year, 10 months ago

**Selected Answer: C**

cross many EC2 instances and across multiple Availability Zones = EFS

upvoted 2 times

Question #838 *Topic 1*

A company is running a highly sensitive application on Amazon EC2 backed by an Amazon RDS database. Compliance regulations mandate that all personally identifiable information (PII) be encrypted at rest.

Which solution should a solutions architect recommend to meet this requirement with the LEAST amount of changes to the infrastructure?

A. Deploy AWS Certificate Manager to generate certificates. Use the certificates to encrypt the database volume.

B. Deploy AWS CloudHSM, generate encryption keys, and use the keys to encrypt database volumes.

C. Configure SSL encryption using AWS Key Management Service (AWS KMS) keys to encrypt database volumes.

D. Configure Amazon Elastic Block Store (Amazon EBS) encryption and Amazon RDS encryption with AWS Key Management Service (AWS KMS) keys to encrypt instance and database volumes. Most Voted

**Correct Answer:** D

*Community vote distribution*

D (100%)

## Comments

**boluwatito** Highly Voted 2 years, 1 month ago

**Selected Answer: D**

Amazon RDS relies on Amazon EBS volumes for storage.
By configuring Amazon EBS encryption, the underlying storage volumes are encrypted.

upvoted 7 times

**JA2018** Most Recent 1 year, 6 months ago

**Selected Answer: D**

- Option D requires the least infrastructure modification as it leverages existing features of EBS and RDS to enable encryption with KMS keys.

- This is the most straightforward way to meet the compliance requirement without significant changes to the existing setup.

upvoted 1 times

**mk168898** 1 year, 7 months ago

SSL/Certificate => encrypt in transit, so A and C are wrong.
so i feel the answer is between B and D.

upvoted 2 times

**sandordini** 2 years, 1 month ago

**Selected Answer: D**

Encryption should be KMS, SSL is for transit not at rest...
Even though the question never mentioned any EBS volumes whatsoever, I would still go for D....

upvoted 4 times

**zinabu** 2 years, 2 months ago

answer:C

upvoted 2 times

**zinabu** 2 years, 2 months ago

answer:C

upvoted 2 times

Question #839 *Topic 1*

A company runs an AWS Lambda function in private subnets in a VPC. The subnets have a default route to the internet through an Amazon EC2 NAT instance. The Lambda function processes input data and saves its output as an object to Amazon S3.

Intermittently, the Lambda function times out while trying to upload the object because of saturated traffic on the NAT instance's network. The company wants to access Amazon S3 without traversing the internet.

Which solution will meet these requirements?

A. Replace the EC2 NAT instance with an AWS managed NAT gateway.

B. Increase the size of the EC2 NAT instance in the VPC to a network optimized instance type.

C. Provision a gateway endpoint for Amazon S3 in the VPUpdate the route tables of the subnets accordingly. **Most Voted**

D. Provision a transit gateway. Place transit gateway attachments in the private subnets where the Lambda function is running.

**Correct Answer:** C

*Community vote distribution*

C (92%) 8%



## Comments

**mk168898** 1 year, 7 months ago
without internet => gateway endpoints
upvoted 3 times

**Scheldon** 2 years ago
**Selected Answer: C**
Answer C

"The company wants to access Amazon S3 without traversing the internet." so we cannot use any NAT like in answer A & B. Transit Gateways is allowing reach Direct Connect or VPN connection from VPC. Hence C need to be a good answer
upvoted 4 times

**boubie44** 2 years, 1 month ago
why not D? i don't understand
upvoted 2 times

**DanielWuTRT** 1 year, 11 months ago
Complexity and cost are high and too complicated for scenarios where only S3 access is required.
upvoted 2 times

**waldirlsantos** 2 years, 1 month ago
**Selected Answer: C**
The Key words are "Without traversig the internet". So, the awser is C.
https://docs.aws.amazon.com/pt_br/vpc/latest/privatelink/gateway-endpoints.html
upvoted 4 times

**AlvinC2024** 2 years, 2 months ago
**Selected Answer: C**
By provisioning a gateway endpoint for Amazon S3 in the VPC, you enable the Lambda function running in the private subnets to access S3 directly without needing to go through the NAT instance or traverse the internet. This solution helps alleviate the network congestion issue and reduces latency since the traffic between Lambda and S3 stays within the AWS network. Additionally, updating the route tables of the subnets to route S3 traffic through the gateway endpoint ensures that the Lambda function can seamlessly communicate with S3 without encountering timeouts caused by network saturation on the NAT instance.
upvoted 3 times

**dds69** 2 years, 2 months ago

Selected Answer: A

NAT gateways are highly available and can automatically scale up to meet increased traffic demands.

upvoted 1 times

**sandordini** 2 years, 1 month ago

And uses the internet... So it can be a good solution, but not here, as: Without traversing the internet

upvoted 4 times

**hpmargathia** 2 years, 2 months ago

A

https://aws.amazon.com/about-aws/whats-new/2015/12/introducing-amazon-vpc-nat-gateway-a-managed-nat-service/

upvoted 1 times

Question #840 Topic 1

A news company that has reporters all over the world is hosting its broadcast system on AWS. The reporters send live broadcasts to the broadcast system. The reporters use software on their phones to send live streams through the Real Time Messaging Protocol (RTMP).

A solutions architect must design a solution that gives the reporters the ability to send the highest quality streams. The solution must provide accelerated TCP connections back to the broadcast system.

What should the solutions architect use to meet these requirements?

A. Amazon CloudFront

B. AWS Global Accelerator **Most Voted**

C. AWS Client VPN

D. Amazon EC2 instances and AWS Elastic IP addresses

**Correct Answer:** B

*Community vote distribution*

B (94%) 6%



## Comments

**Mikado211** Highly Voted 2 years, 2 months ago

HTTP(S) -> Cloudfront
Other TCP -> AWS Global Accelerator

upvoted 5 times

**Mikado211** 2 years, 2 months ago

So the answer is B :)

upvoted 3 times

**mk168898** Most Recent 1 year, 7 months ago

I see TCP/UDP related connection => AWS Global accelerator

upvoted 2 times

**elmyth** 1 year, 9 months ago

Selected Answer: A

Looks like the question is very old, but before the right answer was Cloudfront. Now AWS says that "All RTMP workloads should begin migrating to a standard CloudFront Web distribution and use one of several HTTP streaming protocols such as HTTP Live Streaming (HLS), Dynamic Adaptive Streaming over HTTP (DASH), Microsoft Smooth Streaming (MSS), or HTTP Dynamic Streaming (HDS)."
https://repost.aws/questions/QUoUZgHZh7SEWlnQUPlBmVNQ/announcement-rtmp-support-discontinuing-on-december-31-2020

upvoted 1 times

**Scheldon** 2 years ago

Selected Answer: B

AnswerB

We can eliminate C and D, A is for Web apps hence B should be ok

additionally https://docs.aws.amazon.com/global-accelerator/latest/dg/introduction-how-it-works.html

upvoted 2 times

**sandordini** 2 years, 1 month ago

Selected Answer: B

Upload, TCP > AWS Global Accelerator

upvoted 4 times

**Drew3000** 2 years, 2 months ago
Last time I made it to the last question, they added more questions 2 mins later.
upvoted 2 times

**Awsbeginner87** 2 years, 2 months ago
They added 40 more questions today 😔
upvoted 4 times

**chasingsummer** 2 years, 2 months ago
**Selected Answer: B**
Can't believe I finally made it to the last question. Good luck to everyone!
upvoted 4 times

**TruthWS** 2 years, 2 months ago
OptionB
upvoted 2 times

**Kaula** 2 years, 2 months ago
Where are questions 841-848?
I am I missing something?
upvoted 2 times

**Kaula** 2 years, 2 months ago
**Selected Answer: B**
B makes sense not A since CloudFront is CDN
upvoted 2 times

**dds69** 2 years, 2 months ago
**Selected Answer: B**
Global accelerator provides the acceleration for TCP
upvoted 4 times

Question #841 *Topic 1*

A company uses Amazon EC2 instances and Amazon Elastic Block Store (Amazon EBS) to run its self-managed database. The company has 350 TB of data spread across all EBS volumes. The company takes daily EBS snapshots and keeps the snapshots for 1 month. The daily change rate is 5% of the EBS volumes.

Because of new regulations, the company needs to keep the monthly snapshots for 7 years. The company needs to change its backup strategy to comply with the new regulations and to ensure that data is available with minimal administrative effort.

Which solution will meet these requirements MOST cost-effectively?

A. Keep the daily snapshot in the EBS snapshot standard tier for 1 month. Copy the monthly snapshot to Amazon S3 Glacier Deep Archive with a 7-year retention period.

B. Continue with the current EBS snapshot policy. Add a new policy to move the monthly snapshot to Amazon EBS Snapshots Archive with a 7-year retention period. **Most Voted**

C. Keep the daily snapshot in the EBS snapshot standard tier for 1 month. Keep the monthly snapshot in the standard tier for 7 years. Use incremental snapshots.

D. Keep the daily snapshot in the EBS snapshot standard tier. Use EBS direct APIs to take snapshots of all the EBS volumes every month. Store the snapshots in an Amazon S3 bucket in the Infrequent Access tier for 7 years.

**Correct Answer:** B

*Community vote distribution*

B (50%) | A (45%) | 3%

## Comments

**Scheldon** Highly Voted 2 years ago

**Selected Answer: B**

AnswerB
The problem is that we need to choose best solution which is most cost-effective and have minimal administrative effort.
Glacier is the best choice for 1st look, but there is one problem with that solution. From what I know there is no easy way to copy from EBS to Glacier and additionally curent strategy is to make incremental snapshots. To copy file from EBS to (s3) Glacier we would need to run linux to which we will mount EBS and we will need copy everything to S3 and then move to glaceir deep archive. And what is more you will have only incremental snapshot. Hence every solution which will say copy/move to S3 is not minimal administrative effort. Not mentionig that you will not have full snapshot
https://repost.aws/questions/QUsaCoBAfbR6WMOz6BH3vqHA/move-ebs-to-glacier

upvoted 9 times

**MatAlves** 1 year, 8 months ago

I first thought the same, but NO: "Which solution will meet these requirements MOST cost-effectively?" There is no mention to "operational overhead" or "minimal changes", etc. The question is asking purely "what is cheaper". So, A is the answer.

"We charge you $0.0125 per GB-month of stored data and $0.03 per GB retrieved. "

"All Storage / Month $0.0036 per GB
S3 Glacier Deep Archive *** - For long-term data archiving that is accessed once or twice in a year and can be restored within 12 hours"

upvoted 2 times

**elmyth** 1 year, 7 months ago

"to ensure that data is available with minimal administrative effort." - there are 2 conditions

upvoted 4 times

**rondelldell** Highly Voted 2 years, 2 months ago

**Selected Answer: A**

How much does EBS snapshots archive cost?
Pricing and billing. Archived snapshots are billed at a rate of $0.0125 per GB-month. For example, if you archive a 100 GiB snapshot, you are billed $1.25 (100 GiB * $0.0125) per month.

What is the cost of Glacier?
Even though uploading data to Amazon S3 Glacier is free, there is a pricing method for upload requests, which is $0.03 per 1,000 requests. Transferring data out of S3 Glacier to the same region is free; however, there is a cost for transferring data to a different region.
• $0.0036 per GB / Month

upvoted 7 times

**ln2718** Most Recent 9 months, 3 weeks ago

Selected Answer: B

You can't copy EBS snapshots to your own S3/Glacier Deep Archive objects; use EBS Snapshots Archive for long-term, low-cost storage.

upvoted 1 times

**TEC65** 1 year, 1 month ago

Selected Answer: B

snapshots archive designed for low cost snapshots

upvoted 1 times

**0c21057** 1 year, 1 month ago

Selected Answer: B

The correct answer is B: Continue with the current EBS snapshot policy. Add a new policy to move the monthly snapshot to Amazon EBS Snapshots Archive with a 7-year retention period.

upvoted 1 times

**gigilla** 1 year, 2 months ago

Selected Answer: A

A: Amazon S3 Glacier Deep Archive

upvoted 1 times

**nadeerm** 1 year, 3 months ago

Selected Answer: B

Cost Efficiency : Amazon EBS Snapshots Archive is designed to provide a cost-effective option for long-term storage of EBS snapshots. By moving monthly snapshots to the archive tier, you will significantly reduce the cost

upvoted 1 times

**tch** 1 year, 3 months ago

Selected Answer: B

Amazon EBS Snapshots Archive is a storage tier that you can use for low-cost, long-term storage of your rarely-accessed snapshots that do not need frequent or fast retrieval.

Move to S3... more job, so A is not a good idea...

Only 5% daily change.. would consider cheaper than S3 for long run... (MOST cost-effectively)

I don't think in the exam need to calculate cost per $, how to remember?

Just add new policy to move the snapshot to Amazon EBS Snapshots Archive (minimal administrative effort)

upvoted 1 times

**Dantecito** 1 year, 3 months ago

Selected Answer: A

A or B but I will choose A because Is the MOST cost-effectively and companies will always choose to save money.

upvoted 1 times

**Salilgen** 1 year, 5 months ago

Selected Answer: B

"EBS Snapshots are stored incrementally."
"The snapshots are automatically saved to Amazon Simple Storage Service (Amazon S3) for long-term retention"
"Save up to 75% in snapshot storage costs by using EBS Snapshots Archive for the long-term retention (over 90 days) of seldom-accessed snapshots."
Moreover, copy files (option A) requires more administrative effort than automatic policy.

https://aws.amazon.com/ebs/snapshots/#:~:text=available%20and%20durable-,Amazon%20EBS%20Snapshots%20are%20a%20convenient%20way%20to%20back%20up,availability%20of%20your%20EBS%20Snapshots.

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

Selected Answer: A

A - S3 Glacier Deep Archive is designed for long-term data archiving with a retention period of up to 7 years or more at an extremely low cost (the most cost-effective S3 storage class).
B - OK but EBS Snapshots Archive costs more.
C - Keep the monthly snapshot in the standard tier for 1 month, OK. For 7 years??? You'll pay so much money for that. The standard EBS snapshot tie is designed for frequent access.
D - Storing snapshots directly in Amazon S3 (even in the Infrequent Access tier) is not ideal for EBS snapshots because it doesn't leverage EBS snapshot mechanisms, which are designed for efficient storage and management of EBS backup data.

upvoted 2 times

**ARV14** 1 year, 6 months ago

Selected Answer: B

https://repost.aws/knowledge-center/ebs-copy-snapshot-data-s3-create-volume. =>I want to copy an Amazon Elastic Block Store (Amazon EBS) snapshot to my Amazon Simple Storage Service (Amazon S3) bucket. I also want to create Amazon EBS volumes from data that's stored in my S3 bucket.

Looks like B

Short description
When you create an EBS snapshot, it's automatically stored in an Amazon S3 bucket that AWS manages. You can copy snapshots within the same AWS Region, or from one Region to another. However, you can't copy snapshots to S3 buckets that you manage.

upvoted 1 times

**MatAlves** 1 year, 8 months ago

Selected Answer: B

If we consider both cost and administrative effort more closely:

EBS Snapshots Archive may be easier to manage but could incur higher costs over 7 years compared to S3 Glacier.
S3 Glacier, despite its complexity for initial transfers, could end up being more cost-effective in the long run, especially for large data volumes.
Ultimately, if minimizing costs is a primary concern and the organization can handle the initial complexity of transferring snapshots, using S3 Glacier (Option A) could still be worth considering.

So, while Option B is easier, if cost is a significant factor, Option A might be the better choice despite the additional administrative effort involved. It's a trade-off that depends on the organization's priorities regarding cost and operational simplicity.

upvoted 4 times

**MatAlves** 1 year, 8 months ago

"We charge you $0.0125 per GB-month of stored data and $0.03 per GB retrieved. "

https://aws.amazon.com/blogs/aws/new-amazon-ebs-snapshots-archive/

"All Storage / Month $0.0036 per GB
S3 Glacier Deep Archive *** - For long-term data archiving that is accessed once or twice in a year and can be restored within 12 hours"

https://aws.amazon.com/s3/pricing/

2nd time reviewing this question and, yeah, "A" is the option that better meets the "cost" requirement.

upvoted 2 times

**dhewa** 1 year, 9 months ago

Selected Answer: A

Keyword here is cost. S3 Glacier deep archive is significantly cheaper than keeping snapshots in the EBS snapshot standard tier or even the EBS Snapshots Archive.

upvoted 4 times

**Abdullah2004** 1 year, 9 months ago

Selected Answer: A

A is correct answer

upvoted 3 times

**Johnoppong101** 1 year, 9 months ago

Selected Answer: B

Choose EBS Snapshot Archive when:

Data is associated with EBS volumes.
You need to maintain point-in-time copies of your EBS volumes.
You require faster restore times than S3 Glacier Archive.
You need to comply with regulations requiring EBS snapshot retention.

upvoted 3 times

**Zahran23** 1 year, 11 months ago

**Selected Answer: A**
Option (B) is incorrect due to the following:
Archiving is recommended for monthly, quarterly, or yearly snapshots. Archiving daily incremental snapshots of a single volume can lead to higher costs when compared to keeping them in the standard tier.

https://docs.aws.amazon.com/ebs/latest/userguide/snapshot-archive.html

upvoted 5 times

**Cpso** 1 year, 6 months ago

the company needs to keep the monthly snapshots for 7 years. not daily

upvoted 1 times

Question #842 *Topic 1*

A company runs an application on several Amazon EC2 instances that store persistent data on an Amazon Elastic File System (Amazon EFS) file system. The company needs to replicate the data to another AWS Region by using an AWS managed service solution.

Which solution will meet these requirements MOST cost-effectively?

A. Use the EFS-to-EFS backup solution to replicate the data to an EFS file system in another Region.

B. Run a nightly script to copy data from the EFS file system to an Amazon S3 bucket. Enable S3 Cross-Region Replication on the S3 bucket.

C. Create a VPC in another Region. Establish a cross-Region VPC peer. Run a nightly rsync to copy data from the original Region to the new Region.

D. Use AWS Backup to create a backup plan with a rule that takes a daily backup and replicates it to another Region. Assign the EFS file system resource to the backup plan. Most Voted

**Correct Answer:** D

*Community vote distribution*

D (70%) A (30%)

## Comments

**sandordini** Highly Voted 2 years, 1 month ago

Selected Answer: D

NOTA: EFS-to-EFS backup: You must deploy this solution in the same AWS Region as your source Amazon EFS Folesystem
Not B, C: Not a managed AWS Solution
D: AWS backup will do the job, and is managed service.

upvoted 8 times

**EdricHoang** Highly Voted 1 year, 11 months ago

Selected Answer: D

A is oke but it says "'AWS managed service solution"
So I go for D

upvoted 5 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

Selected Answer: D

While EFS-to-EFS replication (Option A) is great for real-time replication, Option D (AWS Backup) is the more cost-effective choice for long-term retention and administrative simplicity.

Also, I copied this from somewhere:
EFS-to-EFS Replication might not technically be an "AWS Managed Service" in the same sense as services like AWS Backup or Amazon RDS, but it is a solution supported by AWS. The process is automated in the sense that AWS handles the replication between EFS file systems, but it does not provide the same level of managed service (like backup solutions) that abstracts away storage, retention, and backup policy management.

However, the EFS-to-EFS solution is often classified under the broader umbrella of AWS managed solutions, since it directly integrates with EFS, but AWS Backup may provide a more clear-cut "AWS managed" approach with better control over backup policies and retention.

upvoted 1 times

**Cpso** 1 year, 6 months ago

Selected Answer: D

A-D : managed solution
Cost D<A

upvoted 1 times

**Madushanka** 1 year, 7 months ago

Answer D: AWS Backup is a managed service that handles backup operations. If AWS Backup is not available in your region, you can consider using EFS-to-EFS backup
upvoted 2 times

**mk168898** 1 year, 7 months ago
AWS backup is the only managed service so D
upvoted 2 times

**NSA_Poker** 1 year, 11 months ago
**Selected Answer: D**
(A,B,C) are eliminated. They are NOT managed service solutions.
(D) is correct. 'AWS Backup offers a cost-effective, FULLY MANAGED, policy-based service that simplifies data protection at scale.'

https://aws.amazon.com/getting-started/hands-on/amazon-efs-backup-and-restore-using-aws-backup/
upvoted 3 times

**Rahulmaddy** 2 years ago
**Selected Answer: A**
A is cheaper and less complicated to impliment compared to D
upvoted 3 times

**Nm55569** 2 years ago
**Selected Answer: A**
Aws backup is not replicating. Efs replication is and it's managed in that you configure it and then it does the replication - no further actions requirec
It's also cheapest since it's free, you just pay for data transfer and storage
https://docs.aws.amazon.com/efs/latest/ug/efs-replication.html
https://aws.amazon.com/blogs/aws/new-replication-for-amazon-elastic-file-system-efs/
upvoted 4 times

**Scheldon** 2 years ago
**Selected Answer: D**
AnswerD

I think the most cost effective would solution presented in C, but hence in question it's clearly wrote that we should use AWS Managed Services Solution, hence I think we have no other choice then to choose AWS backup (Option D)
upvoted 2 times

**Scheldon** 2 years ago
https://docs.aws.amazon.com/managedservices/latest/userguide/features.html
upvoted 2 times

**expouser** 2 years ago
A: Replication is available in all AWS Regions in which EFS is available. To use replication in a Region that is disabled by default, you must first opt in to the Region. For more information, see Managing AWS Regions in the AWS General Reference Reference Guide. If you later opt out of a Region, Amazon EFS pauses all replication activities for the Region. To resume replication activities for the Region, you need to again opt in to the AWS Region. https://docs.aws.amazon.com/efs/latest/ug/efs-replication.html
upvoted 1 times

**0bdf3af** 2 years ago
A. you can replicate to another Region
upvoted 1 times

**BatVanyo** 2 years, 1 month ago
**Selected Answer: D**
To me "an AWS managed service solution" automatically translates to AWS Backup.
...Can't say if this is cheaper than EFS replication tho.
upvoted 3 times

**xBUGx** 2 years, 2 months ago
**Selected Answer: A**
To replicate data from an Amazon Elastic File System (EFS) file system to another AWS Region, the MOST cost-effective solution would be to use EFS Replication. Here's why:

EFS Replication:
EFS Replication allows you to natively create a copy of your file system in an AWS Region or Availability Zone (AZ) of your choice.
It automatically and transparently copies your data from the source file system to the destination, maintaining an RPO (Recovery Point Objective) of 15 minutes for most file systems.
This solution is specifically designed for replicating EFS data across Regions, ensuring data resilience and protection.
There are no additional costs for using replication failback, and you pay for the usual replication and file system changes as described in Amazon EFS

pricing12.
EFS Replication is available in all AWS Regions where EFS is available1.

upvoted 3 times

**boluwatito** 2 years, 2 months ago

But it is not a managed service

upvoted 3 times

Question #843 *Topic 1*

An ecommerce company is migrating its on-premises workload to the AWS Cloud. The workload currently consists of a web application and a backend Microsoft SQL database for storage.

The company expects a high volume of customers during a promotional event. The new infrastructure in the AWS Cloud must be highly available and scalable.

Which solution will meet these requirements with the LEAST administrative overhead?

A. Migrate the web application to two Amazon EC2 instances across two Availability Zones behind an Application Load Balancer. Migrate the database to Amazon RDS for Microsoft SQL Server with read replicas in both Availability Zones.

B. Migrate the web application to an Amazon EC2 instance that runs in an Auto Scaling group across two Availability Zones behind an Application Load Balancer. Migrate the database to two EC2 instances across separate AWS Regions with database replication.

C. Migrate the web application to Amazon EC2 instances that run in an Auto Scaling group across two Availability Zones behind an Application Load Balancer. Migrate the database to Amazon RDS with Multi-AZ deployment. **Most Voted**

D. Migrate the web application to three Amazon EC2 instances across three Availability Zones behind an Application Load Balancer. Migrate the database to three EC2 instances across three Availability Zones.

**Correct Answer:** C

*Community vote distribution*

C (83%) A (17%)

## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

Initially I chose A but then again, "highly available and scalable" should indicate multi az and auto scaling group, so C is the only valid option.

upvoted 1 times

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC

I would de-couple as many elements as possible with appopriate redundancy (HA), hence Auto-Scale in 2 AZ for EC2, Database in Multi-AZ and ALB infront of EC2. It will allow to increase amount of servers in case of need and will prevent from service unavailability in case if something fail ;)

upvoted 3 times

**zinabu** 2 years, 1 month ago

**Selected Answer: C**

yes ''C'' but it was better if it says Amazon RDS for Microsoft SQL Multi-AZ. any ways

upvoted 3 times

**KennethNg923** 1 year, 12 months ago

Agree, Options Only C has auto scaling and RDS, and RDS support sql server, so it could only be C

upvoted 2 times

**rondelldell** 2 years, 2 months ago

**Selected Answer: C**

only c

upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago

Selected Answer: A

Option a

upvoted 2 times

Question #844 *Topic 1*

A company has an on-premises business application that generates hundreds of files each day. These files are stored on an SMB file share and require a low-latency connection to the application servers. A new company policy states all application-generated files must be copied to AWS. There is already a VPN connection to AWS.

The application development team does not have time to make the necessary code modifications to move the application to AWS.

Which service should a solutions architect recommend to allow the application to copy files to AWS?

A. Amazon Elastic File System (Amazon EFS)

B. Amazon FSx for Windows File Server

C. AWS Snowball

D. AWS Storage Gateway **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (74%) B (26%)

## Comments

**FlyingHawk** 1 year, 5 months ago

**Selected Answer: D**

Only B and D support smb share, so A and C are out, the requirement is to copy the files at smb shares to AWS automatically. Storage Gateway automatically copies files to AWS (e.g., Amazon S3) in the background, meeting the requirement to copy all application-generated files to AWS.

upvoted 3 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: D**

D is good to go. AWS Storage Gateway provides seamless integration between on-premises environments and AWS, and specifically, File Gateway can be used to expose an SMB interface that on-premises applications can use.

For B: While FSx does support SMB, it's optimized for AWS cloud workloads, not necessarily for on-premises systems. If you want an on-premises system to directly interact with an SMB share in AWS, you would typically need a service that integrates with the on-premises network seamlessly.

upvoted 3 times

**ckhemani** 1 year, 6 months ago

**Selected Answer: D**

Amazon FSx for Windows File Server is not used of transfer data. if it was Amazon Fsx Gateway, then it would have been been correct.

They are looking solution for copying not moving to which filesystem.

upvoted 4 times

**dhewa** 1 year, 9 months ago

**Selected Answer: D**

B is incorrect in this case because the development team doesn't have time to modify the application.

upvoted 2 times

**EdricHoang** 1 year, 10 months ago

**Selected Answer: B**

Amazon FSx for Windows File Server also support SMB and hybrid solution. And, no modification is needed.

upvoted 4 times

**Nm55569** 2 years ago

**Selected Answer: B**

Why not "Amazon FSx for Windows File Server"?
upvoted 3 times

**ike001** 1 year, 11 months ago
because we require a hybrid solution here
upvoted 1 times

**EdricHoang** 1 year, 10 months ago
Amazon FSx for Windows File Server also support hybrid solution
upvoted 1 times

**XXXXXlNN** 1 year, 8 months ago
..I am wondering where it says FSx for Windows File Server also support hybrid solution?
upvoted 1 times

**Scheldon** 2 years ago
**Selected Answer: D**
AnswerD

StorageGateway

https://docs.aws.amazon.com/storagegateway/
upvoted 3 times

**jcck202020** 2 years, 1 month ago
AWS Storage Gateway provides a set of hybrid cloud storage services that offer on-premises access to virtually unlimited cloud storage. The File Gateway configuration of AWS Storage Gateway supports the SMB protocol (and NFS), enabling on-premises applications to seamlessly store and retrieve files in Amazon S3 using existing file system protocols. It fits perfectly for applications that need to continue operating without modification, while also adhering to the new policy of copying files to AWS.
Given these descriptions, Option D (AWS Storage Gateway) is the recommended service. It allows for a smooth integration by maintaining the existing SMB file-sharing capabilities and connects seamlessly to AWS through the VPN, enabling daily file transfers without significant changes to application code or infrastructure.
upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago
**Selected Answer: D**
AWS Storage Gateway service enables hybrid storage between on-premises environments and the AWS Cloud. It provides low-latency performance by caching frequently accessed data on premises, while storing data securely and durably in Amazon cloud storage services.
upvoted 2 times

**Kaula** 2 years, 2 months ago
**Selected Answer: D**
https://docs.aws.amazon.com/storagegateway/
upvoted 3 times

Question #845 *Topic 1*

A company has 15 employees. The company stores employee start dates in an Amazon DynamoDB table. The company wants to send an email message to each employee on the day of the employee's work anniversary.

Which solution will meet these requirements with the MOST operational efficiency?

A. Create a script that scans the DynamoDB table and uses Amazon Simple Notification Service (Amazon SNS) to send email messages to employees when necessary. Use a cron job to run this script every day on an Amazon EC2 instance.

B. Create a script that scans the DynamoDB table and uses Amazon Simple Queue Service (Amazon SQS) to send email messages to employees when necessary. Use a cron job to run this script every day on an Amazon EC2 instance.

C. Create an AWS Lambda function that scans the DynamoDB table and uses Amazon Simple Notification Service (Amazon SNS) to send email messages to employees when necessary. Schedule this Lambda function to run every day. **Most Voted**

D. Create an AWS Lambda function that scans the DynamoDB table and uses Amazon Simple Queue Service (Amazon SQS) to send email messages to employees when necessary. Schedule this Lambda function to run every day.

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**In2718** 10 months, 1 week ago

**Selected Answer: C**

If it's only 15 employees the correct answer is E - Google Calendar)) LOL

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: C**

Operational efficiency, script need to be run every time, and send email need to use SNS, so it should be C

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC

Deploying full instance is an overkill. Lambda should be enough + SNS to sent email. And it should be quite cheap

upvoted 3 times

**Mikado211** 2 years, 1 month ago

**Selected Answer: C**

SNS for sending mails
Lambda to scan the database + send the message to the SNS topic.

Using a script on a EC2 will add maintenance on both the EC2 and the script + cronjobs are not reliable and can be hard to monitor properly.

SO answer C !

upvoted 3 times

**Mikado211** 2 years, 1 month ago

SNS for sending mails
Lambda to scan the database + send the message to the SNS topic.

Using a script on a EC2 will add maintenance on both the EC2 and the script + cronjobs are not reliable and can be hard to monitor properly.

upvoted 2 times

Question #846 *Topic 1*

A company's application is running on Amazon EC2 instances within an Auto Scaling group behind an Elastic Load Balancing (ELB) load balancer. Based on the application's history, the company anticipates a spike in traffic during a holiday each year. A solutions architect must design a strategy to ensure that the Auto Scaling group proactively increases capacity to minimize any performance impact on application users.

Which solution will meet these requirements?

A. Create an Amazon CloudWatch alarm to scale up the EC2 instances when CPU utilization exceeds 90%.

B. Create a recurring scheduled action to scale up the Auto Scaling group before the expected period of peak demand.
**Most Voted**

C. Increase the minimum and maximum number of EC2 instances in the Auto Scaling group during the peak demand period.

D. Configure an Amazon Simple Notification Service (Amazon SNS) notification to send alerts when there are autoscaling:EC2_INSTANCE_LAUNCH events.

**Correct Answer:** B

*Community vote distribution*

B (80%) A (15%) 5%

## Comments

**Cpso** 1 year, 6 months ago

**Selected Answer: C**

objective is to optimize autoscaling but being fix number of ec2: "design a strategy to ensure that the Auto Scaling group proactively increases capacity to minimize any performance impact on application users."

upvoted 1 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: B**

the company anticipates a spike in traffic during a holiday each year -> schedule action

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: B**

AnswerB

https://medium.com/@damadhav/aws-scaling-reactive-vs-proactive-vs-predictive-2701ad6d48c9

upvoted 4 times

**7ce90e0** 2 years, 1 month ago

**Selected Answer: B**

it says proactively

upvoted 4 times

**zinabu** 2 years, 1 month ago

**Selected Answer: B**

it needs a scheduled action for the yearly holiday peak traffic

upvoted 2 times

**Mikado211** 2 years, 1 month ago

**Selected Answer: B**

Since we know when we will have a peak of activity. A scheduled scaling is a good idea.

upvoted 3 times

**zinabu** 2 years, 1 month ago

selected answer: B
it needs a scheduled action for the yearly holiday peak traffic

upvoted 2 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: A**

The answer IS A

upvoted 3 times

Question #847 *Topic 1*

A company uses Amazon RDS for PostgreSQL databases for its data tier. The company must implement password rotation for the databases.

Which solution meets this requirement with the LEAST operational overhead?

A. Store the password in AWS Secrets Manager. Enable automatic rotation on the secret. **Most Voted**

B. Store the password in AWS Systems Manager Parameter Store. Enable automatic rotation on the parameter.

C. Store the password in AWS Systems Manager Parameter Store. Write an AWS Lambda function that rotates the password.

D. Store the password in AWS Key Management Service (AWS KMS). Enable automatic rotation on the AWS KMS key.

**Correct Answer:** A

*Community vote distribution*

A (100%)



## Comments

**Scheldon** 1 year, 6 months ago

**Selected Answer: A**

AnswerA
"In Secrets Manager, you can set up automatic rotation for your secrets."
https://docs.aws.amazon.com/secretsmanager/latest/userguide/rotating-secrets.html

upvoted 4 times

**88f8032** 1 year, 7 months ago

**Selected Answer: A**

A. AWS Secrets Manager

upvoted 2 times

**jcck202020** 1 year, 8 months ago

Option A (Store the password in AWS Secrets Manager and enable automatic rotation on the secret) is the best solution. It meets the requirements with the least operational overhead by leveraging built-in features specifically designed for managing and rotating database credentials securely.

upvoted 3 times

Question #848 *Topic 1*

A company runs its application on Oracle Database Enterprise Edition. The company needs to migrate the application and the database to AWS. The company can use the Bring Your Own License (BYOL) model while migrating to AWS. The application uses third-party database features that require privileged access.

A solutions architect must design a solution for the database migration.

Which solution will meet these requirements MOST cost-effectively?

A. Migrate the database to Amazon RDS for Oracle by using native tools. Replace the third-party features with AWS Lambda.

B. Migrate the database to Amazon RDS Custom for Oracle by using native tools. Customize the new database settings to support the third-party features. **Most Voted**

C. Migrate the database to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS). Customize the new database settings to support the third-party features.

D. Migrate the database to Amazon RDS for PostgreSQL by using AWS Database Migration Service (AWS DMS). Rewrite the application code to remove the dependency on third-party features.

**Correct Answer:** B

*Community vote distribution*

B (100%)



## Comments

**Scheldon** 1 year, 6 months ago

**Selected Answer: B**

AnswerB

BYOL + third-party features/applications = RDS Custom. Hence customer is using Oracle so we should use RDS customer for Oracle
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/rds-custom.html
https://aws.amazon.com/blogs/aws/amazon-rds-custom-for-oracle-new-control-capabilities-in-database-environment/
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/working-with-custom-oracle.html

upvoted 3 times

**zinabu** 1 year, 7 months ago

**Selected Answer: B**

Amazon RDS Custom for Oracle by using native tools= to support the third-party features.

upvoted 1 times

**Hkayne** 1 year, 7 months ago

**Selected Answer: B**

B IS suitable for this use case : thé use or BYOL and the use or third party features with privileged access

upvoted 1 times

**jcck202020** 1 year, 8 months ago

Considering the requirements and the need to use Oracle Database features with privileged access and BYOL, Option B (Migrate the database to Amazon RDS Custom for Oracle by using native tools. Customize the new database settings to support the third-party features) is the most cost-effective and suitable solution. It allows for significant customization needed to accommodate specific third-party features while leveraging existing Oracle licenses.

upvoted 2 times

Question #849 *Topic 1*

A large international university has deployed all of its compute services in the AWS Cloud. These services include Amazon EC2, Amazon RDS, and Amazon DynamoDB. The university currently relies on many custom scripts to back up its infrastructure. However, the university wants to centralize management and automate data backups as much as possible by using AWS native options.

Which solution will meet these requirements?

A. Use third-party backup software with an AWS Storage Gateway tape gateway virtual tape library.

B. Use AWS Backup to configure and monitor all backups for the services in use. **Most Voted**

C. Use AWS Config to set lifecycle management to take snapshots of all data sources on a schedule.

D. Use AWS Systems Manager State Manager to manage the configuration and monitoring of backup tasks.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

**Mikado211** Highly Voted 1 year, 8 months ago

**Selected Answer: B**

Centralized management of backups == AWS Backup

upvoted 5 times

**Scheldon** Most Recent 1 year, 6 months ago

**Selected Answer: B**

AnswerB

AWS native tool that will support EC2, RDS and DynamoDB = AWS Backup

https://docs.aws.amazon.com/aws-backup/latest/devguide/working-with-supported-services.html

upvoted 3 times

**Hkayne** 1 year, 7 months ago

**Selected Answer: B**

Automate backups = AWS Backup

upvoted 3 times

Question #850 *Topic 1*

A company wants to build a map of its IT infrastructure to identify and enforce policies on resources that pose security risks. The company's security team must be able to query data in the IT infrastructure map and quickly identify security risks.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon RDS to store the data. Use SQL to query the data to identify security risks.

B. Use Amazon Neptune to store the data. Use SPARQL to query the data to identify security risks. Most Voted

C. Use Amazon Redshift to store the data. Use SQL to query the data to identify security risks.

D. Use Amazon DynamoDB to store the data. Use PartiQL to query the data to identify security risks.

**Correct Answer:** B

*Community vote distribution*

B (90%) 10%

## Comments

**NSA_Poker** 1 year, 11 months ago

Selected Answer: B

Visualize your AWS Infrastructure with Amazon Neptune and AWS Config
https://aws.amazon.com/blogs/database/visualize-your-aws-infrastructure-with-amazon-neptune-and-aws-config/

Using Amazon Neptune for Security Graphs
https://aws.amazon.com/neptune/security-graphs-on-aws/#:~:text=Using%20Amazon%20Neptune%20for%20Security%20Graphs

upvoted 2 times

**Rahulmaddy** 2 years ago

Selected Answer: D

Option B is very tough to impliment for such a simple usecase.

upvoted 1 times

**Scheldon** 2 years ago

Selected Answer: B

AnswerB

Neptune is the only Graph database from all options which seems to be most suitable.
I'm not sure about SPARQL but hence Neptun is the best option hence SPARQL should be to.
https://docs.aws.amazon.com/neptune/latest/userguide/intro.html

upvoted 3 times

**jcck202020** 2 years, 1 month ago

Option B (Use Amazon Neptune to store the data. Use SPARQL to query the data) is the most suitable choice. Neptune is purpose-built for storing and querying graph data, making it a natural fit for representing and querying the complex relationships inherent in an IT infrastructure map. Additionally, SPARQL is a powerful and efficient query language for graph databases, facilitating quick identification of security risks.

upvoted 4 times

**dds69** 2 years, 2 months ago

Selected Answer: B

Using Amazon Neptune with SPARQL, a query language for graph databases, allows the security team to easily query the data in the IT infrastructure map to identify security risks. SPARQL is specifically designed for querying graph data and allows for complex queries to traverse relationships between resources efficiently.

upvoted 4 times

Question #851 *Topic 1*

A large company wants to provide its globally located developers separate, limited size, managed PostgreSQL databases for development purposes. The databases will be low volume. The developers need the databases only when they are actively working.

Which solution will meet these requirements MOST cost-effectively?

A. Give the developers the ability to launch separate Amazon Aurora instances. Set up a process to shut down Aurora instances at the end of the workday and to start Aurora instances at the beginning of the next workday.

B. Develop an AWS Service Catalog product that enforces size restrictions for launching Amazon Aurora instances. Give the developers access to launch the product when they need a development database.

C. Create an Amazon Aurora Serverless cluster. Develop an AWS Service Catalog product to launch databases in the cluster with the default capacity settings. Grant the developers access to the product. **Most Voted**

D. Monitor AWS Trusted Advisor checks for idle Amazon RDS databases. Create a process to terminate identified idle RDS databases.

**Correct Answer:** C

*Community vote distribution*

C (76%) B (24%)



## Comments

**NSA_Poker** Highly Voted 1 year, 11 months ago

**Selected Answer: C**

(A,B,D) eliminated. Aurora instances & Amazon RDS use On-Demand or Reserved INSTANCES. These are more expensive than a serverless solution.

(C) is correct. Amazon Aurora Serverless automatically starts up, shuts down & scales capacity up or down based on your application's needs; you pay only for capacity consumed.

upvoted 10 times

**kelmryan1** Highly Voted 2 years, 1 month ago

Yes but its asking for the most cost effective. B would cause frustration for developers if it was terminated unexpectedly. The answer should be C so developers can easily access when they are needed and auto scales based on demand

upvoted 6 times

**NSA_Poker** 1 year, 11 months ago

You're correct for the wrong reasons. An instance being 'terminated unexpectedly' & it having 'easily access" are qualities related to high availability & not cost effectiveness.

(B) is wrong because it's more $$$.

upvoted 2 times

**In2718** Most Recent 10 months, 1 week ago

**Selected Answer: B**

I don't think they need a cluster, so go with answer B

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

A - Aurora = Overkill and expensive.

B - Just like A.

C - Aurora Serverless scales based on usage and can automatically scale down to zero when the database is not in use. This is ideal for a development environment where databases are only needed during active work hours.

D - Money is not an issue here I think. The real problem is - there's no guarantee that the database would be automatically available when the developers need it. The need for a manual process to monitor and terminate idle databases introduces operational overhead and the potential for human error.

upvoted 3 times

**mk168898** 1 year, 7 months ago

Selected Answer: C

when i see "managed PostgreSQL databases", i immediately look for serverless

upvoted 3 times

**b3b5fdd** 1 year, 8 months ago

Selected Answer: B

I think the answer correct is B as the database should be managed by the developer and Each developer needs a separate DB.

upvoted 1 times

**Johnoppong101** 1 year, 9 months ago

Selected Answer: B

Requirement: Each developer needs a separate DB. In a cluster, DB is shared among developers.

upvoted 1 times

**Salilgen** 1 year, 5 months ago

"There's no limit to the number of databases you can have in an Aurora Serverless cluster"
https://aws.amazon.com/blogs/database/best-practices-for-working-with-amazon-aurora-serverless/

upvoted 1 times

**Johnoppong101** 1 year, 9 months ago

Guys, listen to the question, each developer -> a separate managed PostgreSQL DB.
this is the task. Then Most cost-effective comes. If you get a cheaper thing that does not fulfull the task, you are wrong.
Question: Can each developer get a separate DB for himself from the cluster?

upvoted 1 times

**ike001** 1 year, 11 months ago

Answer C: We know we require Service Catalog. This option provides workflow no how it works

upvoted 2 times

**Scheldon** 2 years ago

AnswerC

Using Aurora Serverless solution + AWS Service Catalog features seems to be a good idea

https://aws.amazon.com/rds/aurora/serverless/
https://aws.amazon.com/servicecatalog/features/

upvoted 2 times

**f07ed8f** 2 years ago

Selected Answer: B

Only option B can have limited size database

upvoted 1 times

**NSA_Poker** 1 year, 11 months ago

"when creating your Aurora database cluster, specify the desired range of database capacity or use the defaults, and connect your applications."
This way we limit the capacity that we pay for.

https://aws.amazon.com/rds/aurora/serverless/

upvoted 5 times

**Sergiuss95** 2 years, 1 month ago

Selected Answer: C

I thin is c

upvoted 3 times

**Hkayne** 2 years, 1 month ago

Selected Answer: B

With AWS Service Catalog, you can meet your compliance requirements while making sure your customers can quickly deploy the cloud resources they need.

upvoted 2 times

Question #852 *Topic 1*

A company is building a web application that serves a content management system. The content management system runs on Amazon EC2 instances behind an Application Load Balancer (ALB). The EC2 instances run in an Auto Scaling group across multiple Availability Zones. Users are constantly adding and updating files, blogs, and other website assets in the content management system.

A solutions architect must implement a solution in which all the EC2 instances share up-to-date website content with the least possible lag time.

Which solution meets these requirements?

A. Update the EC2 user data in the Auto Scaling group lifecycle policy to copy the website assets from the EC2 instance that was launched most recently. Configure the ALB to make changes to the website assets only in the newest EC2 instance.

B. Copy the website assets to an Amazon Elastic File System (Amazon EFS) file system. Configure each EC2 instance to mount the EFS file system locally. Configure the website hosting application to reference the website assets that are stored in the EFS file system. **Most Voted**

C. Copy the website assets to an Amazon S3 bucket. Ensure that each EC2 instance downloads the website assets from the S3 bucket to the attached Amazon Elastic Block Store (Amazon EBS) volume. Run the S3 sync command once each hour to keep files up to date.

D. Restore an Amazon Elastic Block Store (Amazon EBS) snapshot with the website assets. Attach the EBS snapshot as a secondary EBS volume when a new EC2 instance is launched. Configure the website hosting application to reference the website assets that are stored in the secondary EBS volume.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

**mk168898** 1 year, 7 months ago

**Selected Answer: B**

website content must be in sync with all EC2 instances, so use EFS

upvoted 1 times

**ike001** 1 year, 11 months ago

B is the answer

upvoted 1 times

**Scheldon** 2 years ago

**Selected Answer: B**

AnswerB

Looks like this is the only reasonable solution from all presented

upvoted 2 times

**sandordini** 2 years, 1 month ago

**Selected Answer: B**

CMS is usually EFS

upvoted 4 times

Question #853 Topic 1

A company's web application consists of multiple Amazon EC2 instances that run behind an Application Load Balancer in a VPC. An Amazon RDS for MySQL DB instance contains the data. The company needs the ability to automatically detect and respond to suspicious or unexpected behavior in its AWS environment. The company already has added AWS WAF to its architecture.

What should a solutions architect do next to protect against threats?

A. Use Amazon GuardDuty to perform threat detection. Configure Amazon EventBridge to filter for GuardDuty findings and to invoke an AWS Lambda function to adjust the AWS WAF rules. Most Voted

B. Use AWS Firewall Manager to perform threat detection. Configure Amazon EventBridge to filter for Firewall Manager findings and to invoke an AWS Lambda function to adjust the AWS WAF web ACL.

C. Use Amazon Inspector to perform threat detection and to update the AWS WAF rules. Create a VPC network ACL to limit access to the web application.

D. Use Amazon Macie to perform threat detection and to update the AWS WAF rules. Create a VPC network ACL to limit access to the web application.

**Correct Answer:** A

*Community vote distribution*

A (100%)

## Comments

**Awsbeginner87** Highly Voted 1 year, 8 months ago

Selected Answer: A

Malicious or suspicious activity - think of GuardDuty

upvoted 5 times

**Scheldon** Most Recent 1 year, 6 months ago

Selected Answer: A

AnswerA

Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior across your AWS environment.
https://aws.amazon.com/guardduty/features/

upvoted 4 times

**zinabu** 1 year, 7 months ago

Selected Answer: A

Gard duty for automatic treat detection

upvoted 3 times

**Hkayne** 1 year, 7 months ago

Selected Answer: A

Threat detection means guarduty

upvoted 4 times

Question #854 Topic 1

A company is planning to run a group of Amazon EC2 instances that connect to an Amazon Aurora database. The company has built an AWS CloudFormation template to deploy the EC2 instances and the Aurora DB cluster. The company wants to allow the instances to authenticate to the database in a secure way. The company does not want to maintain static database credentials.

Which solution meets these requirements with the LEAST operational effort?

A. Create a database user with a user name and password. Add parameters for the database user name and password to the CloudFormation template. Pass the parameters to the EC2 instances when the instances are launched.

B. Create a database user with a user name and password. Store the user name and password in AWS Systems Manager Parameter Store. Configure the EC2 instances to retrieve the database credentials from Parameter Store.

C. Configure the DB cluster to use IAM database authentication. Create a database user to use with IAM authentication. Associate a role with the EC2 instances to allow applications on the instances to access the database. Most Voted

D. Configure the DB cluster to use IAM database authentication with an IAM user. Create a database user that has a name that matches the IAM user. Associate the IAM user with the EC2 instances to allow applications on the instances to access the database.

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**Hkayne** Highly Voted 1 year, 7 months ago

Selected Answer: C

Using IAM database authentication and associate a role with the EC2 instances is the least operational effort.

upvoted 7 times

**Scheldon** Highly Voted 1 year, 6 months ago

Selected Answer: C

AnswerC

Customer would like to not manage password rotation from my understanding, hence A and B are not the best solution here. I don't think we can associate IAM user with EC2 instance, but we can associate IAM role.
In summary C
https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_use_switch-role-ec2.html
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/iam-roles-for-amazon-ec2.html
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/UsingWithRDS.IAMDBAuth.html

upvoted 5 times

Question #855 *Topic 1*

A company wants to configure its Amazon CloudFront distribution to use SSL/TLS certificates. The company does not want to use the default domain name for the distribution. Instead, the company wants to use a different domain name for the distribution.

Which solution will deploy the certificate without incurring any additional costs?

A. Request an Amazon issued private certificate from AWS Certificate Manager (ACM) in the us-east-1 Region.

B. Request an Amazon issued private certificate from AWS Certificate Manager (ACM) in the us-west-1 Region.

C. Request an Amazon issued public certificate from AWS Certificate Manager (ACM) in the us-east-1 Region. **Most Voted**

D. Request an Amazon issued public certificate from AWS Certificate Manager (ACM) in the us-west-1 Region.

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

Yes, C. Just memorize that. For CloudFront distributions with custom domain names, ACM public certificates must be requested from us-east-1.

upvoted 4 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: C**

Have to use east-1 region for ACM, and it should be public SSL/TLS for domain, so it should be C

upvoted 4 times

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC

Per AWS "Public SSL/TLS certificates provisioned through AWS Certificate Manager are free. You pay only for the AWS resources you create to run your application."
https://aws.amazon.com/certificate-manager/pricing/?nc=sn&loc=3

But hence AWS is recommending to use US east 1 I think I would go with C
Note
We recommend that you use ACM to provision, manage, and deploy SSL/TLS certificates on AWS managed resources. You must request an ACM certificate in the US East (N. Virginia) Region.
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/cnames-and-https-procedures.html

upvoted 3 times

**sandordini** 2 years, 1 month ago

**Selected Answer: C**

browsers trust public certificates automatically by default > C or D
To use an ACM certificate with Amazon CloudFront, you must request or import the certificate in the US East (N. Virginia) region [Nowhere is it statec why is this though...] > C

upvoted 2 times

**BatVanyo** 2 years, 1 month ago

**Selected Answer: C**

The certificate has to be public. The certificate has to be issued in us-east-1:
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/cnames-and-https-requirements.html
"To use an ACM certificate with CloudFront, make sure you request (or import) the certificate in the US East (N. Virginia) Region (us-east-1)."

upvoted 3 times

**rondelldell** 2 years, 1 month ago

**Selected Answer: C**

https://aws.amazon.com/certificate-manager/pricing/
AWS Certificate Manager Pricing
Public SSL/TLS certificates provisioned through AWS Certificate Manager are free. You pay only for the AWS resources you create to run your application.
If you manage AWS Private Certificate Authority (CA) through ACM, refer to the AWS Private CA Pricing page for more details and examples.

upvoted 2 times

**boluwatito** 2 years, 2 months ago

**Selected Answer: C**

It is c

upvoted 3 times

**boluwatito** 2 years, 2 months ago

Should be c, it is a public certificate

upvoted 2 times

**JackyCCK** 2 years, 2 months ago

CloudFront should have a private cert and browser use public cert aiming to achieve non-repudiation. Ans should be A

upvoted 1 times

**cloudee** 2 years, 2 months ago

**Selected Answer: C**

This should be C. Private CA is not free

upvoted 2 times

**Awsbeginner87** 2 years, 2 months ago

Why not D.evrn option D is public CA

upvoted 2 times

Question #856 *Topic 1*

A company creates operations data and stores the data in an Amazon S3 bucket. For the company's annual audit, an external consultant needs to access an annual report that is stored in the S3 bucket. The external consultant needs to access the report for 7 days.

The company must implement a solution to allow the external consultant access to only the report.

Which solution will meet these requirements with the MOST operational efficiency?

A. Create a new S3 bucket that is configured to host a public static website. Migrate the operations data to the new S3 bucket. Share the S3 website URL with the external consultant.

B. Enable public access to the S3 bucket for 7 days. Remove access to the S3 bucket when the external consultant completes the audit.

C. Create a new IAM user that has access to the report in the S3 bucket. Provide the access keys to the external consultant. Revoke the access keys after 7 days.

D. Generate a presigned URL that has the required access to the location of the report on the S3 bucket. Share the presigned URL with the external consultant. Most Voted

**Correct Answer:** D

*Community vote distribution*

D (100%)

## Comments

**Hkayne** Highly Voted 1 year, 7 months ago

**Selected Answer: D**

Pre-signed URLs are used to provide short-term access to a private object in your S3 bucket. They work by appending an AWS Access Key, expiration time, and Sigv4 signature as query parameters to the S3 object.

upvoted 6 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

**Selected Answer: D**

A - Yeah, too much trouble.
B - Even if you remove access after the 7 days, the bucket would be publicly accessible during the time frame, which could lead to accidental data exposure.
C - A little better than A, but you gotta create a new IAM user, provide access keys, and manually revok them after 7 days. I'm not even sure I can remember that a week later...
D - No need to manage IAM users or keys. Also, everything you need to do can be done within 10 minutes.

upvoted 2 times

**Scheldon** 1 year, 6 months ago

**Selected Answer: D**

AnswerD
When using CLI to create presigned URL we can setup 7days (max) for URL expiration from the time of creation.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/ShareObjectPreSignedURL.html

upvoted 4 times

Question #857 *Topic 1*

A company plans to run a high performance computing (HPC) workload on Amazon EC2 Instances. The workload requires low-latency network performance and high network throughput with tightly coupled node-to-node communication.

Which solution will meet these requirements?

A. Configure the EC2 instances to be part of a cluster placement group. **Most Voted**

B. Launch the EC2 instances with Dedicated Instance tenancy.

C. Launch the EC2 instances as Spot Instances.

D. Configure an On-Demand Capacity Reservation when the EC2 instances are launched.

**Correct Answer:** A

*Community vote distribution*

A (100%)



## Comments

**Awsbeginner87** Highly Voted 2 years, 2 months ago

**Selected Answer: A**

Tightly coupled, low-latency,hpc - cluster placement group

upvoted 7 times

**victor78** Most Recent 1 year, 11 months ago

I THINK IT'S C,D

upvoted 1 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: A**

tightly coupled node-to-node communication -> placement group

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: A**

AnswerA

I was thinking it will be D but after some research I think it will be A

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html

upvoted 3 times

**Linuslin** 2 years ago

**Selected Answer: A**

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html

upvoted 1 times

**zinabu** 2 years, 1 month ago

**Selected Answer: A**

he key word here is "tightly coupled node-to-node communication" which means we need to Configure the EC2 instances in a cluster placement group.

upvoted 1 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: A**

all points to a cluster placement group

upvoted 3 times

**AlvinC2024** 2 years, 2 months ago

**Selected Answer: A**

The answer should be A.

upvoted 2 times

Question #858 *Topic 1*

A company has primary and secondary data centers that are 500 miles (804.7 km) apart and interconnected with high-speed fiber-optic cable. The company needs a highly available and secure network connection between its data centers and a VPC on AWS for a mission-critical workload. A solutions architect must choose a connection solution that provides maximum resiliency.

Which solution meets these requirements?

A. Two AWS Direct Connect connections from the primary data center terminating at two Direct Connect locations on two separate devices

B. A single AWS Direct Connect connection from each of the primary and secondary data centers terminating at one Direct Connect location on the same device

C. Two AWS Direct Connect connections from each of the primary and secondary data centers terminating at two Direct Connect locations on two separate devices **Most Voted**

D. A single AWS Direct Connect connection from each of the primary and secondary data centers terminating at one Direct Connect location on two separate devices

**Correct Answer:** C

*Community vote distribution*

C (100%)

## Comments

**Hkayne** Highly Voted 1 year, 6 months ago

**Selected Answer: C**

The most resilient is C

upvoted 5 times

**Scheldon** Most Recent 1 year, 6 months ago

**Selected Answer: C**

AnswerC

Maximum resilient solution will be to have 2 independet connections to AWS from each location (maximum redundancy)

upvoted 3 times

**Sri4321** 1 year, 6 months ago

**Selected Answer: C**

Redundant Connections: Having two Direct Connect connections from each data center provides redundancy in case one connection fails.
Diverse Direct Connect Locations: Terminating the connections at two different Direct Connect locations further eliminates the risk of a single point c failure due to issues at a specific location.
Separate Devices: Using separate devices at each Direct Connect location adds another layer of redundancy, preventing a single device failure from impacting connectivity.

upvoted 4 times

Question #859 Topic 1

A company runs several Amazon RDS for Oracle On-Demand DB instances that have high utilization. The RDS DB instances run in member accounts that are in an organization in AWS Organizations.

The company's finance team has access to the organization's management account and member accounts. The finance team wants to find ways to optimize costs by using AWS Trusted Advisor.

Which combination of steps will meet these requirements? (Choose two.)

A. Use the Trusted Advisor recommendations in the management account. Most Voted

B. Use the Trusted Advisor recommendations in the member accounts where the RDS DB instances are running.

C. Review the Trusted Advisor checks for Amazon RDS Reserved Instance Optimization. Most Voted

D. Review the Trusted Advisor checks for Amazon RDS Idle DB Instances.

E. Review the Trusted Advisor checks for compute optimization. Crosscheck the results by using AWS Compute Optimizer.

**Correct Answer:** AC

*Community vote distribution*

AC (71%) 13% Other

## Comments

**Scheldon** Highly Voted 2 years ago

Selected Answer: AC

AnswerAC
We have On-Demand instances of RDS. If DB is used very often then it can occur that using Reserved Instrance can bring some $$ savings.

It is mentioned in AWS docs :)

https://docs.aws.amazon.com/awssupport/latest/user/cost-optimization-checks.html#amazon-ec2-reserved-instances-optimization
upvoted 5 times

**SnaxAttax** Most Recent 6 months, 2 weeks ago

Selected Answer: AC

High utilization means we don't care about idle-- that isn't the problem we are solving.
upvoted 1 times

**In2718** 9 months, 2 weeks ago

Selected Answer: BC

Chat AI -

B. Use the Trusted Advisor recommendations in the member accounts where the RDS DB instances are running.
— The finance team needs to run checks in the accounts where the RDS instances actually reside.

C. Review the Trusted Advisor checks for Amazon RDS Reserved Instance Optimization.
— Trusted Advisor will suggest buying RDS Reserved Instances for cost savings if DBs have high utilization.
upvoted 2 times

**7dcef09** 1 year, 1 month ago

Selected Answer: AC

A is correct
You can set up the organizational view feature to create a report for all member accounts in your AWS organization. For more information, see Organizational view for AWS Trusted Advisor.
upvoted 1 times

**7dcef09** 1 year, 1 month ago
https://docs.aws.amazon.com/awssupport/latest/user/organizational-view.html
upvoted 1 times

**FlyingHawk** 1 year, 5 months ago
**Selected Answer: BC**
A is incorrect, as Trust advisor recommendations are account specific. Trusted Advisor in the management account provides insights only for that account, not for member accounts. Unless you have a centralized tool that aggregates recommendations, this option alone does not meet the requirement. C helps identify opportunities to save money by purchasing Reserved Instances for RDS if usage patterns justify it. This is one of the key cost optimization strategies for highly utilized RDS instances.
upvoted 2 times

**FlyingHawk** 1 year, 5 months ago
"You can use the Trusted Advisor Recommendations page of the Trusted Advisor console to review check results for your AWS account and then follow the recommended steps to fix any issues. " Trust advisor recommendations are account specific.
upvoted 2 times

**FlyingHawk** 1 year, 5 months ago
https://docs.aws.amazon.com/awssupport/latest/user/get-started-with-aws-trusted-advisor.html
upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago
**Selected Answer: AC**
B is OK but less efficient than A.
D is extremely unlikely because "instances have high utilization".
E - The finance team is focused on RDS for Oracle On-Demand DB instances, not compute (EC2) instances, so this check won't be useful for optimizing RDS costs.
upvoted 1 times

**MatAlves** 1 year, 8 months ago
**Selected Answer: AC**
"On-Demand DB instances that have high utilization"

High Utilization = no point in checking for Idle instances
On-Demand = it makes sense to replace On-demand for Reserved instances.
upvoted 4 times

**example_** 1 year, 11 months ago
**Selected Answer: CD**
https://docs.aws.amazon.com/awssupport/latest/user/cost-optimization-checks.html
upvoted 1 times

**victor78** 1 year, 11 months ago
I THINK IT'S C,D
upvoted 2 times

**example_** 1 year, 11 months ago
https://docs.aws.amazon.com/awssupport/latest/user/cost-optimization-checks.html
upvoted 1 times

**Tomrr** 2 years ago
**Selected Answer: AC**
Instances are running On-Demand, they need to check to see if they can save money by switching the reserved instances from on-demand

An important part of using AWS involves balancing your Reserved Instance (RI) purchase against your On-Demand Instance usage. This check provides recommendations on which RIs will help reduce the costs incurred from using On-Demand Instances.
upvoted 3 times

**bujuman** 2 years ago
**Selected Answer: AD**
Cost Optimization: By providing actionable recommendations, Trusted Advisor assists you in identifying areas of overspending and underutilization, like idle RDS DB instances or underused EBS volumes, leading to significant cost savings.
upvoted 1 times

**f07ed8f** 2 years ago
**Selected Answer: AD**
Please check with question 308. It should run in management account and review the Trusted Advisor check for Amazon RDS Idle DB Instances.
upvoted 1 times

**f07ed8f** 2 years ago
I correct myself the answer should be AC as the DB can be switch to on-demand rather than reserved and it would save save the cost.
upvoted 2 times

**rondelldell** 2 years, 2 months ago
**Selected Answer: AC**
https://docs.aws.amazon.com/awssupport/latest/user/cost-optimization-checks.html#amazon-rds-reserved-instance-optimization
upvoted 4 times

**Awsbeginner87** 2 years, 2 months ago
**Selected Answer: AE**
Option A,E
upvoted 1 times

**xBUGx** 2 years, 2 months ago
i dont think AWS Compute Optimizer work with RDS
upvoted 1 times

**xBUGx** 2 years, 2 months ago
**Selected Answer: AD**
use Trusted advisor on management account
upvoted 1 times

**AlvinC2024** 2 years, 2 months ago
**Selected Answer: AC**
https://docs.aws.amazon.com/awssupport/latest/user/organizational-view.html
upvoted 3 times

Question #860 Topic 1

A solutions architect is creating an application. The application will run on Amazon EC2 instances in private subnets across multiple Availability Zones in a VPC. The EC2 instances will frequently access large files that contain confidential information. These files are stored in Amazon S3 buckets for processing. The solutions architect must optimize the network architecture to minimize data transfer costs.

What should the solutions architect do to meet these requirements?

A. Create a gateway endpoint for Amazon S3 in the VPC. In the route tables for the private subnets, add an entry for the gateway endpoint. **Most Voted**

B. Create a single NAT gateway in a public subnet. In the route tables for the private subnets, add a default route that points to the NAT gateway.

C. Create an AWS PrivateLink interface endpoint for Amazon S3 in the VPIn the route tables for the private subnets, add an entry for the interface endpoint.

D. Create one NAT gateway for each Availability Zone in public subnets. In each of the route tables for the private subnets, add a default route that points to the NAT gateway in the same Availability Zone.

**Correct Answer:** A

*Community vote distribution*

A (100%)



## Comments

**Scheldon** 1 year, 6 months ago

**Selected Answer: A**

AnswerA

I think only option A have any sense. It is cheap (no cost), it is secure (traffic is not going to public network).
https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html

upvoted 3 times

**Hkayne** 1 year, 7 months ago

**Selected Answer: A**

Aws gateway will have no cost because the trafic will stay on aws infrastructure.

upvoted 2 times

**Tanidanindo** 1 year, 8 months ago

**Selected Answer: A**

Gateway endpoint will minimize data transfer costs

upvoted 2 times

**Awsbeginner87** 1 year, 8 months ago

**Selected Answer: A**

A- gateway endpoint for S3

upvoted 2 times

**xBUGx** 1 year, 8 months ago

**Selected Answer: A**

gateway endpoint for Amazon S3

upvoted 1 times

**AlvinC2024** 1 year, 8 months ago

**Selected Answer: A**

Gateway endpoint is free https://digitalcloud.training/vpc-interface-endpoint-vs-gateway-endpoint-in-aws/.
upvoted 1 times

Question #861 *Topic 1*

A company wants to relocate its on-premises MySQL database to AWS. The database accepts regular imports from a client-facing application, which causes a high volume of write operations. The company is concerned that the amount of traffic might be causing performance issues within the application.

How should a solutions architect design the architecture on AWS?

A. Provision an Amazon RDS for MySQL DB instance with Provisioned IOPS SSD storage. Monitor write operation metrics by using Amazon CloudWatch. Adjust the provisioned IOPS if necessary. **Most Voted**

B. Provision an Amazon RDS for MySQL DB instance with General Purpose SSD storage. Place an Amazon ElastiCache cluster in front of the DB instance. Configure the application to query ElastiCache instead.

C. Provision an Amazon DocumentDB (with MongoDB compatibility) instance with a memory optimized instance type. Monitor Amazon CloudWatch for performance-related issues. Change the instance class if necessary.

D. Provision an Amazon Elastic File System (Amazon EFS) file system in General Purpose performance mode. Monitor Amazon CloudWatch for IOPS bottlenecks. Change to Provisioned Throughput performance mode if necessary.

**Correct Answer:** A

*Community vote distribution*

A (82%) B (18%)



## Comments

**Hkayne** Highly Voted 2 years, 1 month ago

**Selected Answer: A**

A or B. Can't be B because there is high volume of write no need for Elasticache

upvoted 7 times

**GOTJ** Most Recent 1 year, 3 months ago

**Selected Answer: B**

Even though option "A" makes sense, everybody discarding option "B" is focusing on improving the import process performance itself by using ElastiCache, which is basically correct. However, there is not a clear indicator that the "client-facing application" ONLY performs high volumes of write operations. In fact, this can be pointless (a "write only" database?).

In my opinion, the main concern of the company is how THE REST of the application processes might be affected while this write-intensive operations run in the database. And those operations surely includes reads.

Using an ElastiCache in front of the RDB instance or creating a read replica to redirect reads to might help with performance of THE REST of application when write-intensive imports processes are executing, as the query result might be cached.

My vote goes to "B"

upvoted 1 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: A**

high volume of write operation -> Provisioned IOPS SSD storage

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: A**

Answer A

For sure we cannot choose generar purpose IOPS SSD hence I would choos provisioned one. addtionally it is a good idea to monitor performance with CloudWatch and adjust setup(provisioned IOPS) if there will be a need.

upvoted 4 times

**sandordini** 2 years, 1 month ago

**Selected Answer: B**

The most effective strategy for coping with that limit is to supplement disk-based databases with in-memory caching (Elasticache for Redis, Write-through strategy) I'd go for B...

upvoted 3 times

**JA2018** 1 year, 6 months ago

why I would not choose #B? While ElastiCache can be used for caching read-heavy workloads, it's not the best choice for a database with high write operations as it is primarily designed for fast reads from a cache.

upvoted 1 times

**Mr_Marcus** 2 years ago

If it were changes to existing data, maybe. The scenario specifically says data imports. Going with "A".

upvoted 2 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: A**

Amazon RDS for MySQL DB instance with Provisioned IOPS SSD storage

upvoted 4 times

Question #862 *Topic 1*

A company runs an application in the AWS Cloud that generates sensitive archival data files. The company wants to rearchitect the application's data storage. The company wants to encrypt the data files and to ensure that third parties do not have access to the data before the data is encrypted and sent to AWS. The company has already created an Amazon S3 bucket.

Which solution will meet these requirements?

A. Configure the S3 bucket to use client-side encryption with an Amazon S3 managed encryption key. Configure the application to use the S3 bucket to store the archival files.

B. Configure the S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Configure the application to use the S3 bucket to store the archival files.

C. Configure the S3 bucket to use dual-layer server-side encryption with AWS KMS keys (SSE-KMS). Configure the application to use the S3 bucket to store the archival files.

D. Configure the application to use client-side encryption with a key stored in AWS Key Management Service (AWS KMS). Configure the application to store the archival files in the S3 bucket. Most Voted

**Correct Answer:** D

*Community vote distribution*

D (100%)



## Comments

**xBUGx** Highly Voted 1 year, 8 months ago

**Selected Answer: D**

"ensure that third parties do not have access to the data before the data is encrypted and sent to AWS"

upvoted 8 times

**f07ed8f** Highly Voted 1 year, 6 months ago

**Selected Answer: D**

"Amazon S3 managed encryption key" (SSE-S3) is a server-side encryption. Therefore it is not a client-side encryption. To encrypt the data before sending to S3, it has to be client-side encryption.

upvoted 6 times

**Hkayne** Most Recent 1 year, 6 months ago

**Selected Answer: D**

Must encrypt the data on client side before uploading it to S3

upvoted 3 times

Question #863 *Topic 1*

A company uses Amazon RDS with default backup settings for its database tier. The company needs to make a daily backup of the database to meet regulatory requirements. The company must retain the backups for 30 days.

Which solution will meet these requirements with the LEAST operational overhead?

A. Write an AWS Lambda function to create an RDS snapshot every day.

B. Modify the RDS database to have a retention period of 30 days for automated backups. **Most Voted**

C. Use AWS Systems Manager Maintenance Windows to modify the RDS backup retention period.

D. Create a manual snapshot every day by using the AWS CLI. Modify the RDS backup retention period.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

**Hkayne** Highly Voted 2 years, 1 month ago

**Selected Answer: B**

By default, Amazon RDS creates and saves automated backups of your DB instance securely in Amazon S3 for a user-specified retention period. You can set the backup retention period from 1 to 35 days. The maximum retention period currently available for automated snapshots is 35 days. When automated backups are turned on for your DB Instance, Amazon RDS automatically performs a full, daily snapshot of your data and captures transaction logs.

upvoted 5 times

**EdricHoang** Most Recent 1 year, 11 months ago

**Selected Answer: B**

Its B

"Amazon RDS performs a full daily backup of your data during a backup window that you define when you create the DB instance. You can configure a retention period of up to 35 days for the automated backup."

upvoted 2 times

Question #864 *Topic 1*

A company that runs its application on AWS uses an Amazon Aurora DB cluster as its database. During peak usage hours when multiple users access and read the data, the monitoring system shows degradation of database performance for the write queries. The company wants to increase the scalability of the application to meet peak usage demands.

Which solution will meet these requirements MOST cost-effectively?

A. Create a second Aurora DB cluster. Configure a copy job to replicate the users' data to the new database. Update the application to use the second database to read the data.

B. Create an Amazon DynamoDB Accelerator (DAX) cluster in front of the existing Aurora DB cluster. Update the application to use the DAX cluster for read-only queries. Write data directly to the Aurora DB cluster.

C. Create an Aurora read replica in the existing Aurora DB cluster. Update the application to use the replica endpoint for read-only queries and to use the cluster endpoint for write queries. **Most Voted**

D. Create an Amazon Redshift cluster. Copy the users' data to the Redshift cluster. Update the application to connect to the Redshift cluster and to perform read-only queries on the Redshift cluster.

**Correct Answer:** C

*Community vote distribution*

C (70%) B (30%)

## Comments

**bmmbmm** 6 months ago

**Selected Answer: C**

How B could be an answer?? Amazon DynamoDB Accelerator is for "DynamoDB" literally..

upvoted 1 times

**In2718** 10 months, 1 week ago

**Selected Answer: B**

Follow-up... definitely answer B

"During peak usage hours when multiple users access and read the data"

Gemini -

DAX is a caching service for DynamoDB, and its cost is primarily based on the instance type and the number of nodes in your DAX cluster, charged on a per-hour basis.

DAX is designed to provide microsecond-level response times for read-heavy applications using DynamoDB, offloading read requests from the DynamoDB table and reducing read capacity unit (RCU) consumption.

upvoted 1 times

**Steve122** 8 months, 3 weeks ago

DAX -> Amazon DynamoDB Accelerator!

upvoted 2 times

**In2718** 10 months, 1 week ago

**Selected Answer: B**

The better answer between B and C is this - MOST cost-effectively. The cost for a DAX will cost less than the additional read replicas.

upvoted 1 times

**mk168898** 1 year, 7 months ago

I was thinking between B and C. Saw that DAX is for Dynamo DB which is different from the question Aurora DB.
So i picked C

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: C**

Read Replica for multiple users access and read the data

upvoted 4 times

**ug56c** 1 year, 12 months ago

Answer C, wording is that read queries are slowing down write queries -> we need to optimize for read queries -> we need to add read replicas.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Managing.Performance.html#Aurora.Managing.Performance.ReadScaling

upvoted 4 times

**sheilawu** 2 years ago

**Selected Answer: B**

write queries=>High performance of read & Write Dynamo, I think DAX is a better solution.

upvoted 1 times

**Rhydian25** 1 year, 11 months ago

Dynamo is NoSQL...

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: C**

AnswerC
A - We can create only RO replica of DB hence this is not possible.
D - Redshift is not for a DB from my understanding but more like analitics tools
B - Could be a thinkg but there is no point of Read-Only from DAX and write to DB cluster. beside it is hard to say if it would be cost effective as we are paying per hour not R/W requests as it is with replica.
Hence I would go with C

upvoted 2 times

Question #865 Topic 1

A company's near-real-time streaming application is running on AWS. As the data is ingested, a job runs on the data and takes 30 minutes to complete. The workload frequently experiences high latency due to large amounts of incoming data. A solutions architect needs to design a scalable and serverless solution to enhance performance.

Which combination of steps should the solutions architect take? (Choose two.)

A. Use Amazon Kinesis Data Firehose to ingest the data. Most Voted

B. Use AWS Lambda with AWS Step Functions to process the data.

C. Use AWS Database Migration Service (AWS DMS) to ingest the data.

D. Use Amazon EC2 instances in an Auto Scaling group to process the data.

E. Use AWS Fargate with Amazon Elastic Container Service (Amazon ECS) to process the data. Most Voted

**Correct Answer:** AE

*Community vote distribution*

AE (100%)

## Comments

**xBUGx** Highly Voted 2 years, 2 months ago

Selected Answer: AE

Lambda maxed to 15mins

upvoted 5 times

**JoeTromundo** 1 year, 8 months ago

But through Step Functions it would be possible to divide a long task, lasting more than 15 minutes, into multiple Lambda invocations with a maximum duration of 15 minutes.

upvoted 1 times

**JA2018** 1 year, 6 months ago

but that will introduce additional overhead.

upvoted 1 times

**mk168898** Most Recent 1 year, 7 months ago

Eliminate B because lambda max handle 15minutes.

upvoted 2 times

**Scheldon** 2 years ago

Selected Answer: AE

AnswerAE

Amazon Kinesis data firehose for data ingesting, and hence output cannot go to EC2, hence Fargate with ECS.

upvoted 3 times

**Hkayne** 2 years, 1 month ago

Selected Answer: AE

A is correct for ingesting data.
B or E both choises are serverless but the difference is the lambda maximum execution time is 15 minutes. So the right option is E.
A and E

upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago

Selected Answer: AE

A-ingesting real-time data
E- serverless option ECS+fargate

upvoted 4 times

**AlvinC2024** 2 years, 2 months ago

**Selected Answer: AE**

The maximum run time for lambda is 15 mins.

upvoted 3 times

Question #866 *Topic 1*

A company runs a web application on multiple Amazon EC2 instances in a VPC. The application needs to write sensitive data to an Amazon S3 bucket. The data cannot be sent over the public internet.

Which solution will meet these requirements?

A. Create a gateway VPC endpoint for Amazon S3. Create a route in the VPC route table to the endpoint. **Most Voted**

B. Create an internal Network Load Balancer that has the S3 bucket as the target.

C. Deploy the S3 bucket inside the VPCreate a route in the VPC route table to the bucket.

D. Create an AWS Direct Connect connection between the VPC and an S3 regional endpoint.

**Correct Answer:** A

*Community vote distribution*

A (100%)



## Comments

**mk168898** 1 year, 7 months ago

not allowed to access internet, and need to access S3 => gateway VPC endpoint

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: A**

AnswerA

upvoted 2 times

**sandordini** 2 years, 1 month ago

**Selected Answer: A**

no internet, S3 > gateway VPC endpoint

upvoted 3 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: A**

VPC endpoint

upvoted 2 times

**Mikado211** 2 years, 2 months ago

**Selected Answer: A**

"data cannot be sent over the public internet." == VPC Endpoint

upvoted 2 times

**Awsbeginner87** 2 years, 2 months ago

**Selected Answer: A**

Option A

upvoted 2 times

Question #867 *Topic 1*

A company runs its production workload on Amazon EC2 instances with Amazon Elastic Block Store (Amazon EBS) volumes. A solutions architect needs to analyze the current EBS volume cost and to recommend optimizations. The recommendations need to include estimated monthly saving opportunities.

Which solution will meet these requirements?

A. Use Amazon Inspector reporting to generate EBS volume recommendations for optimization.

B. Use AWS Systems Manager reporting to determine EBS volume recommendations for optimization.

C. Use Amazon CloudWatch metrics reporting to determine EBS volume recommendations for optimization.

D. Use AWS Compute Optimizer to generate EBS volume recommendations for optimization. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (100%)



## Comments

**Scheldon** Highly Voted 2 years ago

**Selected Answer: D**

AnswerD

AWS Compute Optimizer helps avoid overprovisioning and underprovisioning four types of AWS resources—Amazon Elastic Compute Cloud (EC2) instance types, Amazon Elastic Block Store (EBS) volumes, Amazon Elastic Container Service (ECS) services on AWS Fargate, and AWS Lambda functions—based on your utilization data.
https://aws.amazon.com/compute-optimizer/

upvoted 5 times

**KennethNg923** Most Recent 1 year, 12 months ago

**Selected Answer: D**

Since need analyze the current EBS volume cost and to recommend optimizations, so have to use AWS compute optimizer

upvoted 2 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: D**

Get recommendations to optimize your use of AWS resources

upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago

**Selected Answer: D**

AWS Compute Optimizer helps avoid overprovisioning and underprovisioning four types of AWS resources—Amazon Elastic Compute Cloud (EC2) instance types, Amazon Elastic Block Store (EBS) volumes, Amazon Elastic Container Service (ECS) services on AWS Fargate, and AWS Lambda functions—based on your utilization data.

upvoted 2 times

Question #868 *Topic 1*

A global company runs its workloads on AWS. The company's application uses Amazon S3 buckets across AWS Regions for sensitive data storage and analysis. The company stores millions of objects in multiple S3 buckets daily. The company wants to identify all S3 buckets that are not versioning-enabled.

Which solution will meet these requirements?

B. Use Amazon S3 Storage Lens to identify all S3 buckets that are not versioning-enabled across Regions. **Most Voted**

C. Enable IAM Access Analyzer for S3 to identify all S3 buckets that are not versioning-enabled across Regions.

D. Create an S3 Multi-Region Access Point to identify all S3 buckets that are not versioning-enabled across Regions.

**Correct Answer:** B

*Community vote distribution*

B (100%)



## Comments

**joseantoniopolo** Highly Voted 1 year, 8 months ago

Selected Answer: B

https://aws.amazon.com/blogs/aws/s3-storage-lens/

upvoted 10 times

**bmmbmm** Most Recent 6 months ago

Selected Answer: B

S3 Storage Lens's Data-protection metrics provide insights for data protection features, such as encryption and S3 Versioning. For example, you can identify buckets that are not using default encryption with AWS Key Management Service keys (SSE-KMS) or S3 Versioning.
from
https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage_lens_basics_metrics_recommendations.html

upvoted 1 times

**nadeerm** 1 year, 3 months ago

Selected Answer: B

Answer is B
Option A: Set up an AWS CloudTrail event that has a rule to identify all S3 buckets that are not versioning-enabled across Regions.

upvoted 2 times

**LeonSauveterre** 1 year, 5 months ago

Selected Answer: B

FYI, the missing Option A: Set up an AWS CloudTrail event that has a rule to identify all S3 buckets that are not versioning-enabled across Regions.

A - CloudTrail is meant for audit logs, not for analyzing bucket configurations directly.
B - Storage Lens can provide an overview of all S3 buckets and their versioning status across Regions. Those built-in features that can directly tackle your problems should be your top preferences.
C - IAM Access Analyzer is designed for access management, not storage configuration analysis.
D - S3 Multi-Region Access Points allow applications to access S3 buckets across Regions using a single global endpoint. So not relevant as well.

upvoted 4 times

**sandordini** 1 year, 7 months ago

Correct answer: A
A: You can use an AWS Config managed rule to identify Amazon S3 buckets that do not have versioning enabled.

upvoted 2 times

**JA2018** 1 year, 6 months ago

Potential Option A could "Use AWS Config rules to identify all S3 buckets that are not versioning-enabled across Regions "

upvoted 1 times

**sandordini** 1 year, 7 months ago

Question #889 contains all the answers including A, which is cloudtrail and obviously wrong.
On the other hand S3 Storage Lens:
You can use the Versioning-enabled bucket count metric to see which buckets use S3 Versioning. Then, you can take action in the S3 console to enable S3 Versioning for other buckets. So correct answer: B

upvoted 8 times

**Awsbeginner87** 1 year, 8 months ago

Where is option A

upvoted 3 times

**xBUGx** 1 year, 8 months ago

where is option A?

upvoted 2 times

Question #869 Topic 1

A company wants to enhance its ecommerce order-processing application that is deployed on AWS. The application must process each order exactly once without affecting the customer experience during unpredictable traffic surges.

Which solution will meet these requirements?

A. Create an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Put all the orders in the SQS queue. Configure an AWS Lambda function as the target to process the orders. Most Voted

B. Create an Amazon Simple Notification Service (Amazon SNS) standard topic. Publish all the orders to the SNS standard topic. Configure the application as a notification target.

C. Create a flow by using Amazon AppFlow. Send the orders to the flow. Configure an AWS Lambda function as the target to process the orders.

D. Configure AWS X-Ray in the application to track the order requests. Configure the application to process the orders by pulling the orders from Amazon CloudWatch.

**Correct Answer:** A

*Community vote distribution*

A (100%)

## Comments

**bmmbmm** 6 months ago

Selected Answer: A

We now have to know how much do AWS loves SQS+Lambda strategy after all this time.

upvoted 2 times

**mk168898** 1 year, 7 months ago

Must process at least once => SQS

upvoted 2 times

**MatAlves** 1 year, 8 months ago

Selected Answer: A

"Standard queues support at-least-once message delivery, and FIFO queues support exactly-once message processing and high-throughput mode."

https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/welcome.html#sqs-benefits

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

Selected Answer: A

must process each order exactly once -> FIFO queue

upvoted 2 times

**Scheldon** 2 years ago

Selected Answer: A

AnswerA

SQS with FIFO queue will allow to read every customer order in order in which they came and only once.

upvoted 2 times

**Hkayne** 2 years, 1 month ago

Selected Answer: A

FIFO queue is the solution

upvoted 2 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: A**

SQS and FIFO

upvoted 2 times

**Kaula** 2 years, 2 months ago

**Selected Answer: A**

FIFO > SQS

upvoted 2 times

**Mikado211** 2 years, 2 months ago

**Selected Answer: A**

The application must process each order exactly once == SQS + FIFO

upvoted 2 times

Question #870 *Topic 1*

A company has two AWS accounts: Production and Development. The company needs to push code changes in the Development account to the Production account. In the alpha phase, only two senior developers on the development team need access to the Production account. In the beta phase, more developers will need access to perform testing.

Which solution will meet these requirements?

A. Create two policy documents by using the AWS Management Console in each account. Assign the policy to developers who need access.

B. Create an IAM role in the Development account. Grant the IAM role access to the Production account. Allow developers to assume the role.

C. Create an IAM role in the Production account. Define a trust policy that specifies the Development account. Allow developers to assume the role. **Most Voted**

D. Create an IAM group in the Production account. Add the group as a principal in a trust policy that specifies the Production account. Add developers to the group.

**Correct Answer:** C

*Community vote distribution*

C (74%) D (21%) 5%

## Comments

**bmmbmm** 6 months ago

**Selected Answer: C**

Option D is incorrect because an IAM group cannot be used as a principal in an IAM role trust policy.
A trust policy can only specify principals such as: AWS accounts, IAM users, IAM roles and Federated identities.
IAM groups are not valid principals, so they cannot be granted permission to assume a role.
Therefore, the trust relationship in Option D is not technically possible.

upvoted 1 times

**FlyingHawk** 1 year, 5 months ago

**Selected Answer: C**

1. Creating an IAM role in the Production account with a trust relationship to the Development account.

2. Grant the sts:AssumeRole permission to the specific developers (or groups) who need access to the Production account.
3. During the beta phase, you can easily add more developers to the IAM group or policy in the Development account that allows them to assume the role in the Production account.
4. In the Alpha phase, grant AssumeRole Permission to the groups of Two Senior Developers only

upvoted 3 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

A - We need secure cross-account access between the Development and Production accounts. Option A did nothing about that.
B - This setup violates the AWS best practice for cross-account access, which recommends creating roles in the account being accessed (aka Production env).
C - As more developers need access, you can grant permissions in the Development account without modifying the role in Production. Also, access is granted through temporary credentials generated when the role is assumed, reducing the risk of long-term credential exposure.
D - Groups cannot establish trust between accounts. It didn't provide any mechanism for Development account users to access the Production account.

upvoted 2 times

**LeonSauveterre** 1 year, 5 months ago

To "assume" a role in AWS means that you (a developer) temporarily take on the permissions associated with that role. This is done using the AWS Security Token Service (STS), which generates temporary credentials.

Brief steps of option C:
1. Sets up a trust policy that allows entities from the Development account to assume the role. Attaches permissions for the required resources (in this case, access to specific services).
2. Attach an IAM policy to your user or role, allowing you to call "sts:AssumeRole".
3. Assume the role by doing this:

aws sts assume-role \
--role-arn "arn:aws:iam::PRODUCTION_ACCOUNT_ID:role/ROLE_NAME" \
--role-session-name "MySessionName"

4. Use the temporary credentials to access the resources in the Production account during the session duration.
upvoted 1 times

**AMEJack** 1 year, 6 months ago
**Selected Answer: C**
It should be C, groups can't be used in trust policy.
upvoted 2 times

**Mayank0502** 1 year, 11 months ago
**Selected Answer: D**
answer should be D
upvoted 1 times

**f07ed8f** 2 years ago
**Selected Answer: C**
https://docs.aws.amazon.com/IAM/latest/UserGuide/tutorial_cross-account-with-roles.html
upvoted 3 times

**TwinSpark** 2 years ago
**Selected Answer: D**
Weird question, but D is actually the only one that allow you to select which developer got access and when, so will go for D
upvoted 1 times

**KennethNg923** 1 year, 12 months ago
Agree, as C will let any developers assume the role without control
upvoted 1 times

**KennethNg923** 1 year, 12 months ago
I check here: https://docs.aws.amazon.com/IAM/latest/UserGuide/tutorial_cross-account-with-roles.html, and yes it should be use IAM role, I correct my choice to C
upvoted 4 times

**03beafc** 2 years, 1 month ago
**Selected Answer: A**
you can't assign groups as principals, b and c don't specify only the senior devs, a is the only one that works here
upvoted 1 times

**03beafc** 2 years, 1 month ago
edit, none of these answers are right....
upvoted 1 times

**Mikado211** 2 years, 1 month ago
**Selected Answer: D**
If you want ALL the developers to assume the role in the production, then C using a trust policy to assume the role in production is perfect BUT

You could allow users in development account to assume the role in production, but in the end you will maintain potentially a big trust policy depending of the total number of users.

Here you want only some developers to connect to the production (others will follow without knowing if they all can connect and without knowing the number) so managing a separate group will give you a little more maintenance but will allow you to have different rights between the users.

I'd say D
upvoted 1 times

**802c4ff** 2 years, 1 month ago
**Selected Answer: C**
https://docs.aws.amazon.com/IAM/latest/UserGuide/tutorial_cross-account-with-roles.html
upvoted 3 times

**xBUGx** 2 years, 2 months ago

**Selected Answer: D**

i think D is better

upvoted 1 times

Question #871 *Topic 1*

A company wants to restrict access to the content of its web application. The company needs to protect the content by using authorization techniques that are available on AWS. The company also wants to implement a serverless architecture for authorization and authentication that has low login latency.

The solution must integrate with the web application and serve web content globally. The application currently has a small user base, but the company expects the application's user base to increase.

Which solution will meet these requirements?

A. Configure Amazon Cognito for authentication. Implement Lambda@Edge for authorization. Configure Amazon CloudFront to serve the web application globally. **Most Voted**

B. Configure AWS Directory Service for Microsoft Active Directory for authentication. Implement AWS Lambda for authorization. Use an Application Load Balancer to serve the web application globally.

C. Configure Amazon Cognito for authentication. Implement AWS Lambda for authorization. Use Amazon S3 Transfer Acceleration to serve the web application globally.

D. Configure AWS Directory Service for Microsoft Active Directory for authentication. Implement Lambda@Edge for authorization. Use AWS Elastic Beanstalk to serve the web application globally.

**Correct Answer:** A

*Community vote distribution*

A (100%)

## Comments

**Salilgen** 1 year, 5 months ago

**Selected Answer: A**

https://aws.amazon.com/blogs/networking-and-content-delivery/authorizationedge-how-to-use-lambdaedge-and-json-web-tokens-to-enhance-web-application-security/

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

Authen: Cognito
Globally: Lambda@Edge + cloudfront

upvoted 3 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: A**

Serve content globally means the use of Cloudfront

upvoted 4 times

**Danges** 2 years, 2 months ago

**Selected Answer: A**

Implementación a nivel global ==> AWS Cloud Front

upvoted 2 times

Question #872 *Topic 1*

A development team uses multiple AWS accounts for its development, staging, and production environments. Team members have been launching large Amazon EC2 instances that are underutilized. A solutions architect must prevent large instances from being launched in all accounts.

How can the solutions architect meet this requirement with the LEAST operational overhead?

A. Update the IAM policies to deny the launch of large EC2 instances. Apply the policies to all users.

B. Define a resource in AWS Resource Access Manager that prevents the launch of large EC2 instances.

C. Create an IAM role in each account that denies the launch of large EC2 instances. Grant the developers IAM group access to the role.

D. Create an organization in AWS Organizations in the management account with the default policy. Create a service control policy (SCP) that denies the launch of large EC2 instances, and apply it to the AWS accounts. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (100%)

## Comments

**Hkayne** Highly Voted 2 years, 1 month ago

**Selected Answer: D**

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps.html

upvoted 6 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

**Selected Answer: D**

A - You would need to update and maintain separate IAM policies in each account, which is too much trouble.
B - AWS Resource Access Manager (RAM) is primarily for resource sharing and does not directly restrict resource launches.
C - IAM roles are better suited for granting access rather than imposing restrictions. You would also need to create and maintain these roles in every account, even more trouble than option A.
D - SCPs are powerful here because they apply at the root account level, meaning that even if a developer has direct IAM permissions to launch large instances, the SCP will override and prevent it.

Also, there's no need to create or manage multiple IAM policies or roles across accounts. Once the SCP is defined and applied, it enforces the restriction automatically.

upvoted 1 times

**744fdad** 1 year, 10 months ago

why is it not A? If the goal is only to prevent launch of EC2s

upvoted 2 times

**78b9037** 1 year, 5 months ago

Updating IAM policies must be done separately in each account, and policies for each user must be maintained, which will cause high operation overhead, so D.

upvoted 1 times

**example_** 1 year, 11 months ago

**Selected Answer: D**

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps_examples.html

upvoted 3 times

Question #873 *Topic 1*

A company has migrated a fleet of hundreds of on-premises virtual machines (VMs) to Amazon EC2 instances. The instances run a diverse fleet of Windows Server versions along with several Linux distributions. The company wants a solution that will automate inventory and updates of the operating systems. The company also needs a summary of common vulnerabilities of each instance for regular monthly reviews.

What should a solutions architect recommend to meet these requirements?

A. Set up AWS Systems Manager Patch Manager to manage all the EC2 instances. Configure AWS Security Hub to produce monthly reports.

B. Set up AWS Systems Manager Patch Manager to manage all the EC2 instances. Deploy Amazon Inspector, and configure monthly reports. Most Voted

C. Set up AWS Shield Advanced, and configure monthly reports. Deploy AWS Config to automate patch installations on the EC2 instances.

D. Set up Amazon GuardDuty in the account to monitor all EC2 instances. Deploy AWS Config to automate patch installations on the EC2 instances.

**Correct Answer:** B

*Community vote distribution*

B (94%) 6%

## Comments

**TwinSpark** Highly Voted 2 years ago

Selected Answer: B

AWS Security Hub performs security best practice checks, aggregates alerts, and enables automated remediation.
Amazon Inspector scan workload for vulnerabilities
Guardduty threat detection for malicious activity in all account
AWS Shield DDOS
Regarding security B and D can be right (maybe D a little too much). For patching B is the only valid option.

upvoted 6 times

**mk168898** Most Recent 1 year, 7 months ago

Inspector is for checking vulnerabilities, so B

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

Selected Answer: B

Inspector for Common vulnerabilities so it is B

upvoted 2 times

**maxhg** 2 years, 1 month ago

inspector for instances and software vulnerabilities

upvoted 4 times

**Hkayne** 2 years, 1 month ago

Selected Answer: B

AWS Systems Manager Patch Manager to automate the process of installing security-related updates for both the operating system and applications
Amazon Inspector for Automated and continual vulnerability management at scale

upvoted 2 times

**Alagong** 2 years, 2 months ago

Selected Answer: A

Create an Auto Scaling group and an ELB in the DR Region, configuring the DynamoDB table as a global table, and setting up DNS failover to the new ELB. This approach allows for quick failover since the infrastructure is already in place and only DNS needs to be updated to redirect traffic.

upvoted 1 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: B**

Inspector for vulnerability scanning

upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago

**Selected Answer: B**

Option b

upvoted 3 times

Question #874 *Topic 1*

A company hosts its application in the AWS Cloud. The application runs on Amazon EC2 instances in an Auto Scaling group behind an Elastic Load Balancing (ELB) load balancer. The application connects to an Amazon DynamoDB table.

For disaster recovery (DR) purposes, the company wants to ensure that the application is available from another AWS Region with minimal downtime.

Which solution will meet these requirements with the LEAST downtime?

A. Create an Auto Scaling group and an ELB in the DR Region. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new DR Region's ELB. **Most Voted**

B. Create an AWS CloudFormation template to create EC2 instances, ELBs, and DynamoDB tables to be launched when necessary. Configure DNS failover to point to the new DR Region's ELB.

C. Create an AWS CloudFormation template to create EC2 instances and an ELB to be launched when necessary. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new DR Region's ELB.

D. Create an Auto Scaling group and an ELB in the DR Region. Configure the DynamoDB table as a global table. Create an Amazon CloudWatch alarm with an evaluation period of 10 minutes to invoke an AWS Lambda function that updates Amazon Route 53 to point to the DR Region's ELB.

**Correct Answer:** A

*Community vote distribution*

A (92%) 8%

## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: A**

A - By creating an Auto Scaling group and ELB in the DR Region upfront, you ensure that the infrastructure is ready to handle traffic immediately in case of a failure in the primary Region. This minimizes downtime during a disaster.
B & C - Waiting for the CloudFormation stack to create instances, ELBs, and DynamoDB tables delays recovery.
D - CloudWatch alarms and Lambda functions will introduce additional complexity and potential delays due to the evaluation period (>= 10 minutes).

For A, simply create an Auto Scaling group and ELB in the DR Region. Use the same configurations as the primary Region. Just make sure that the AMI used for EC2 instances includes all necessary application code and dependencies.

upvoted 2 times

**mk168898** 1 year, 7 months ago

B and C seems to take a while to launch, so it is not minimal down time

upvoted 2 times

**muhammadahmer36** 1 year, 10 months ago

**Selected Answer: A**

Downtime I would choose Auto scaling + DNS failover rather than use cloud formation create infrastructure in DR region or Auto scaling + Lambda

upvoted 4 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: A**

Downtime I would choose Auto scaling + DNS failover rather than use cloud formation create infrastructure in DR region or Auto scaling + Lambda

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: A**

AnswerA
Hence there is no information that solution need to be cost effective and the main requirement is minimal downtime i would go with AUTOSCALING in DR region with 1 ELB and 1 server there but in case of need amount of servers can be increased automatically. Hence at least 1 server and ELB will be waiting and DynamoDB thanks to global table will be active in the same DR region as well, hence we need to inform users, using DNS about new destination, hence DNS failover to the ELB in DR region is the best solution here.

upvoted 4 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: A**

With dynamo global tables, we just need to create an ELB and a ASG in the DR region resources. This resources will be used only if the main region fail over.

upvoted 2 times

**Alagong** 2 years, 2 months ago

**Selected Answer: A**

Create an Auto Scaling group and an ELB in the DR Region, configuring the DynamoDB table as a global table, and setting up DNS failover to the new ELB. This approach allows for quick failover since the infrastructure is already in place and only DNS needs to be updated to redirect traffic.

upvoted 4 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: A**

Least downtime. C does not offer minimal downtime

upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago

**Selected Answer: C**

Option C

upvoted 2 times

Question #875 *Topic 1*

A company runs an application on Amazon EC2 instances in a private subnet. The application needs to store and retrieve data in Amazon S3 buckets. According to regulatory requirements, the data must not travel across the public internet.

What should a solutions architect do to meet these requirements MOST cost-effectively?

A. Deploy a NAT gateway to access the S3 buckets.

B. Deploy AWS Storage Gateway to access the S3 buckets.

C. Deploy an S3 interface endpoint to access the S3 buckets.

D. Deploy an S3 gateway endpoint to access the S3 buckets. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (87%) 13%



## Comments

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: D**

About option C - S3 Interface Endpoint
1. Creates a private network interface (ENI) in your VPC that connects to Amazon S3.
2. Provides granular IAM-based controls at the operation level.
3. Typically much more expensive due to the cost associated with the network interface and data processing charges.

About option D - S3 Gateway Endpoint:
1. Creates a route entry in the VPC route table for accessing S3 without crossing the public internet.
2. Simpler and more cost-effective for high-volume S3 access, as there are no additional data processing charges or interface costs.
3. Doesn't support granular IAM controls per operation, but bucket policies can be used for access control so that's good.

upvoted 1 times

**JA2018** 1 year, 6 months ago

**Selected Answer: C**

Why I opt for Option C?

#Private Access:
It allows direct access to S3 buckets from a private subnet without data traversing the public internet, meeting the regulatory requirement.

#Cost-Effective:
Compared to other options, utilizing an S3 interface endpoint is generally the most cost-efficient way to achieve private access to S3.

#Summary:
When data needs to stay within a private network, use an S3 interface endpoint to access S3 buckets.
Always consider the specific regulatory requirements and cost implications when choosing a solution.

upvoted 1 times

**mk168898** 1 year, 7 months ago

no internet and need access to s3 => s3 gateway endpoint

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: D**

Gateway endpoint free, so definitely interface end point expensive than it

upvoted 4 times

**Scheldon** 2 years ago

**Selected Answer: D**

AnswerD
https://docs.aws.amazon.com/AmazonS3/latest/userguide/privatelink-interface-endpoints.html
taking into consideration that in both cases (s3 Instance Endpoint and S3Gateway endpoint), network traffic remains on the AWS network we need tc think about other data which we have. For example application is in AWS cloud hence there is no need for access from on-premises. in that situation S3 Gateway endpoint seems to be better (and it is for free)
https://docs.aws.amazon.com/AmazonS3/latest/userguide/privatelink-interface-endpoints.html#types-of-vpc-endpoints-for-s3

upvoted 4 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: D**

D for sure.

upvoted 2 times

**BatVanyo** 2 years, 1 month ago

**Selected Answer: D**

Gateway endpoints are free.

upvoted 3 times

**awsshare** 2 years, 2 months ago

**Selected Answer: D**

Sorry, I think D is the correct option. Gateway endpoint is cheaper than Interface endpoint

upvoted 2 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: D**

Gateway endpoint for S3

upvoted 4 times

**awsshare** 2 years, 2 months ago

**Selected Answer: C**

should be C

upvoted 2 times

Question #876 Topic 1

A company hosts an application on Amazon EC2 instances that run in a single Availability Zone. The application is accessible by using the transport layer of the Open Systems Interconnection (OSI) model. The company needs the application architecture to have high availability.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

A. Configure new EC2 instances in a different Availability Zone. Use Amazon Route 53 to route traffic to all instances.

B. Configure a Network Load Balancer in front of the EC2 instances. Most Voted

C. Configure a Network Load Balancer for TCP traffic to the instances. Configure an Application Load Balancer for HTTP and HTTPS traffic to the instances.

D. Create an Auto Scaling group for the EC2 instances. Configure the Auto Scaling group to use multiple Availability Zones. Configure the Auto Scaling group to run application health checks on the instances. Most Voted

E. Create an Amazon CloudWatch alarm. Configure the alarm to restart EC2 instances that transition to a stopped state.

**Correct Answer:** BD

*Community vote distribution*

BD (100%)

## Comments

**Scheldon** Highly Voted 2 years ago

Selected Answer: BD

AnswerBD
There is no information about the ports numbers and only that means that we need to use NLB, ALB is only for HTTP hence there is no point of using that solution as connection can be done via any port for exampl 4000
Autoscaling with instances in multiple AZ is the best solution. it will allow run new EC2 if it fail and in case if whole AZ will go down we will have 2nd one.

upvoted 8 times

**LuongTo** Most Recent 1 year, 7 months ago

C is more comprehensive than B. I voted CD

upvoted 2 times

**mk168898** 1 year, 7 months ago

accessible by transport layer means not by application layer means no need ALB

upvoted 2 times

**AbhiBK** 1 year, 9 months ago

Answer is A & D - By deploying EC2 instances in multiple Availability Zones (Option A), you ensure that your application remains available even if one Availability Zone experiences an outage. This setup provides redundancy and fault tolerance.

upvoted 1 times

**muhammadahmer36** 1 year, 10 months ago

Selected Answer: BD

as it mention "The application is accessible by using the transport layer" which is TCP, so no more information or reason to use ALB as well, so I will go for B+D

upvoted 3 times

**KennethNg923** 1 year, 12 months ago

Selected Answer: BD

as it mention "The application is accessible by using the transport layer" which is TCP, so no more information or reason to use ALB as well, so I will go for B+D

upvoted 3 times

**sandordini** 2 years, 1 month ago

**Selected Answer: BD**

No word about the HTTP/application layer, only OSI 4 - TCP > B, an NLB should be enough
D: for Autoscaling.

upvoted 4 times

**Tanidanindo** 2 years, 2 months ago

**Selected Answer: BD**

transport layer means just NLB.

upvoted 3 times

**Awsbeginner87** 2 years, 2 months ago

**Selected Answer: BD**

B- since network layer operates at layer 4 i.e transport layer
D- for hHA

upvoted 2 times

**Awsbeginner87** 2 years, 2 months ago

Edited-D option for HA

upvoted 2 times

**xBUGx** 2 years, 2 months ago

**Selected Answer: BD**

question says the application is running on Transport Layer. i dont think there is need for ALB

upvoted 2 times

Question #877 *Topic 1*

A company uses Amazon S3 to host its static website. The company wants to add a contact form to the webpage. The contact form will have dynamic server-side components for users to input their name, email address, phone number, and user message.

The company expects fewer than 100 site visits each month. The contact form must notify the company by email when a customer fills out the form.

Which solution will meet these requirements MOST cost-effectively?

A. Host the dynamic contact form in Amazon Elastic Container Service (Amazon ECS). Set up Amazon Simple Email Service (Amazon SES) to connect to a third-party email provider.

B. Create an Amazon API Gateway endpoint that returns the contact form from an AWS Lambda function. Configure another Lambda function on the API Gateway to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic. **Most Voted**

C. Host the website by using AWS Amplify Hosting for static content and dynamic content. Use server-side scripting to build the contact form. Configure Amazon Simple Queue Service (Amazon SQS) to deliver the message to the company.

D. Migrate the website from Amazon S3 to Amazon EC2 instances that run Windows Server. Use Internet Information Services (IIS) for Windows Server to host the webpage. Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail.

**Correct Answer:** B

*Community vote distribution*

B (100%)

## Comments

**mk168898** 1 year, 7 months ago

i see need to notify => SNS

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: B**

as it said "The company expects fewer than 100 site visits each month" and Lambda charge per each time calling it, so i will go for B

upvoted 3 times

**EdricHoang** 1 year, 12 months ago

**Selected Answer: B**

Limitation is 100 submission a month, not frequent -> Lambda

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: B**

AnswerB

API gateway + Lambda seems to be the best option expecially when SNS is in use which is specially created to push messaged to the subscribers (company appropraiate team in this situation)

upvoted 3 times

**Hkayne** 2 years, 1 month ago

**Selected Answer: B**

B is the right answer

upvoted 2 times

Question #878 *Topic 1*

A company creates dedicated AWS accounts in AWS Organizations for its business units. Recently, an important notification was sent to the root user email address of a business unit account instead of the assigned account owner. The company wants to ensure that all future notifications can be sent to different employees based on the notification categories of billing, operations, or security.

Which solution will meet these requirements MOST securely?

A. Configure each AWS account to use a single email address that the company manages. Ensure that all account owners can access the email account to receive notifications. Configure alternate contacts for each AWS account with corresponding distribution lists for the billing team, the security team, and the operations team for each business unit.

B. Configure each AWS account to use a different email distribution list for each business unit that the company manages. Configure each distribution list with administrator email addresses that can respond to alerts. Configure alternate contacts for each AWS account with corresponding distribution lists for the billing team, the security team, and the operations team for each business unit.

C. Configure each AWS account root user email address to be the individual company managed email address of one person from each business unit. Configure alternate contacts for each AWS account with corresponding distribution lists for the billing team, the security team, and the operations team for each business unit.

D. Configure each AWS account root user to use email aliases that go to a centralized mailbox. Configure alternate contacts for each account by using a single business managed email distribution list each for the billing team, the security team, and the operations team. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (59%) A (28%) 10%

## Comments

**Linuslin** Highly Voted 2 years ago

**Selected Answer: D**

D is correct answer.
Configuring each AWS account's root user to use email aliases that go to a centralized mailbox ensures that sensitive notifications are not directly sent to individual email addresses. Instead, they are directed to a centralized mailbox, reducing the risk of unauthorized access to sensitive information. Additionally, configuring alternate contacts for each account using a single business-managed email distribution list for the billing team the security team, and the operations team ensures that notifications are appropriately routed to the respective teams based on the categories of billing, operations, or security. This approach centralizes control and reduces the likelihood of misconfiguration or unauthorized access to sensitive notifications.

upvoted 13 times

**Linuslin** 2 years ago

Option A doesn't provide the same level of security because it relies on a single email address managed by the company, which could be compromised, and it requires all account owners to access this email account, potentially leading to access control issues.

Option B presents a similar issue as Option A, as it relies on different email distribution lists but still requires administrator email addresses that may be vulnerable to compromise.

Option C also has security concerns because it associates the root user email address with individual employees, which may not be ideal for security and access control purposes.

So, Option D is the most secure solution for ensuring that future notifications can be sent to different employees based on the notification categories of billing, operations, or security.

upvoted 10 times

**d401c0d** Highly Voted 2 years, 1 month ago

**Selected Answer: A**
Question mentions the email was sent to a business unit account instead of an account owner. Thus, A mentions all account owners to have access t( email account.
upvoted 6 times

**46a8566** Most Recent 4 months, 3 weeks ago
**Selected Answer: C**
"Single business managed email distribution list each" = same team list for ALL accounts. Not business-unit specific. Less granular control.
upvoted 1 times

**In2718** 9 months, 2 weeks ago
**Selected Answer: A**
Chat AI -
A. Single managed email for root + alternate contacts per team (distribution lists).
✓ Root is controlled centrally (secure).
✓ Alternate contacts ensure correct routing.
✓ Scales across business units.
→ This is AWS best practice.
upvoted 1 times

**GOTJ** 1 year, 3 months ago
**Selected Answer: B**
Option B is correct, according to AWS Prescriptive Guidance (see link below):

When setting up primary and alternate contacts for your AWS account, use an email distribution list instead of an individual's email address. Using ar email distribution list makes sure that ownership and reachability are preserved as individuals in your organization come and go. Set alternate contacts for billing, operations, and security notifications, and use appropriate email distribution lists accordingly. AWS uses these email addresses tc contact you, so it is important you retain access to them.

https://docs.aws.amazon.com/prescriptive-guidance/latest/aws-startup-security-baseline/acct-01.html
upvoted 4 times

**bujuman** 1 year, 6 months ago
**Selected Answer: D**
The most secure way should use "Update the alternate contacts for any AWS account in your organization"
https://docs.aws.amazon.com/accounts/latest/reference/manage-acct-update-contact-alternate.html
upvoted 3 times

**MrIndia2022** 1 year, 7 months ago
question- option D uses a single distribution list for all business units vs Option B which uses separate, independent distribution lists for each business unit. This involves more segregation of emails and hence is more secure. I am confused between option B and d. Pl help
upvoted 1 times

**MatAlves** 1 year, 8 months ago
**Selected Answer: D**
Both B and D use Distribution Lists as part of the of the solution:
B - "configure each DL with administrator email addresses that can respond to alerts"
D - "configure alternate contacts for each account by using single business managed DL each" per team.

I confess the wording isn't amazing, but between B and D, the latter is the one that properly addresses the issue involving root user email address.
upvoted 3 times

**jadeconstancy** 1 year, 11 months ago
**Selected Answer: D**
correct answer is D
upvoted 4 times

**sheilawu** 2 years ago
**Selected Answer: A**
https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_accounts_update_primary_email.html
upvoted 4 times

Question #879 Topic 1

A company runs an ecommerce application on AWS. Amazon EC2 instances process purchases and store the purchase details in an Amazon Aurora PostgreSQL DB cluster.

Customers are experiencing application timeouts during times of peak usage. A solutions architect needs to rearchitect the application so that the application can scale to meet peak usage demands.

Which combination of actions will meet these requirements MOST cost-effectively? (Choose two.)

A. Configure an Auto Scaling group of new EC2 instances to retry the purchases until the processing is complete. Update the applications to connect to the DB cluster by using Amazon RDS Proxy. Most Voted

B. Configure the application to use an Amazon ElastiCache cluster in front of the Aurora PostgreSQL DB cluster.

C. Update the application to send the purchase requests to an Amazon Simple Queue Service (Amazon SQS) queue. Configure an Auto Scaling group of new EC2 instances that read from the SQS queue. Most Voted

D. Configure an AWS Lambda function to retry the ticket purchases until the processing is complete.

E. Configure an Amazon AP! Gateway REST API with a usage plan.

**Correct Answer:** AC

*Community vote distribution*

AC (47%) BC (44%) 9%

## Comments

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: AC**

A) uses RDS Proxy which is mainly for connection pooling and availability issues. Proxy is for too many connections(, not for performance: read replicas, caching)
B is caching which is designed for solving read-issues. (Here we have timeouts, and connection issues.)
C: SQS is good method for decoupling.

upvoted 9 times

**Abdullah_Cloud** Highly Voted 2 years, 1 month ago

**Selected Answer: BC**

i think it's BC

upvoted 6 times

**In2718** Most Recent 10 months, 1 week ago

**Selected Answer: AC**

The answers should be A and C for the reasons stated in the discussion

upvoted 1 times

**0dd48f2** 11 months ago

**Selected Answer: BC**

B - Caching to improve the read-issues
C- SQS decoupling to avoid retrying the purchase
Even though option A could help by using RDS Proxy for connection pooling, the first part of this option relates to retry the purchase which is not necessary because of SQS.

upvoted 1 times

**7dcef09** 1 year, 1 month ago

**Selected Answer: BC**

BC are better answers.
B improves perfomance which mitigates the application timeouts during times of peak usage.

C decouples the workload.
upvoted 1 times

**tch** 1 year, 2 months ago

**Selected Answer: BC**

application timeouts... not database connection timeout
upvoted 1 times

**nadeerm** 1 year, 3 months ago

**Selected Answer: BC**

Amazon ElastiCache (Option B): ElastiCache can cache frequently accessed data, reducing the load on the Aurora PostgreSQL database. This improve read performance and reduces latency, which helps handle peak traffic more efficiently.

Amazon SQS with Auto Scaling (Option C): By sending purchase requests to an SQS queue, the application can decouple the front-end from the back-end processing. An Auto Scaling group of EC2 instances can process messages from the queue, ensuring that the system scales dynamically based on demand. This approach ensures that purchase requests are processed asynchronously, reducing the likelihood of timeouts during peak usage.
upvoted 1 times

**ieffiong** 1 year, 3 months ago

**Selected Answer: BC**

Combining A&C just leads to redundant auto scaling groups especially when you have SQS configured already. The question talks about combination of actions.
upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: AC**

A - RDS Proxy improves scalability and database connection pooling.
B - Only helpful for read-heavy workloads. This question indicates "purchase transactions" (writes), so the main bottleneck is still there.
C - SQS allows for better handling of bursts in traffic by queueing requests (which is also true for so many questions before #879).
D - Cost-prohibitive at scale due to high invocation rates. And lambda is not suitable for long-running operations like database transactions.
E - Irrelevant.
upvoted 2 times

**GOTJ** 1 year, 3 months ago

Even though your RDS proxy approach is correct, by choosing "AC" you are configuring two ASGs: one to process the purchase (option C) and one for... Retry the purchase (?) (option A) This is not cost effective

In my opinion, designing architectures specifically to "retry" transactions (options AD), specially with timeout issues, is something that everyone should avoid, as it can easily degeneres into the "snowball" effect
upvoted 2 times

**EllenLiu** 1 year, 5 months ago

**Selected Answer: AB**

why not choose B? C is wired design.
upvoted 1 times

**Cpso** 1 year, 6 months ago

**Selected Answer: AB**

As a developer choose both A&C is wired? while we use SQS solution to buffer request before handle by EC2 fleet. why we need EC2 in front of SQS from A.? what of thier propose.
upvoted 2 times

**XXXXXlNN** 1 year, 7 months ago

BC.
Not idea why retry helps in this scenario besides it adds more complexity into the current design and also doesn't resolve the avalibility issue...
upvoted 4 times

**b3b5fdd** 1 year, 8 months ago

**Selected Answer: BC**

B and C!
upvoted 1 times

**MatAlves** 1 year, 8 months ago

**Selected Answer: BC**

A - simply pointless.

B- You're already using SQS (C), so why using ec2 to "retry the purchase"? They will stay in the queue until the purchase is processed. Otherwise, they will simply return to the queue.

C - This decouples the application from direct database calls, allowing the processing of purchase requests to scale independently and manage load more effectively.

upvoted 2 times

**pujithacg8** 1 year, 10 months ago

when we have SQS in option C why do you have to retry it again

I think the answer is B and C

upvoted 4 times

**EdricHoang** 1 year, 12 months ago

**Selected Answer: AC**

Combine SQS and auto-scaling EC2:
https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-using-sqs-queue.html

upvoted 3 times

Question #880 *Topic 1*

A company that uses AWS Organizations runs 150 applications across 30 different AWS accounts. The company used AWS Cost and Usage Report to create a new report in the management account. The report is delivered to an Amazon S3 bucket that is replicated to a bucket in the data collection account.

The company's senior leadership wants to view a custom dashboard that provides NAT gateway costs each day starting at the beginning of the current month.

Which solution will meet these requirements?

A. Share an Amazon QuickSight dashboard that includes the requested table visual. Configure QuickSight to use AWS DataSync to query the new report.

B. Share an Amazon QuickSight dashboard that includes the requested table visual. Configure QuickSight to use Amazon Athena to query the new report. **Most Voted**

C. Share an Amazon CloudWatch dashboard that includes the requested table visual. Configure CloudWatch to use AWS DataSync to query the new report.

D. Share an Amazon CloudWatch dashboard that includes the requested table visual. Configure CloudWatch to use Amazon Athena to query the new report.

**Correct Answer:** B

*Community vote distribution*

B (100%)



## Comments

**mk168898** 1 year, 7 months ago

dashboard => quicksight, S3 query => athena

upvoted 2 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: B**

QuickSight for dashboard and Athena for query each month so it is B

upvoted 3 times

**sandordini** 2 years, 1 month ago

**Selected Answer: B**

Senior Leadership, custom dashboard, visualization: Quicksight Dashboard
S3 query: Athena

upvoted 3 times

**d401c0d** 2 years, 1 month ago

**Selected Answer: B**

B. Share an Amazon QuickSight dashboard that includes the requested table visual. Configure QuickSight to use Amazon Athena to query the new report.

QuickSight works well with Athena and it can interact S3

upvoted 2 times

**Mikado211** 2 years, 2 months ago

**Selected Answer: B**

You definitely use Athena to request S3.
Both cloudwatch and quicksight can interact with S3.
Since we are taking about "The company's senior leadership" I'd tend to use quicksight for a better format.

upvoted 4 times

Question #881 Topic 1

A company is hosting a high-traffic static website on Amazon S3 with an Amazon CloudFront distribution that has a default TTL of 0 seconds. The company wants to implement caching to improve performance for the website. However, the company also wants to ensure that stale content is not served for more than a few minutes after a deployment.

Which combination of caching methods should a solutions architect implement to meet these requirements? (Choose two.)

A. Set the CloudFront default TTL to 2 minutes. Most Voted

B. Set a default TTL of 2 minutes on the S3 bucket.

C. Add a Cache-Control private directive to the objects in Amazon S3.

D. Create an AWS Lambda@Edge function to add an Expires header to HTTP responses. Configure the function to run on viewer response.

E. Add a Cache-Control max-age directive of 24 hours to the objects in Amazon S3. On deployment, create a CloudFront invalidation to clear any changed files from edge caches. Most Voted

**Correct Answer:** AE

*Community vote distribution*

AE (54%) AC (41%) 2%

## Comments

**BBR01** Highly Voted 2 years, 1 month ago

Selected Answer: AE

AE.
By default, each file automatically expires after 24 hours, but you can change the default behavior in two ways:

1. To change the cache duration for all files that match the same path pattern, you can change the CloudFront settings for Minimum TTL, Maximum TTL, and Default TTL for a cache behavior.

2. To change the cache duration for an individual file, you can configure your origin to add a Cache-Control header with the max-age or s-maxage directive, or an Expires header to the file.
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Expiration.html#expiration-individual-objects

upvoted 9 times

**MatAlves** Highly Voted 1 year, 8 months ago

Selected Answer: AC

You simply can't have A and E in the same approach:

"Default TTL applies only when your origin does not add HTTP headers such as Cache-Control max-age, Cache-Control s-maxage, or Expires to objects."

C - Cache-Control private directive specifies that the response is intended for a single user and should not be cached by shared caches - it can still b cached, but only on a client device.

This combination of steps would provide the best solution for the case.

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/distribution-web-values-specify.html#DownloadDistValuesDefaultTTL

upvoted 7 times

**Sergantus** 1 year, 7 months ago

Setting Cache-Control: private would prevent CloudFront from caching the content entirely, which is not the goal outlined, as it wants to use caching. After some time with updates, the caching performance will degrade for the entire solution as more and more objects get that directive.

upvoted 2 times

**In2718** Most Recent 9 months, 2 weeks ago

**Selected Answer: AE**

A. Set the CloudFront default TTL to 2 minutes.
Enables edge caching (improves performance) for any objects that don't send explicit cache headers.

E. Add Cache-Control: max-age=86400 (24h) on S3 objects and, on deployment, run a CloudFront invalidation for changed files.
Gives strong caching for performance, while invalidations ensure changed content stops being served within a few minutes after deploy (the time it takes invalidations to propagate).

upvoted 1 times

**tch** 1 year, 2 months ago

**Selected Answer: AE**

Controlling how long Amazon S3 content is cached by Amazon CloudFront
https://docs.aws.amazon.com/whitepapers/latest/build-static-websites-aws/controlling-how-long-amazon-s3-content-is-cached-by-amazon-cloudfront.html

upvoted 1 times

**tch** 1 year, 2 months ago

Set maximum TTL value
Specify cache-control headers

upvoted 1 times

**robotgeek** 1 year, 4 months ago

**Selected Answer: A**

This question has gaps:
B) Can not be done in relation to S3 catching, this is a CloudFront specific configuration
C) Question does not make any comment in the sense of "private" objects... any asumption in that sense is your personal oppinion, not the question stating it
D) Collisions with A, why would you do something like that programatically?
E) Can not be because why would you put 24h when it is not requested, also why would you invalidate a deployment that is already TTL=0, everything is up to date in the caches with that config

upvoted 1 times

**Salilgen** 1 year, 5 months ago

**Selected Answer: AC**

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Expiration.html#ExpirationDownloadDist

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: AE**

A - Setting the default TTL to 2 minutes allows CloudFront to cache content for a short period.
B - TTL is not a concept applied directly to S3 buckets. TTL and caching behaviors are managed through Cache-Control headers and CloudFront settings.
C - The *private directive* lets caching occur locally only, not in shared caches like CloudFront.
D - Ensures content is served fresh by tightly controlling cache expiration, not even "a few minutes".
E - Using a long Cache-Control max-age directive improves performance, while targeted CloudFront invalidations ensure that updated content is served immediately after deployment. This is common practice.

upvoted 3 times

**Cpso** 1 year, 6 months ago

**Selected Answer: AE**

browser will always get cache-control value set by S3 not cloudfront.
cloudfront will use parameter ttl override the cache-control header for decision time to cache on cloudfront.

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Expiration.html?utm_source=chatgpt.com#ExpirationDownloadDist

upvoted 2 times

**Cpso** 1 year, 6 months ago

**Selected Answer: AE**

A - make browser to cache 2 minutes. after 2 minute browser will ask cloudfront

E: cloudfront would not ask S3 for 24hour . invalidation will clearing this cacheing.

so A+E is most effective.

upvoted 2 times

**JA2018** 1 year, 6 months ago

**Selected Answer: AD**

#A - Set the CloudFront default TTL to 2 minutes: This directly controls how long content is cached at the CloudFront edge locations, allowing for a short caching window to meet the requirement of not serving stale content for more than a few minutes.

#D - Create a Lambda@Edge function to add an Expires header: By using a Lambda@Edge function triggered on the "viewer response", you can dynamically set an Expires header with a precise expiration time on each request, ensuring fine-grained control over caching behavior.

upvoted 1 times

**JA2018** 1 year, 6 months ago

#B - Set a default TTL of 2 minutes on the S3 bucket: This would not be as effective as setting the TTL at the CloudFront level as the caching would happen at the origin server (S3) instead of the edge locations, potentially impacting performance.

# C - Add a Cache-Control private directive: This directive would prevent browser caching, which is not the desired outcome in this scenario.

# E - Add a Cache-Control max-age directive of 24 hours with CloudFront invalidation: Setting a large max-age on objects in S3 and invalidating on deployment would not provide the necessary granular control for keeping content fresh within a few minutes.

upvoted 1 times

**1ba9aa0** 1 year, 10 months ago

**Selected Answer: AC**

A-C,

Because A-E is not possible following this link: https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/distribution-web-values-specify.html#DownloadDistValuesDefaultTTL
"Default TTL applies only when your origin does not add HTTP headers such as Cache-Control max-age, Cache-Control s-maxage, or Expires to objects."

upvoted 3 times

**EdricHoang** 1 year, 11 months ago

**Selected Answer: AC**

If the content still keep client's cache in 24h, its wrong (answer E)

upvoted 3 times

**ug56c** 1 year, 12 months ago

**Selected Answer: AE**

If your minimum TTL is greater than 0, CloudFront uses the cache policy’s minimum TTL, even if the Cache-Control: no-cache, no-store, and/or private directives are present in the origin headers.
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Expiration.html

upvoted 5 times

**Nm55569** 2 years ago

**Selected Answer: AE**

"However, the company also wants to ensure that stale content is not served for more than a few minutes after a deployment."
After a deployment

upvoted 4 times

**Sergantus** 1 year, 7 months ago

Exactly, it was not outlined that the user shouldn't see the stale content, only that it's not served.

upvoted 1 times

**Scheldon** 2 years ago

**Selected Answer: AC**

Answer (AC)

Per table on URL
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Expiration.html#expiration-individual-objects
answer E is incorrect because if we will change cache-control max-age to 24h it will means that customer browser will cache web for 24h and customer want to be sure that it will be not longer then few min.
Expires header (answer D) from my understanding can be used only on full folder of web not as lambda function which will reply to customer reqeusts.
We are setting Default TTL for CloudFront (answer A) not on S3 (answer B) and it will say CloudFront to cache web for 2min.

upvoted 4 times

**Scheldon** 2 years ago

Adding Cache-control private (answerC) will work per customer wish but only if we will add them to the objects which are changed very often or if we will set minimum TTL.
In the 1 situation User Browser will not store files which we designate to be often changed and mentioned files will be downloaded every time from origin.
In the 2 situation, Cloud front will cache web files for min TTL time but customer browser will not store them.
Taking all that in to account I would go with AC

upvoted 3 times

**Linuslin** 2 years ago

**Selected Answer: AE**

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Expiration.html#expiration-individual-objects

https://stackoverflow.com/questions/43343759/confused-with-minimum-maximum-and-default-ttl-in-cloudfront

upvoted 2 times

**02ffe1c** 2 years, 1 month ago

**Selected Answer: DE**

Since it don't want to cache more than a minute, A cannot be an answer

upvoted 1 times

**mk168898** 1 year, 7 months ago

no where in the question did it say 1 minute. You mean more than a few minutes?

upvoted 1 times

Question #882 Topic 1

A company runs its application by using Amazon EC2 instances and AWS Lambda functions. The EC2 instances run in private subnets of a VPC. The Lambda functions need direct network access to the EC2 instances for the application to work.

The application will run for 1 year. The number of Lambda functions that the application uses will increase during the 1-year period. The company must minimize costs on all application resources.

Which solution will meet these requirements?

A. Purchase an EC2 Instance Savings Plan. Connect the Lambda functions to the private subnets that contain the EC2 instances.

B. Purchase an EC2 Instance Savings Plan. Connect the Lambda functions to new public subnets in the same VPC where the EC2 instances run.

C. Purchase a Compute Savings Plan. Connect the Lambda functions to the private subnets that contain the EC2 instances. **Most Voted**

D. Purchase a Compute Savings Plan. Keep the Lambda functions in the Lambda service VPC.

**Correct Answer:** C

*Community vote distribution*

C (80%) 13% 7%

## Comments

**Guru4Cloud** Highly Voted 2 years, 2 months ago

**Selected Answer: C**

Compute Savings Plan: This plan offers significant discounts on Lambda functions compared to on-demand pricing. Since the application will run for a year, a sustained use discount like Compute Savings Plan is ideal.
Private Subnets: Lambda functions in private subnets can directly access EC2 instances within the VPC without needing internet access, reducing security risks and potential egress costs.

upvoted 7 times

**0c21057** Most Recent 1 year, 1 month ago

**Selected Answer: C**

Compute Savings Plans provide the most flexibility and help to reduce your costs by up to 66%. These plans automatically apply to EC2 instance usage regardless of instance family, size, AZ, Region, OS or tenancy, and also apply to Fargate or Lambda usage. For example, with Compute Savings Plans, you can change from C4 to M5 instances, shift a workload from EU (Ireland) to EU (London), or move a workload from EC2 to Fargate or Lambda at any time and automatically continue to pay the Savings Plans price.

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: C**

Why Compute Savings Plan: Covers both EC2 and Lambda, providing flexibility as Lambda usage increases.
Why Private Subnets: Ensures network access between Lambda and EC2 instances.

!!
An EC2 Instance Savings Plan doesn't cover Lambda. Also, Lambda functions in public subnets cannot directly communicate with EC2 instances in private subnets.

upvoted 2 times

**mk168898** 1 year, 7 months ago

A and B is not because it is talkign about EC2, but the question is asking for Lambda

upvoted 2 times

**jacinml** 1 year, 8 months ago

**Selected Answer: C**

Compute savings include lamdba and EC2, Ec2 savings only EC2 instances.
https://aws.amazon.com/savingsplans/compute-pricing/

upvoted 2 times

**Lin878** 1 year, 11 months ago

I confuse this Question. Instance saving plan is cheaper than compute saving plan.
https://aws.amazon.com/savingsplans/compute-pricing/

upvoted 1 times

**navneet_sh** 1 year, 11 months ago

But composite saving plan discount Automatically applies to Lambda.

upvoted 2 times

**navneet_sh** 1 year, 11 months ago

Sorry I mean Compute Savings Plan.

upvoted 2 times

**sheilawu** 2 years ago

**Selected Answer: A**

In this question has point out "access EC2 instances" within VPC,=> Lambda VPC to an ENI (Elastic network interface) in your account VPC.=>No charge.
Therefore I stick with A, Not D.

upvoted 2 times

**JohnYu** 1 year, 7 months ago

An EC2 Instance Savings Plan is limited to savings on EC2 instances only. It does not provide cost savings for Lambda functions. Therefore, it is not the best choice given that the number of Lambda functions is expected to increase.

upvoted 3 times

**sheilawu** 2 years ago

I am sorry I mean C

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: D**

AnswerD

https://docs.aws.amazon.com/lambda/latest/dg/foundation-networking.html

upvoted 1 times

Question #883 *Topic 1*

A company has deployed a multi-account strategy on AWS by using AWS Control Tower. The company has provided individual AWS accounts to each of its developers. The company wants to implement controls to limit AWS resource costs that the developers incur.

Which solution will meet these requirements with the LEAST operational overhead?

A. Instruct each developer to tag all their resources with a tag that has a key of CostCenter and a value of the developer's name. Use the required-tags AWS Config managed rule to check for the tag. Create an AWS Lambda function to terminate resources that do not have the tag. Configure AWS Cost Explorer to send a daily report to each developer to monitor their spending.

B. Use AWS Budgets to establish budgets for each developer account. Set up budget alerts for actual and forecast values to notify developers when they exceed or expect to exceed their assigned budget. Use AWS Budgets actions to apply a DenyAll policy to the developer's IAM role to prevent additional resources from being launched when the assigned budget is reached. **Most Voted**

C. Use AWS Cost Explorer to monitor and report on costs for each developer account. Configure Cost Explorer to send a daily report to each developer to monitor their spending. Use AWS Cost Anomaly Detection to detect anomalous spending and provide alerts.

D. Use AWS Service Catalog to allow developers to launch resources within a limited cost range. Create AWS Lambda functions in each AWS account to stop running resources at the end of each work day. Configure the Lambda functions to resume the resources at the start of each work day.

**Correct Answer:** B

*Community vote distribution*

B (72%) C (28%)

## Comments

**Scheldon** Highly Voted 2 years ago

**Selected Answer: B**

AnswerB

https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-controls.html
Taking into consideration that AWS Budgets is allowing to will inform you that you exceeded budged and execute actions like for example IAM actions to prevent running new resources in cloud, I think this option is a good and resonable move. In case of need budged can be always increasec and "chains" disabled.

upvoted 5 times

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: B**

My first instinct says B, but Im concerned about the central management abilities of AWS Budgets. It seems that even though it is not planned to be used primarily to control other accounts its still possible:
"You can use actions to define an explicit response that you want to take when a budget exceeds its action threshold. You can trigger these alerts on actual or forecasted cost and usage budgets.
1. The management account sets the budget and threshold for the member account using budget filters.
2. When the budget threshold is breached, a budget action applies a restrictive SCP on the OU.

So hopefully B :D

upvoted 5 times

**In2718** Most Recent 9 months, 2 weeks ago

**Selected Answer: B**

AWS Budgets lets you set a per-account spend cap with alerts and Budgets actions that automatically apply a deny policy (e.g., attach a restrictive IAM policy/SCP to the developer's role) once the budget is hit. That both enforces a limit and requires minimal build/ops work.

upvoted 1 times

**vmia159** 12 months ago

Selected Answer: C

No answers is correct but I need to answer to leave comment. This question properly does not count towards the final mark.

Since what the developer have is AWS Account from the AWS Organisation, deny developer's IAM role does not make any sense as they can use the root account and all the service is still running and cost money. What you actually need is SCP and stackset to deny their action and terminate all the service they create.

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

Selected Answer: B

A - Tags help monitor costs, and AWS Config can enforce tagging rules, but manually tagging is TOO MUCH WORK.
B - Preventive (aka not reactive) cost control, good. Too provocative? Yeah, maybe. Developers might experience disruption if they hit their budget limits unexpectedly, and then the cost a little over the budget might not seem like a problem anymore... BUT this is actually a preventive approach with "the LEAST operational overhead".
C - "Daily reports" might be too late.
D - It can limit resource usage during non-working hours, but overall, you gotta deal with high operational overhead.

upvoted 2 times

**LeonSauveterre** 1 year, 5 months ago

The gist is, when you reach the limit, you must be stopped. It's the only way to prevent unexpected expenses. Reports, monitors, analyses, logs, charts, they DON'T work.

upvoted 1 times

**MatAlves** 1 year, 8 months ago

Selected Answer: B

As beautifully explained in this article:

https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-controls.html

upvoted 2 times

**NSA_Poker** 1 year, 11 months ago

Selected Answer: B

(A) is eliminated. 'send a daily report to each developer' can be ignored.
(C) is eliminated. 'detect anomalous spending' won't stop the spending.
(D) is eliminated. 'stop running resources at the end of each work day' won't stop developers from mining bitcoin ($$$) the next day.
(B) is correct. 'actions to apply a DenyAll policy..' is the only solution that will 'implement controls to limit AWS resource costs that the developers incur.'

upvoted 3 times

**MatAlves** 1 year, 8 months ago

I'd definitely be mining bitcoin the next day...

upvoted 3 times

**f07ed8f** 2 years ago

Selected Answer: C

Seem AWS Budgets does not have DenyAll function but only

Apply a custom Deny IAM policy that restricts the ability for a user, group, or role to provision additional Amazon EC2 resources

upvoted 2 times

**BBR01** 2 years, 1 month ago

Selected Answer: C

B and D are too aggressive.
A - "Instruct each developer", nope, too much operational work.

upvoted 4 times

Question #884 Topic 1

A solutions architect is designing a three-tier web application. The architecture consists of an internet-facing Application Load Balancer (ALB) and a web tier that is hosted on Amazon EC2 instances in private subnets. The application tier with the business logic runs on EC2 instances in private subnets. The database tier consists of Microsoft SQL Server that runs on EC2 instances in private subnets. Security is a high priority for the company.

Which combination of security group configurations should the solutions architect use? (Choose three.)

A. Configure the security group for the web tier to allow inbound HTTPS traffic from the security group for the ALB. Most Voted

B. Configure the security group for the web tier to allow outbound HTTPS traffic to 0.0.0.0/0.

C. Configure the security group for the database tier to allow inbound Microsoft SQL Server traffic from the security group for the application tier. Most Voted

D. Configure the security group for the database tier to allow outbound HTTPS traffic and Microsoft SQL Server traffic to the security group for the web tier.

E. Configure the security group for the application tier to allow inbound HTTPS traffic from the security group for the web tier. Most Voted

F. Configure the security group for the application tier to allow outbound HTTPS traffic and Microsoft SQL Server traffic to the security group for the web tier.

**Correct Answer:** ACE

*Community vote distribution*

ACE (100%)

## Comments

EdricHoang Highly Voted 1 year, 12 months ago

Selected Answer: ACE

Security group is stateful, just need allow Inbound.

upvoted 6 times

KennethNg923 1 year, 12 months ago

Agree since security group is stateful so allow inbound is enough

upvoted 2 times

sk1974 Most Recent 1 year, 3 months ago

Selected Answer: ACE

Since SG's are stateful , allow inbound only. Ignore all outbound roles

upvoted 1 times

Scheldon 2 years ago

Selected Answer: ACE

AnswerACE:

Security Group is protecting instances, it's statefull. by defoult is allowing for outgoing traffic but not incomming.
hence we need to allow for inboud traffic. path looks like below
ALB >>HTTPS>> WEB tier >>HTTPS>> Application >>SQL traffic>> SQL DB
hence we need allow for
incoming https traffic on web tier
then
incomming http on app tier
and on the end for
incomming sql traffic on DB tier

upvoted 4 times

**sandordini** 2 years, 1 month ago

**Selected Answer: ACE**

ALB >>HTTPS>> WEB tier >>HTTPS>> Application >>SQL traffic>> SQL DB

upvoted 4 times

Question #885 *Topic 1*

A company has released a new version of its production application. The company's workload uses Amazon EC2, AWS Lambda, AWS Fargate, and Amazon SageMaker.

The company wants to cost optimize the workload now that usage is at a steady state. The company wants to cover the most services with the fewest savings plans.

Which combination of savings plans will meet these requirements? (Choose two.)

A. Purchase an EC2 Instance Savings Plan for Amazon EC2 and SageMaker.

B. Purchase a Compute Savings Plan for Amazon EC2, Lambda, and SageMaker.

C. Purchase a SageMaker Savings Plan. **Most Voted**

D. Purchase a Compute Savings Plan for Lambda, Fargate, and Amazon EC2. **Most Voted**

E. Purchase an EC2 Instance Savings Plan for Amazon EC2 and Fargate.

**Correct Answer:** CD

*Community vote distribution*

CD (92%) 8%



## Comments

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: CD**

It's pretty obvious, although it's called: Machine Learning Savings Plans for Amazon SageMaker (C)
For the compute workloads we need a compute savings plan, that covers all the 3 compute options we use here (EC2, Lambda and Fargate) (D)

upvoted 8 times

**andreea10** Most Recent 10 months ago

**Selected Answer: BC**

While this covers Lambda, Fargate, and EC2, it still doesn't cover SageMaker, if you choose D, you'd still need a SageMaker Savings Plan to cover SageMaker usage

upvoted 1 times

**mk168898** 1 year, 7 months ago

just pick the 2 options that doesn't overlap

upvoted 4 times

**mattyu** 2 years ago

**Selected Answer: CD**

no doubt

upvoted 3 times

**Scheldon** 2 years ago

Answer CD

https://aws.amazon.com/savingsplans/ml-pricing/
https://aws.amazon.com/savingsplans/compute-pricing/

upvoted 3 times

Question #886 *Topic 1*

A company uses a Microsoft SQL Server database. The company's applications are connected to the database. The company wants to migrate to an Amazon Aurora PostgreSQL database with minimal changes to the application code.

Which combination of steps will meet these requirements? (Choose two.)

A. Use the AWS Schema Conversion Tool (AWS SCT) to rewrite the SQL queries in the applications.

B. Enable Babelfish on Aurora PostgreSQL to run the SQL queries from the applications. **Most Voted**

C. Migrate the database schema and data by using the AWS Schema Conversion Tool (AWS SCT) and AWS Database Migration Service (AWS DMS). **Most Voted**

D. Use Amazon RDS Proxy to connect the applications to Aurora PostgreSQL.

E. Use AWS Database Migration Service (AWS DMS) to rewrite the SQL queries in the applications.

**Correct Answer:** BC

*Community vote distribution*

BC (100%)



## Comments

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: BC**

B: Babelfish for Aurora PostgreSQL is a new capability for Amazon Aurora PostgreSQL-Compatible Edition that enables Aurora to understand commands from applications written for Microsoft SQL Server.
C: Is just obvious: Use Data Migration Tool for the migration, Schema Conversion tool for the Schema conversion.

upvoted 7 times

**pranavsharma1604** 2 years ago

https://aws.amazon.com/rds/aurora/babelfish/

upvoted 3 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

**Selected Answer: BC**

A - AWS SCT does not directly rewrite SQL queries in the application code. This would require manual updates.
B - Babelfish allows Aurora PostgreSQL to understand SQL Server's native T-SQL language and protocol.
C - Convert the schema & migrate the data effectively, good.
D - Amazon RDS Proxy improves database scalability and availability (if the question mentions something like having trouble conneting during peek hours or whatnot then RDS Proxy is likely the answer) but does not address the compatibility issues between SQL Server and Aurora PostgreSQL.
E - This is not enough. Refer to option C.

upvoted 3 times

**KennethNg923** 1 year, 12 months ago

**Selected Answer: BC**

DMS + SCT is correct, but " rewrite the SQL queries in the applications." is wrong so A + E are out.
Then only left B + C -> DMS + SCT + Babekfish (for SQL Server)

upvoted 2 times

**Scheldon** 2 years ago

**Selected Answer: BC**

AnswerBC
DMS will allow for DATABASE migration and use AWS Schema Conversion Tool (AWS SCT) to create some or all of the target tables, indexes, views, triggers, and so on.
https://docs.aws.amazon.com/dms/latest/userguide/Welcome.html
To minimalize amount of code which need to me changes we need to use babelfish
https://aws.amazon.com/rds/aurora/babelfish/

upvoted 2 times

**pranavsharma1604** 2 years ago

**Selected Answer: BC**

https://aws.amazon.com/rds/aurora/babelfish/

upvoted 4 times

Question #887 Topic 1

A company plans to rehost an application to Amazon EC2 instances that use Amazon Elastic Block Store (Amazon EBS) as the attached storage.

A solutions architect must design a solution to ensure that all newly created Amazon EBS volumes are encrypted by default. The solution must also prevent the creation of unencrypted EBS volumes.

Which solution will meet these requirements?

A. Configure the EC2 account attributes to always encrypt new EBS volumes. **Most Voted**

B. Use AWS Config. Configure the encrypted-volumes identifier. Apply the default AWS Key Management Service (AWS KMS) key.

C. Configure AWS Systems Manager to create encrypted copies of the EBS volumes. Reconfigure the EC2 instances to use the encrypted volumes.

D. Create a customer managed key in AWS Key Management Service (AWS KMS). Configure AWS Migration Hub to use the key when the company migrates workloads.

**Correct Answer:** A

*Community vote distribution*

A (61%) B (39%)

## Comments

**Scheldon** Highly Voted 2 years ago

**Selected Answer: A**

AnswerA

The task is to force automatic encryption for every new EBS volume and prevent possibility of creation any unencrypted volume hence:

https://docs.aws.amazon.com/ebs/latest/userguide/work-with-ebs-encr.html#ebs-encryption_key_mgmt

To enable encryption by default for a Region

Open the Amazon EC2 console at https://console.aws.amazon.com/ec2/.

From the navigation bar, select the Region.

From the navigation pane, select EC2 Dashboard.

In the upper-right corner of the page, choose Account Attributes, Data protection and security.

Choose Manage.

Select Enable. You keep the AWS managed key with the alias alias/aws/ebs created on your behalf as the default encryption key, or choose a symmetric customer managed encryption key.

Choose Update EBS encryption.

upvoted 10 times

**bmmbmm** Most Recent 6 months ago

**Selected Answer: A**

A can be set for an account level, enabling "Encrypt new EBS volumes by default" don't have to configure every each instance. Option B : Config is only for monitoring and detecting, cannot prevent the creation. Remediation might work, but still allows creation in the future.

upvoted 1 times

**andreea10** 10 months ago

**Selected Answer: A**

AWS Config can check if a volume is unencrypted after it's created and mark it as noncompliant, but it does not prevent the creation of unencrypted volumes

upvoted 1 times

**5ac21bc** 1 year ago

**Selected Answer: B**

"EBS volumes are encrypted by default"....I would choose B over A

upvoted 1 times

**FlyingHawk** 1 year, 4 months ago

**Selected Answer: A**

https://docs.aws.amazon.com/ebs/latest/userguide/encryption-by-default.html
You can configure your AWS account to enforce the encryption of the new EBS volumes and snapshot copies that you create. For example, Amazon EBS encrypts the EBS volumes created when you launch an instance and the snapshots that you copy from an unencrypted snapshot. For examples o transitioning from unencrypted to encrypted EBS resources, see https://docs.aws.amazon.com/ebs/latest/userguide/ebs-encryption.html#encrypt-unencrypted: To enable encryption by default for a Region
%aws ec2 enable-ebs-encryption-by-default --region region

In the upper-right corner of the page, choose Account Attributes, Data protection and security.

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: B**

A - This automatically encrypts all newly created volumes, but can't prevent unencrypted volumes.
B - AWS Config can enforce and automate compliance by using a rule to check whether all EBS volumes are encrypted.
C - Creating encrypted copies of volumes after they are created? Why bother to do this
D - What does Migration Hub have to do with this... I don't understand.

upvoted 2 times

**Xandero** 1 year, 6 months ago

**Selected Answer: A**

https://docs.aws.amazon.com/ebs/latest/userguide/encryption-by-default.html

upvoted 1 times

**EdricHoang** 1 year, 11 months ago

**Selected Answer: B**

"The solution must also prevent the creation of unencrypted EBS volumes."
For prevention future actions, I go for AWS config. You can setup Encryption in EC2, but Its manual process, what happen if you add one or more EC2?

upvoted 3 times

**Scheldon** 2 years ago

AnswerA

https://docs.aws.amazon.com/ebs/latest/userguide/work-with-ebs-encr.html#ebs-encryption_key_mgmt
To enable encryption by default for a Region
Open the Amazon EC2 console at https://console.aws.amazon.com/ec2/.

From the navigation bar, select the Region.

From the navigation pane, select EC2 Dashboard.

In the upper-right corner of the page, choose Account Attributes, Data protection and security.

Choose Manage.

Select Enable. You keep the AWS managed key with the alias alias/aws/ebs created on your behalf as the default encryption key, or choose a symmetric customer managed encryption key.

Choose Update EBS encryption.

upvoted 1 times

**0bdf3af** 2 years ago

A. https://repost.aws/knowledge-center/ebs-automatic-encryption

upvoted 2 times

**Isomas** 2 years ago

**Selected Answer: B**

As it needs to prevent creation of Unencrypted EBS volume

upvoted 3 times

**viejito** 2 years, 1 month ago

B es correcto , AWS Config para identificar automáticamente los volúmenes de EBS no cifrados y aplicar una acción correctiva.A,C,D : incorrectas , no cumplen con el cifrado automático

upvoted 3 times

**JA2018** 1 year, 6 months ago

From Google Translate

B is OK, AWS Config to automatically identify unencrypted EBS volumes and apply corrective action. A,C,D: Incorrect, do not comply with

upvoted 1 times

Question #888 *Topic 1*

An ecommerce company wants to collect user clickstream data from the company's website for real-time analysis. The website experiences fluctuating traffic patterns throughout the day. The company needs a scalable solution that can adapt to varying levels of traffic.

Which solution will meet these requirements?

A. Use a data stream in Amazon Kinesis Data Streams in on-demand mode to capture the clickstream data. Use AWS Lambda to process the data in real time. **Most Voted**

B. Use Amazon Kinesis Data Firehose to capture the clickstream data. Use AWS Glue to process the data in real time.

C. Use Amazon Kinesis Video Streams to capture the clickstream data. Use AWS Glue to process the data in real time.

D. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to capture the clickstream data. Use AWS Lambda to process the data in real time.

**Correct Answer:** A

*Community vote distribution*

A (93%) 7%

## Comments

**bora4motion** 1 year, 1 month ago

**Selected Answer: A**

realtime - data streams, near real time - firehose. a is the correct answer.

upvoted 2 times

**LeonSauveterre** 1 year, 5 months ago

**Selected Answer: A**

A - Kinesis Data Streams in on-demand mode automatically scales based on traffic, making it an ideal choice for fluctuating traffic patterns.
B - Kinesis Data Firehose is used for NEAR real-time data delivery. Also, AWS Glue is better suited for ETL and batch processing rather than real-time analysis.
C - Kinesis Video Streams is designed for video or audio streaming data, not clickstream data.
D - You can process data, but can't capture raw data first.

Also, AWS Lambda provides real-time data processing without the need for managing infrastructure.

upvoted 1 times

**mk168898** 1 year, 7 months ago

click stream => kinesis data streams

upvoted 2 times

**MatAlves** 1 year, 8 months ago

B - C are out = Glue doesn't support real-time data processing.
D - Why would you use Kinesis data ANALYTICS to ingest clickstream data instead of Amazon Kinesis DATA STREAM?

upvoted 2 times

**Salilgen** 1 year, 5 months ago

https://docs.aws.amazon.com/glue/latest/dg/streaming-chapter.html

upvoted 1 times

**tch** 1 year, 3 months ago

this is AWS Glue Streaming... not AWS Glue

upvoted 1 times

**EdricHoang** 1 year, 12 months ago

**Selected Answer: A**

This one is very tricky, need to read the context carefully:
"The company needs a scalable solution that can adapt to varying levels of traffic" -> Both Firehouse and Stream are scalable. But, Firehose is automatic where Stream is not. However, the question does NOT say it should be automatic and Glue is not support real-time analysis.
Thats why go for A.
B is very close to.

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: A**

AnswerA

Apache Flink (previously known as Amazon Kinesis Data Analytics) seems to not allowing sent data directly to Lambda...
Glue is allowing to integrate data from couple of sources in to one.
Hence I think A is correct answer
https://docs.aws.amazon.com/lambda/latest/dg/with-kinesis.html
https://aws.amazon.com/kinesis/data-streams/features/?nc=sn&loc=2

upvoted 2 times

**f07ed8f** 2 years ago

**Selected Answer: A**

Seem AWS Glue does not support process data in real time. I vote for A

upvoted 2 times

**f07ed8f** 2 years ago

**Selected Answer: B**

Both Kinesis Data Streams and Firehose are scalable but Firehose offers automated scaling. I vote fore B

upvoted 1 times

**sandordini** 2 years, 1 month ago

**Selected Answer: A**

I think Apache Flink (previously known as Amazon Kinesis Data Analytics) would also be fine, but as here it wants to combine it with Lambda, I would rather opt for Kinesis Data Streams + Lambda, so A, because of the figure on this page:
https://aws.amazon.com/kinesis/

upvoted 4 times

Question #889 *Topic 1*

A global company runs its workloads on AWS. The company's application uses Amazon S3 buckets across AWS Regions for sensitive data storage and analysis. The company stores millions of objects in multiple S3 buckets daily. The company wants to identify all S3 buckets that are not versioning-enabled.

Which solution will meet these requirements?

A. Set up an AWS CloudTrail event that has a rule to identify all S3 buckets that are not versioning-enabled across Regions.

B. Use Amazon S3 Storage Lens to identify all S3 buckets that are not versioning-enabled across Regions. **Most Voted**

C. Enable IAM Access Analyzer for S3 to identify all S3 buckets that are not versioning-enabled across Regions.

D. Create an S3 Multi-Region Access Point to identify all S3 buckets that are not versioning-enabled across Regions.

**Correct Answer:** B

*Community vote distribution*

B (100%)



## Comments

**mk168898** 1 year, 7 months ago

sotrage lens used to identify non versioning

upvoted 1 times

**Scheldon** 2 years ago

**Selected Answer: B**

AnswerB

S3 Sorage Lens "can also identify buckets that aren't following data-protection best practices, such as using S3 Replication or S3 Versioning. "

https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage_lens_basics_metrics_recommendations.html

upvoted 3 times

**sandordini** 2 years, 1 month ago

**Selected Answer: B**

You can use the Versioning-enabled bucket count metric to see which buckets use S3 Versioning. Then, you can take action in the S3 console to enable S3 Versioning for other buckets.

upvoted 2 times

Question #890 Topic 1

A company needs to optimize its Amazon S3 storage costs for an application that generates many files that cannot be recreated. Each file is approximately 5 MB and is stored in Amazon S3 Standard storage.

The company must store the files for 4 years before the files can be deleted. The files must be immediately accessible. The files are frequently accessed in the first 30 days of object creation, but they are rarely accessed after the first 30 days.

Which solution will meet these requirements MOST cost-effectively?

A. Create an S3 Lifecycle policy to move the files to S3 Glacier Instant Retrieval 30 days after object creation. Delete the files 4 years after object creation. **Most Voted**

B. Create an S3 Lifecycle policy to move the files to S3 One Zone-Infrequent Access (S3 One Zone-IA) 30 days after object creation. Delete the files 4 years after object creation.

C. Create an S3 Lifecycle policy to move the files to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days after object creation. Delete the files 4 years after object creation.

D. Create an S3 Lifecycle policy to move the files to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days after object creation. Move the files to S3 Glacier Flexible Retrieval 4 years after object creation.

**Correct Answer:** A

*Community vote distribution*

A (64%) C (33%) 2%

## Comments

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: A**

Although it's not stated what is meant by 'rarely accessed', this scenario would primarily be a candidate for the Glacier Instant Retrieval tier as the storage price would be more than 3 times lower compared to Standard IA. In the specific case of files being more frequently retrieved than quarterly, it can qualify for consideration of Standard IA.
Actually, we don't have the required info, so we have to guess what they are thinking.. which is pretty lame, to be honest..

upvoted 14 times

**sheilawu** 1 year, 12 months ago

S3 Glacier Instant Retrieval only retain the file for maximun 90 days, that is why A is not correct answer.
So C is right answer.

upvoted 2 times

**Edwars** 1 year, 11 months ago

No, it isn't. It is just the opposite.

Objects archived to S3 Glacier Flexible Retrieval have a minimum of 90 days of storage. If an object is deleted, overwritten, or transitioned before 90 days, a pro-rated charge equal to the storage charge for the remaining days will be incurred.

https://aws.amazon.com/s3/faqs/?nc=sn&loc=7

upvoted 4 times

**MatAlves** 1 year, 8 months ago

Hell no.

S3 Glacier Instant Retrieval is designed for long-lived, rarely accessed data that is retained for months or years. Objects that are archived to S3 Glacier Instant Retrieval have a minimum of 90 days of storage

upvoted 2 times

**Dantecito** 1 year, 3 months ago

The 90-day minimum storage duration means that if you delete, overwrite, or transition an object before 90 days, AWS will charge you for the remaining unused storage days.
It's a bit confusing but it's not like you can only have the files for 90 days, all glaciers are intended to retain data for archival purposes and 4 years as the question says is archival.
upvoted 2 times

**bmmbmm** Most Recent 6 months ago
Selected Answer: A
Glacier Instant Retrieval : Storage cost - low(way cheaper than Standard IA), access cost : optimized
Standard IA : Storage cost - average, access cost : low
so, access some times -> Standard IA is better, rarely accessed -> Glacier is better
upvoted 1 times

**tch** 1 year, 3 months ago
Selected Answer: A
B is out!
store the files for 4 years before the files can be deleted - for long term we should use S3 Glacier Instant Retrieval.

files are frequently accessed in the first 30 days of object creation, it actually store in Amazon S3 Standard storage
upvoted 1 times

**tch** 1 year, 3 months ago
Each file is approximately 5 MB and is stored in Amazon S3 Standard storage, meaning when file created .. files already store in Amazon S3 Standard storage.. move to move the files to S3 Glacier Instant Retrieval 30 days after object creation is fine...
upvoted 1 times

**a8a1e0e** 1 year, 3 months ago
Selected Answer: C
Option C is correct instead of A because S3 Standard-IA provides lower long-term storage costs with immediate access, whereas S3 Glacier Instant Retrieval has higher retrieval costs, making it less cost-effective for files that may still need to be accessed occasionally.
upvoted 1 times

**dfgdsfgfdgreg** 1 year, 4 months ago
Selected Answer: A
Glacier is much cheaper for storage than standard. The catch is you have to pay for 90 days of storage no matter what, even if the object is deleted. Here we will store data for 4 years so glacier is the cheapest solution.
upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago
Selected Answer: C
A - S3 Glacier Instant Retrieval is designed for archival storage where objects are infrequently accessed but still require millisecond retrieval, so yes actually A will work. It's just that Glacier storage costs more than S3 Standard-IA (given that after 30 days, the files are only RARELY accessed), so this is not the most cost-effective option.
B - For files that "cannot be recreated," durability is crucial, so single Availability Zone is not OK.
C - S3 Standard-IA is optimized for objects accessed less frequently but still requires rapid retrieval and high durability.
D - After 4 years, let's just delete those files.
upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago
Just a thought, if not constrained to these option, I would simply use Intelligent-Tiering to automatically move the files between access tiers (frequent access, infrequent access, etc) based on actual usage. Eventually, I'd delete these files after 4 years as well. I think this approach is better.
upvoted 1 times

**MatAlves** 1 year, 8 months ago
Selected Answer: A
Instant Retrieval = immediately accessible.

Coose the S3 Glacier Instant Retrieval storage class when you need milliseconds access to low cost archive data.

https://aws.amazon.com/s3/faqs/?nc=sn&loc=7
upvoted 3 times

**scaredSquirrel** 1 year, 9 months ago
Selected Answer: A
Glacier Instant Retrieval is much cheaper, and it is intended for archival storage with very low access patterns
upvoted 3 times

**Ucy** 1 year, 10 months ago
Selected Answer: C

While S3 Glacier Instant Retrieval offers immediate access, it has a minimum storage duration policy, Objects stored in S3 Glacier Instant Retrieval have a minimum storage duration of 90 days.

upvoted 2 times

**FrozenCarrot** 1 year, 11 months ago

Selected Answer: A

S3 Glacier Instant Retrieval delivers the fastest access to archive storage, with the same throughput and milliseconds access as the S3 Standard and S3 Standard-IA storage classes.
--- https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/

upvoted 4 times

**EdricHoang** 1 year, 11 months ago

Selected Answer: B

cost effective -> OneZone IA
"The files must be immediately accessible" -> cannot be Glacier

upvoted 1 times

**MatAlves** 1 year, 8 months ago

Instant Retrieval = immediately accessible.

Coose the S3 Glacier Instant Retrieval storage class when you need milliseconds access to low cost archive data.

https://aws.amazon.com/s3/faqs/?nc=sn&loc=7

upvoted 2 times

**Johnoppong101** 1 year, 9 months ago

files can not be recreated -> OneZone IA not accepted

upvoted 2 times

**Mayank0502** 1 year, 11 months ago

Selected Answer: C

this option has most durability

upvoted 1 times

**Lin878** 1 year, 11 months ago

Selected Answer: C

"are rarely accessed after the first 30 days" - not often
I will go with "C".

upvoted 1 times

**345645a** 1 year, 11 months ago

https://aws.amazon.com/es/s3/storage-classes/glacier/instant-retrieval/

upvoted 2 times

**sheilawu** 2 years ago

Selected Answer: C

mmediately accessible =>C

upvoted 1 times

**Scheldon** 2 years ago

Selected Answer: C

AnswerC
We cannot choose B because if that one zone will fail, company will not be able to recreate them.
We cannot choose D because we do not have to store files after 4y hence we can delete them (cost savings)
We cannot choose A - Glacier is less expensive (0,004 per GB) then S3-Standard - IA but is not allowing for instant access which is one of requirements (there is no information that data access shouldn't be accessible immedietly) we have only information that after 30d access to data is less frequently. Hence I think we need to choose S3 Standard - IA (answer C)

upvoted 4 times

**bujuman** 2 years ago

Selected Answer: C

Requirements:
- frequently accessed for 30 days
- lower cost

Features for S3 Standard-IA:
- Infrequently accessed objects
- Milliseconds to acces

According to me, best option for this use case is C
NB: Glacier better suits for lower cost, infrequent access.
upvoted 2 times

Question #891 *Topic 1*

A company runs its critical storage application in the AWS Cloud. The application uses Amazon S3 in two AWS Regions. The company wants the application to send remote user data to the nearest S3 bucket with no public network congestion. The company also wants the application to fail over with the least amount of management of Amazon S3.

Which solution will meet these requirements?

A. Implement an active-active design between the two Regions. Configure the application to use the regional S3 endpoints closest to the user.

B. Use an active-passive configuration with S3 Multi-Region Access Points. Create a global endpoint for each of the Regions.

C. Send user data to the regional S3 endpoints closest to the user. Configure an S3 cross-account replication rule to keep the S3 buckets synchronized.

D. Set up Amazon S3 to use Multi-Region Access Points in an active-active configuration with a single global endpoint. Configure S3 Cross-Region Replication. **Most Voted**

**Correct Answer:** D

*Community vote distribution*

D (100%)



## Comments

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: D**

Using a Multi-region Accesspoint in an Active-Active setup will send data to the closest Region, without accessing the internet: "send remote user data to the nearest S3 bucket with no public network congestion"
Not very easy to read and understand but it's all there: https://docs.aws.amazon.com/AmazonS3/latest/userguide/MultiRegionAccessPoints.html

upvoted 16 times

**LeonSauveterre** Most Recent 1 year, 5 months ago

**Selected Answer: D**

A - The "implementation" is so much more complicated than you think.
B - Active-passive? Then you gotta promote the passive region to active manually in case of failure.
C - This option does nothing to achieve low-latency or failover requirements.
D - It simplifies configuration and management with a single global endpoint, and provides seamless failover and high availability through Multi-Region Access Points and Cross-Region Replication.

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

And yes, the single global endpoint in option D will automatically route traffic to the nearest bucket, which ensures low-latency, which is a plus.

upvoted 1 times

**muhammadahmer36** 1 year, 11 months ago

**Selected Answer: D**

D. Set up Amazon S3 to use Multi-Region Access Points in an active-active configuration with a single global endpoint. Configure S3 Cross-Region Replication.

upvoted 2 times

**ike001** 1 year, 11 months ago

D is correct

upvoted 3 times

**Scheldon** 2 years ago

**Selected Answer: D**

Answer D

https://docs.aws.amazon.com/AmazonS3/latest/userguide/MultiRegionAccessPoints.html

When you create a Multi-Region Access Point, you specify a set of AWS Regions where you want to store data to be served through that Multi-Region Access Point. You can use S3 Cross-Region Replication (CRR) to synchronize data among buckets in those Regions. You can then request or write data through the Multi-Region Access Point global endpoint. Amazon S3 automatically serves requests to the replicated dataset from the closest available Region. Multi-Region Access Points are also compatible with applications that are running in Amazon virtual private clouds (VPCs), including those that are using AWS PrivateLink for Amazon S3.

https://docs.aws.amazon.com/AmazonS3/latest/userguide/replication.html

upvoted 3 times

**1223d0e** 2 years, 1 month ago

To me it looks like C, the requirement is to send the request to the closest region

upvoted 1 times

**trongod05** 1 year, 8 months ago

It's D. In an active-active configuration, requests made to an S3 Multi-Region Access Point's global endpoint automatically route over the AWS global network to the nearest S3 bucket. This allows applications to automatically avoid congested network segments on the public internet, improving application performance and reliability.

upvoted 2 times

Question #892 *Topic 1*

A company is migrating a data center from its on-premises location to AWS. The company has several legacy applications that are hosted on individual virtual servers. Changes to the application designs cannot be made.

Each individual virtual server currently runs as its own EC2 instance. A solutions architect needs to ensure that the applications are reliable and fault tolerant after migration to AWS. The applications will run on Amazon EC2 instances.

Which solution will meet these requirements?

A. Create an Auto Scaling group that has a minimum of one and a maximum of one. Create an Amazon Machine Image (AMI) of each application instance. Use the AMI to create EC2 instances in the Auto Scaling group Configure an Application Load Balancer in front of the Auto Scaling group.

B. Use AWS Backup to create an hourly backup of the EC2 instance that hosts each application. Store the backup in Amazon S3 in a separate Availability Zone. Configure a disaster recovery process to restore the EC2 instance for each application from its most recent backup.

C. Create an Amazon Machine Image (AMI) of each application instance. Launch two new EC2 instances from the AMI. Place each EC2 instance in a separate Availability Zone. Configure a Network Load Balancer that has the EC2 instances as targets.
**Most Voted**

D. Use AWS Mitigation Hub Refactor Spaces to migrate each application off the EC2 instance. Break down functionality from each application into individual components. Host each application on Amazon Elastic Container Service (Amazon ECS) with an AWS Fargate launch type.

**Correct Answer:** C

*Community vote distribution*

A (50%) C (50%)



## Comments

**Vasiliy** Highly Voted 2 years ago

**Selected Answer: A**

Autoscaling with max=1 is what is needed to keep only one instance at a time - it will still fail, but it will spawn exactly one instance in case of failure (we are not allowed to change the design of the app)
Having single instances in different AZ will not help - if one of the AZs is down, the app will still be affected

upvoted 13 times

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: C**

NOT A: Autoscaling with Maximum of 1 EC2 :D
NOT B: Hourly backup... RPO 1hr
C: AMI, Multi-AZ -> Fault tolerant
NOT D: ECS with Fargate, but it needs to run on EC2..

upvoted 12 times

**Scheldon** 1 year, 11 months ago

You cannot change application and based on the story in the past there was only one server/copy of application running at the same time. So we cannot run more then one copy of any application at once.

It is possible to set Min and Max to 1 which will automatically bring up server when it will crash. Taking into consideration that we cannot change application design and load-balancing between regions would probably need that (no information if applications are statefull or stateless) i would go for solution in answer A

upvoted 5 times

**KarlosRC** Most Recent 3 months ago

**Selected Answer: C**
"ensure that the applications are reliable" - we need 2 AZ
upvoted 1 times

**ln2718** 10 months, 1 week ago
**Selected Answer: A**
The question is if "fault tolerant" is considered such with an automatic restart in an autoscaling group of 1? Terrible question, but the rest of the question matches.
upvoted 1 times

**bora4motion** 1 year, 1 month ago
**Selected Answer: C**
needs to be C for fault tolerant.
upvoted 1 times

**Dantecito** 1 year, 3 months ago
**Selected Answer: C**
C is the answerd.
A. Changes in application designs are not the same as changes in infrastructure, also is not fault tolerant and an ALB just for one instance? whats the point of that
upvoted 1 times

**dfgdsfgfdgreg** 1 year, 4 months ago
**Selected Answer: C**
A has a single point of failure = not fault tolerant
upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago
**Selected Answer: C**
A - Creating an AMI of each application instance ensures that the migrated applications are deployed consistently with their existing configurations, yes, but an Auto Scaling group with a minimum and maximum of one provides no redundancy. If the single instance fails, the application becomes unavailable.
B - Backup-based disaster recovery does not provide real-time fault tolerance. It introduces downtime during recovery and does not meet the reliability requirement.
C - Launching two EC2 instances in separate Availability Zones (AZs) provides high availability. If one AZ experiences an outage, the application remains accessible on the instance in the other AZ.
D - "Changes to the application designs cannot be made."
upvoted 2 times

**LeonSauveterre** 1 year, 5 months ago
I copied this about option A. It's got some valid points, but just for your references.
An Application Load Balancer is also not the best fit for all legacy applications, as it operates at the application layer (of 7-layer model) and requires HTTP/HTTPS traffic. So for protocols like FTP, SMTP or other TCP-based customized protocols, an ALB would be unsuitable.
upvoted 1 times

**JA2018** 1 year, 6 months ago
**Selected Answer: C**
I will choose Option C for the following reasons:

#1 AMI creation: Creating an AMI of each application instance allows for easy replication.

#2 Multiple Availability Zones: Launching instances in different Availability Zones provides redundancy in case of regional outages.

#3 Network Load Balancer: A Network Load Balancer distributes traffic across the instances, providing high availability by automatically routing requests to a healthy instance if one fails.
upvoted 2 times

**daveya** 1 year, 6 months ago
What if application is statefull and there's no backup?
upvoted 2 times

**zits88** 1 year, 7 months ago
**Selected Answer: C**
I don't see how people can choose A here when it is talking about reliability and the answer only has a maximum of 1. There's nothing about cost effectiveness so there is no reason this one is better. Definitely C.
upvoted 1 times

**Cpso** 1 year, 6 months ago
we cant change applicaiton logic to be run 2 instance concurrently.

upvoted 1 times

**muhammadahmer36** 1 year, 11 months ago

Selected Answer: A

A. Create an Auto Scaling group that has a minimum of one and a maximum of one. Create an Amazon Machine Image (AMI) of each application instance. Use the AMI to create EC2 instances in the Auto Scaling group Configure an Application Load Balancer in front of the Auto Scaling group.

upvoted 3 times

**EdricHoang** 1 year, 11 months ago

Selected Answer: C

Fault tolerance is not High Availability
Answer A is HA design, not Fault tolerance

upvoted 2 times

**Cpso** 1 year, 6 months ago

exam said we can change application logic. 99.9% of single app usually work strange if you run it 2 process concurrently and random each request to them.

upvoted 1 times

**Lin878** 1 year, 11 months ago

Selected Answer: A

A makes sense.
C is possible but manual intervention.

upvoted 2 times

**emakid** 1 year, 11 months ago

answer is A.
Auto Scaling Group:

Explanation:
Minimum and Maximum of one instance: Ensures that the instance is always running. If the instance fails, Auto Scaling will automatically replace it with a new one, maintaining high availability.
Amazon Machine Image (AMI): Captures the current state of the application instance, ensuring that new instances launched by Auto Scaling will have the same configuration.

Application Load Balancer (ALB):

Load Balancer: Distributes traffic to the instances in the Auto Scaling group, ensuring fault tolerance. Even though there is only one instance, the ALB can help manage incoming traffic and be ready for future scaling if needed.

For C:
While this provides high availability, it does not address fault tolerance as effectively as the Auto Scaling group approach. Without Auto Scaling, if an instance fails, manual intervention is required to launch new instances.

upvoted 4 times

**24b2e9e** 1 year, 11 months ago

A makes sense

upvoted 2 times

**Nm55569** 2 years ago

Selected Answer: A

It's either A or B but A is a better option. The application design cannot be changed so we don't know if it can run across 2 servers.

upvoted 3 times

Question #893 *Topic 1*

A company wants to isolate its workloads by creating an AWS account for each workload. The company needs a solution that centrally manages networking components for the workloads. The solution also must create accounts with automatic security controls (guardrails).

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Control Tower to deploy accounts. Create a networking account that has a VPC with private subnets and public subnets. Use AWS Resource Access Manager (AWS RAM) to share the subnets with the workload accounts. **Most Voted**

B. Use AWS Organizations to deploy accounts. Create a networking account that has a VPC with private subnets and public subnets. Use AWS Resource Access Manager (AWS RAM) to share the subnets with the workload accounts.

C. Use AWS Control Tower to deploy accounts. Deploy a VPC in each workload account. Configure each VPC to route through an inspection VPC by using a transit gateway attachment.

D. Use AWS Organizations to deploy accounts. Deploy a VPC in each workload account. Configure each VPC to route through an inspection VPC by using a transit gateway attachment.

**Correct Answer:** A

*Community vote distribution*

A (81%) B (19%)

## Comments

**bujuman** Highly Voted 2 years ago

**Selected Answer: A**

Statement:
- The solution also must create accounts with automatic security controls (guardrails).

https://docs.aws.amazon.com/controltower/latest/userguide/what-is-control-tower.html
AWS Control Tower provides a pre-packaged set of guardrails (policies) and blueprints (best-practice configurations) to ensure that the environment complies with security and compliance standards. It's designed to simplify the process of creating and managing a multi-account AWS environment while maintaining security and compliance.

upvoted 9 times

**sandordini** Highly Voted 2 years, 1 month ago

**Selected Answer: B**

It's a hard one. I'd go for B
Several accounts in an org, with central mgmt > AWS Organization
Sharing resources among accounts > AWS RAM
AWS Organizations and RAM typically work well together...
Happy to be challenged, of course.

upvoted 6 times

**sandordini** 2 years, 1 month ago

Although automatic security control could be a hint for AWS Control Tower
(set up and operate your multi-account AWS environment with prescriptive controls)

upvoted 1 times

**FlyingHawk** Most Recent 1 year, 4 months ago

**Selected Answer: A**

AWS Control Tower automates the setup of accounts and guardrails, reducing the need for manual configuration.

Centralizing networking in a single account and sharing resources via AWS RAM minimizes the operational effort required to manage networking across multiple accounts.

upvoted 1 times

**LeonSauveterre** 1 year, 5 months ago

Selected Answer: A

A - It works but on first sight I eliminated this option because if you have a large number of workloads, the manual work would be too much. However, other options have bigger issues so this one is pretty great actually.
B - AWS Organizations alone does not provide the automatic guardrails, so it requires manual implementation of security controls and policies.
C - Also a valid choice, especially if more isolation and advanced traffic inspection are needed, but it introduces more complexity than option A and might require more operational oversight, which would not be ideal for "least operational overhead."
D - Like B.

upvoted 1 times

**mk168898** 1 year, 7 months ago

guard rails => AWS control tower

upvoted 2 times

**XXXXXlNN** 1 year, 8 months ago

A
Guardrails >> AWS Control Tower

upvoted 3 times

**dhewa** 1 year, 9 months ago

Selected Answer: A

AWS Control Tower provides built-in guardrails and automates the creation of accounts with security controls.

upvoted 2 times

**muhammadahmer36** 1 year, 11 months ago

Selected Answer: A

A. Use AWS Control Tower to deploy accounts. Create a networking account that has a VPC with private subnets and public subnets. Use AWS Resource Access Manager (AWS RAM) to share the subnets with the workload accounts.

upvoted 2 times

**emakid** 1 year, 11 months ago

Selected Answer: A

It leverages AWS Control Tower for automated account deployment and management, along with AWS RAM for centralized networking management, thus minimizing operational overhead while meeting the company's requirements for workload isolation and automatic security controls.

upvoted 3 times

**stalk98** 2 years ago

Selected Answer: A

answer is A

upvoted 2 times

**Tomrr** 2 years ago

Selected Answer: A

Anser is A, Control Tower has guardrails

AWS Audit Manager provides an AWS Control Tower Guardrails framework to assist you with your audit preparation.

upvoted 2 times

**Scheldon** 2 years ago

Selected Answer: A

Taking into consideration that AWS Control Tower is Orchestrator for AWS Organization which applies guardrails, I think A is a good choose.

https://docs.aws.amazon.com/controltower/latest/userguide/what-is-control-tower.html

upvoted 3 times

**1223d0e** 2 years, 1 month ago

Please explain why the answer is option A

upvoted 1 times

**jackey_feng** 2 years ago

I prefer B which is free. A may cause fee for sevice used while I am not sure about it.

upvoted 1 times